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Faculty of Informatics and Statistics



Cryptocurrency trading – technical analysis

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Declaration

I hereby declare that my bachelor's thesis on topic of „Cryptocurrency trading - technical analysis“ was worked out on my own and I have properly identified all the used literature and other sources and listed them in the attached list.

In Prague on

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Jakub Drobil

Acknowledgement

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Abstract

The subject of the bachelor's thesis "Cryptocurrency trading - technical analysis" is the analysis of the cryptocurrency space, how it is different to the traditional markets trading and exchange wise and what tools and indicators of technical analysis to use when trading it. The bachelor's thesis focuses not only on all of that but also on creating trading strategies based on the tools of technical analysis, using manual and automatic backtesting.

The aim of the work is to acquaint the reader with the basics of blockchain and how it is maintained, the cryptocurrency space and a few cryptocurrencies, the wide differences between cryptocurrency and traditional markets and technical analysis and its application to develop a suitable and ideally profitable trading strategy using various indicators. The data for the analysis is obtained from available internet sources that provide price development of cryptocurrency together with technical indicators. The obtained values are processed in either the MS Excel program or TradingView strategy terminal, where the basic mathematical and logical functions are used. The data is then evaluated using the trading strategy and the optimal variant is found and back tested on new data to verify its profitability.

Keywords

cryptocurrency, financial markets, Bitcoin, technical analysis, trading, trading strategy, blockchain, backtesting

JEL Classification

C5, E42, E59, G1

Abstrakt

Předmětem bakalářské práce „Technická analýza obchodu s kryptoměnami“ je analýza kryptoměnového prostoru a trhů, srovnání v čem se liší od tradičních trhů v jeho obchodních podmínkách a službách nabízených burzami, prezentace nástrojů a indikátorů technické analýzy a jejich aplikace pro obchodování. Bakalářská práce se zaměřuje nejen na to, ale také na tvorbu obchodních strategií na základě nástrojů technické analýzy s využitím ručního a automatického zpětného testování.

Cílem práce je seznámit čtenáře se základy blockchainu a jeho údržbou, s kryptoměnovým prostorem a několika kryptoměnami, širokými rozdíly mezi kryptoměnami a tradičními trhy a technickou analýzou a její aplikací k vývoji vhodné a ideálně ziskové obchodní strategie využívající různé indikátory. Data pro analýzu byla získávána z dostupných internetových zdrojů, které poskytují vývoj cen kryptoměn spolu s technickými ukazateli. Získané hodnoty jsou zpracovány buď v programu MS Excel nebo v terminálu TradingView, kde jsou použity základní matematické a logické funkce. Data jsou následně vyhodnocena v rámci obchodní strategie a je nalezena optimální varianta pro použití, která je následně ověřena.

Klíčová slova

kryptoměny, finanční trhy, Bitcoin, technická analýza, obchodování, obchodní strategie, blockchain, backtestování

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Introduction

Anyone who is interested in what is going on in the world had to notice that the words Bitcoin and cryptocurrency have been popping up a lot in the past few years in social media and even on television. They became a huge topic firstly for investors and later also for the public. Although the technology they present is fantastic the interest in them spiked up mainly due to their fast rise in price, making many people an overnight millionaires and leaving the rest empty handed.

The thesis is divided into 6 main chapters. The first two chapters introduce the blockchain and how it relates to cryptocurrencies, the technical part of cryptocurrencies from the likes of mining and cryptography up to the various ways of entering the crypto space, earning cryptocurrencies, and enjoying the virtual reality that cryptocurrencies present and their real-life use cases. Few chosen cryptocurrencies are then explained in more detail to the reader.

The third and fourth chapter acquaint the reader with the differences between the cryptocurrency and traditional markets in aspects like market cycles, trading and charting software, styles, and hours, how to define strength of a market. A few different cryptocurrency exchanges and traditional market broker will be presented in detail, the differences between them will be made clear and a trader profile for each will be created.

The fifth chapter explains the basics of technical analysis, its tools and indicators offered and a possible way of trading them. At the end of the theoretical part, the basic rules of money and risk management, journaling and position sizing are explained. The practical part forms last chapter of the thesis. It combines the knowledge learned about technical analysis in the previous chapters and creates trading strategies based upon them. They are then back tested manually or automatically for profitability and depending on the results they can be put in use or optimized.

The aim of the work is to acquaint the reader with tools and indicators of technical analysis and to create ideally a profitable trading strategy using them which can be used to trade the cryptocurrency markets.

The data for the analysis is obtained from available internet sources that provide price development of cryptocurrencies together with technical indicators.

This thesis is intended for anyone interested in starting or improving his trading journey across the cryptocurrency and traditional markets or interested in learning about the main differences between them and how to use various technical analysis tools and indicators to trade them. The thesis could also serve as a general introduction to the cryptocurrency space for anyone interested in learning about it.

1 Introduction to cryptocurrencies

In this chapter I would like to present how cryptocurrencies work, briefly talk about their history, development, how they fit in our world, monetary system and how they can be utilized, the technology they run on, its pros and cons and how they affect us in general. To fully understand the topic there are few words I will explain like blockchain and its consensus mechanisms, cryptography, NFT, metaverse and more.

We can find a well-fitting definition of cryptocurrency on Kaspersky website „Cryptocurrency, sometimes called crypto-currency or crypto, is any form of currency that exists digitally or virtually and uses cryptography to secure transactions. Cryptocurrencies don't have a central issuing or regulating authority, instead they are using a decentralized system to record transactions and issue new units.“ (Kaspersky, ©2022) As any other non digital currencies, they work as a medium of exchange. (River, ©2021)

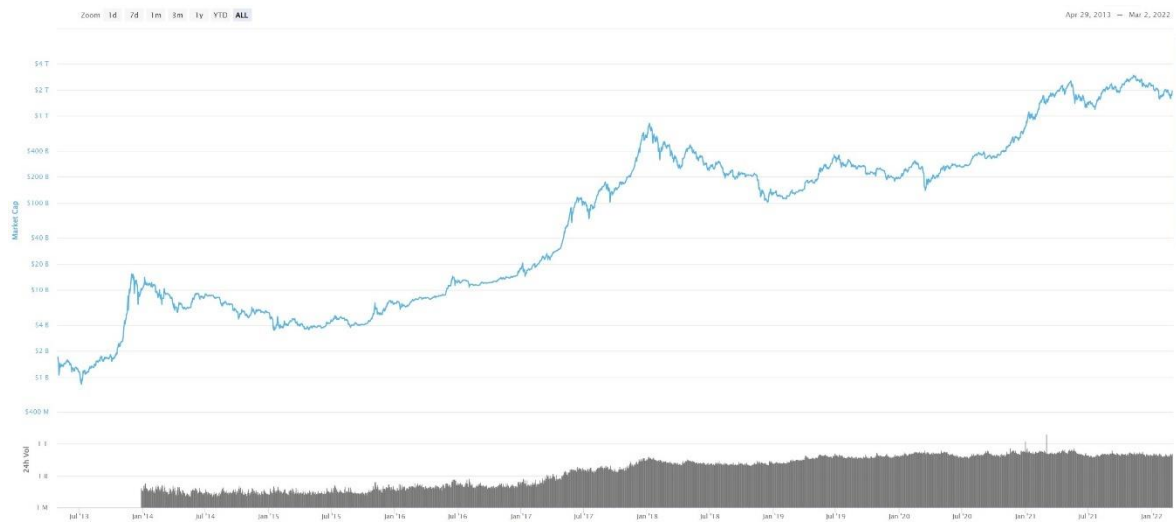
First mentions of cryptocurrencies take us back to the 1980s where American cryptographer David Chaum invented digital cash, which relied on cryptography for its transactions, but that was very far from what came later. In 2008 a so-called Satoshi Nakamoto published a paper titled Bitcoin: A Peer-to-Peer Electronic Cash System in which he outlined a system to create a digital currency that wouldn't require trust in a third party. In 2010 Bitcoin started getting its first adopters but it wasn't until 2017 where it saw its big boom and adoption of institutions. That year Bitcoin went from \$1,000 to \$20,000. (The Guardian, 2021)

Cryptocurrencies became popular very fast for reasons like financial potencial as a new type of investment which made unbelievable amounts of money for its early adopters and as a new piece of technology. Both are why many people and especially the younger generations that grew up with phones and tablets in their hands are driven to them. Some even see the possibility of crypto replacing traditional money but that possibility is still far away. (Supply Chain Game Changer, 2021) Many see cryptocurrencies as a way of controlling their own money, due to their decentralized nature but that is not that true. To be fair the idea created by Satoshi Nakamoto is great but a fully decentralized world without no banks or states sounds as utopia.

As of today 1st, March 2022 there are currently around 18,000 different cryptocurrencies according to the biggest cryptocurrency tracking website. (CoinMarketCap, ©2022) Bitcoin's market cap alone makes up for around 40% of the total cryptocurrency market cap and is worth \$770B which makes it the 9th largest asset in the world after Tesla on the number 8th spot. The number one ranked asset is currently Gold with a market cap of \$12,36T. (ComapaniesMarketCap, ©2022)

For better visualization of its insane growth in such a short time here is a logarithmic chart of the total cryptocurrency market cap dated from April 29, 2013, to March 2, 2022.

Figure 1: Cryptocurrency market cap



own processing according to (CoinMarketCap)

1.1 Blockchain

The basic definition of a blockchain is a transparent „*distributed database shared among nodes of computer network*“ that collects information and puts it in blocks that connect. (Investopedia, 2022) In other words a global public ledger if all transactions.

How blockchain is stored and how it stores its data can be found in the book Mastering Bitcoin „*The blockchain data structure is an ordered, back-linked list of blocks of transactions. The blockchain can be stored as a flat file, or in a simple database.*“ (Antonopoulos, 2017)

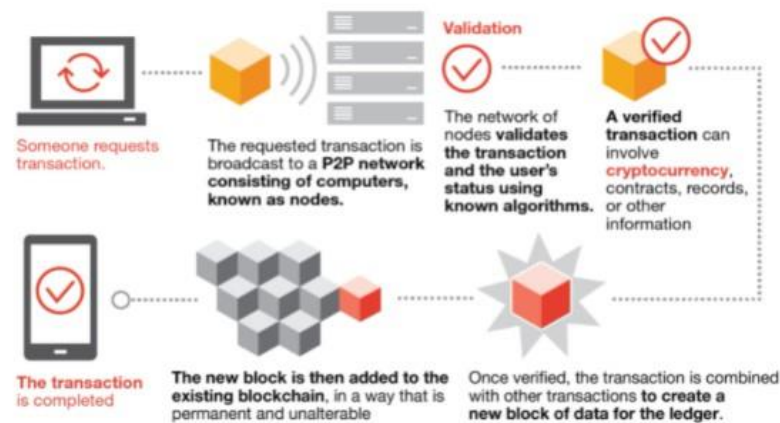
The book also gives a more friendly definition of a blockchain: „*The blockchain is often visualized as a vertical stack, with blocks layered on top of each other and the first block serving as the foundation of the stack.*“ (Antonopoulos, 2017)

The difference between a database and blockchain is that database stores its data into tables whereas blockchain collects information in groups called blocks that have a capacity limit and when filled, are closed, and linked to the previously closed block. That's where it got its name because its structure is a chain of blocks. (Investopedia, 2022)

Each block within the blockchain has its own identity, gained by the SHA256 hash algorithm. Each block within its header has a reference (hash) to the so-called parent one which is the previously closed block. Using the parent block hash, blocks connect each other creating a sequence, a chain that leads to the first block ever created called the genesis block. (Antonopoulos, 2017)

The figure below describes how a blockchain works.

Figure 2: Transactions on the blockchain



(by PwC, ©2022)

Blockchain security

If you would like to modify something in one of the blocks, you would need to modify it in all the other blocks too which would lead to a cascading effect of recalculation of the entire blockchain and that would require a very large computation power, therefore big energy consumption and time. The existence of long chain of blocks is what makes the history of a blockchain basically non-rewritable, and that is what makes Bitcoin that secure. (Antonopoulos, 2017)

1.2 Application of blockchain

Automotive

Consumers could use the blockchain to manage fractional ownership of cars. (PwC, ©2022) This would create a faster way of transferring rights and not only of cars but also property and much more which would lead to saving time and money. People could solve this on their own using a smartphone and an app without the need to go and report everything to the big institutions. There would be less need of paperwork because of that so the state wouldn't need to employ that many officials hence it would save money. Less paperwork means less need of paper which would help our planet and less trees would need to be cut down.

Financial services

Faster and cheaper settlements could save billions of dollars from transaction costs while improving transparency due to the transactions. (PwC, ©2022) Everything in the blockchain is transparent so there would be even less possibility of fraud and all the records would be easier to access.

Voting

Using a blockchain code, people could vote using smartphones, tablets, or computers, resulting in immediately verifiable results. (PwC, ©2022) In today's world everyone owns at least one of those devices, or has the access to them, and since blockchain is transparent the votes would be instantly visible and there could be no arguments like in the last president election in USA where Donald Trump was arguing that some votes were faked, and that the entire election was a lie.

Healthcare

Patients encrypted health information could be shared with multiple providers without the risk of privacy breaches. (PwC, ©2022) One system where anyone can access their medical reports, not only this would be helpful for the patients but the place where this would really shine would be in situations where the doctor needs to make a fast decision. This system would provide him with all the medical records of the injured in seconds and he then could make an informed decision on how to treat him.

e-Government

Blockchain could simplify how entire governments work and increase their efficiency of work by automating basic procedures. Creation of e-Identity which would act as a digital ID could be used when reporting taxes, sending bank reports, business registration or any other actions. This way the state could have a good overview of all its inhabitants. Great example of this is Estonia which already has their entire government automated and running on blockchain and its residents have access to all these time saving tools.

1.3 Blockchain consensus mechanisms

Consensus mechanisms are like sets of rules which the networks must obey, to maintain their credibility, security, and transparency.

Even though there is no central authority which validates and verifies transactions, every transaction in the blockchain is completely secured and verified. This is possible thanks to the various consensus protocols that maintain the blockchain networks. (GeeksforGeeks, 2022)

„A consensus algorithm is a procedure through which all the nodes of the blockchain network reach a common agreement about the present state of the distributed ledger.” (GeeksforGeeks, 2022) This means that the protocol checks that every new block which is added to the blockchain has the correct and true version. This is done by the nodes, they check if the block gives all of them the same response and if it does, it is then verified and can be added to the blockchain.

The blockchain consensus protocol contains few very specific objectives like coming to an agreement, co-operation, equal rights to every node, collaboration, and mandatory

participation of all nodes in the consensus process. In other words, consensus algorithm aims at finding a common agreement which is a win for the network. (GeeksforGeeks, 2022)

Bitcoin's decentralized consensus is made from four processes that are occurring independently on nodes across the network:

1. Independent verification of each transaction, by every full node, based on a comprehensive list of criteria
2. Independent aggregation of those transactions into new blocks by mining nodes, coupled with demonstrated computation through a Proof-of-Work algorithm
3. Independent verification of the new blocks by every node and assembly into a chain
4. Independent selection, by every node, of the chain with the most cumulative computation demonstrated through Proof-of-Work

(Antonopoulos, 2017)

Proof of Work (PoW)

The idea of Proof of Work was first published in 1993 and later applied by Satoshi Nakamoto in his Bitcoin paper. (GeeksforGeeks, 2019) Although Proof of Work consensus is still used by the biggest cryptocurrencies like Bitcoin, Ethereum or Raven coin, it no longer is the most popular consensus mechanism, the majority of cryptocurrencies nowadays runs on the Proof of Stake consensus which doesn't require such computing power and therefore is more environmentally friendly.

How PoW works is well described in an edX blockchain course: *„The Proof of Work consensus algorithm involves solving a computational challenging puzzle in order to create new blocks in the Bitcoin blockchain. Colloquially, the process is known as ‘mining’, and the nodes in the network that engage in mining are known as ‘miners’. The incentive for mining transactions lies in economic payoffs, where competing miners are rewarded with Bitcoins and a small transaction fee.”* (GeeksforGeeks, 2019) With time, the mathematical problem becomes more and more complex, and the computational power needed to solve it greatly increases. The amount of Bitcoins rewarded for solving the puzzle halves every four years which now is due at 2024.

The idea behind Proof of Work is that it's hard to find a solution for a mathematical problem, but it's then very easy to verify if that solution is correct. (GeeksforGeeks, 2019)

Bitcoin's blocks are generated every 10 minutes, on average. It's like its heartbeat and underpins the frequency of currency issuance and the speed of transaction settlement. It has to remain constant for over a period of many decades (there are 21 million Bitcoins, and it is expected that the last block will be mined in around 2140). Over this time, it is expected that computer power will continue to rise at a fast pace and the number of people in mining and their computers will undeniably change. To keep the block generation time at 10 minutes, the difficulty of mining must therefore be adjusted to account for these changes. In fact, the Proof-of-Work target is a dynamic parameter that is periodically adjusted to meet a 10-minute block interval goal. (Antonopoulos, 2017)

How does the consensus adjust to the number of miners and their speed is explained in detail in the book Mastering Bitcoin by Andreas M. Antonopoulos: „Every 2,016 blocks, all nodes retarget the Proof-of-Work. The equation for retargeting measures the time it took to find the last 2,016 blocks and compares that to the expected time of 20,160 minutes (2,016 blocks times the desired 10-minute block interval). The ratio between the actual timespan and desired timespan is calculated and a proportionate adjustment is made to the target.”

In other words: If the miners and their computing power finds a block faster than 10 minutes, the mining difficulty increases. If the block discovery is slower than 10 minutes the difficulty decreases so the balance is maintained. This process is revalued every 20,160 minutes.

Proof of Stake (PoS)

Proof of Stake is another consensus mechanism of the blockchain. Most cryptocurrencies nowadays use it, instead of the Proof of Work one because it is more environmentally friendly and doesn't require such power to be run.

What PoW issues does PoS solve:

- Better transaction speed and fees
- Low barriers of entry
- Electricity consumption and network maintenance
- Miners' lack of interest in contributing value to the network

Proof of stake has its own type of 'miners', they are called validators and are responsible for maintaining the network. To be a validator on the Ethereum network you need to own atleast 32 ETH. (diit, 2020)

Proof of stake works in the way that the validators are selected to check and confirm blocks on the blockchain at random, they aren't competing between each other nor they need to mine blocks and that's why the energy consumption can be much lower than in PoW. The only downside is that the validator can lose a part of his staked coins for going offline, so he needs a stable internet connection 24/7. (Ethereum, 2022)

Proof of Work X Proof of Stake

Let's summarize the differences between PoS and PoW.

The probability of mining a block depends on the computational power the miner has in PoW whereas the probability in PoS depends on how many coins the validator has. (GeeksforGeeks, 2020)

When it comes to the rewards in PoW the first miner to solve the cryptographic puzzle of each block gets the reward whereas in PoS the validator doesn't receive any direct reward, instead he makes his bread by maintaining the network by receiving transaction fees from whoever uses the network. (GeeksforGeeks, 2020)

To hack a PoW blockchain a miner would need to own more than 51% of computing power of the entire network, which is unimaginable considering how big Bitcoin is. In PoS the hacker would need over 51% of all the cryptocurrency stored on the network, which makes it also extremely secure. (GeeksforGeeks, 2020)

The cost of running a PoW network is much higher energy and investment wise because the miners need high tier graphics cards or mining machines and the power to run them 24/7, in the case of PoS a simple server unit is more than enough. (GeeksforGeeks, 2020)

1.4 Cryptography

Cryptography is a science that focuses on methods of maintaining digital security and privacy. Today even the simplest electronic devices use at least a bit of cryptography to protect sensitive information that is inside. In cryptocurrencies it is used to implement the decentralized monetary system and secure it. Simply put cryptography is used to convert data into an incomprehensible pile of code that only chosen parties can read after receiving it, this process is called encryption and decryption. Bitcoin's network doesn't need any encryption since the intention of it is to be an open one which anyone can use and connect through. However, software wallets and such do require encryption because they should only be accessed by their owners and no one else.

Hash Functions

Hash functions is a mathematical function which takes any data as input and produces an output with special characteristic – a large string of letters and numbers. It is used because it can rapidly verify big piles of data. The data is considered verified when its hash is the same as when previously checked. If the data changes the hash will too and because of that the verification will fail. Hashing is how the Proof-of-Work consensus cryptocurrencies work. Bitcoin uses the SHA-256 function for its operations. Hashing is what miners with their equipment do to submit a block to the blockchain, they have to find a hash that is below a certain threshold and that is done by guessing, to do so a large computation power is required. The security of blockchain is strong because each block the miners submit has the hash of the previous one creating a chain, once a block is submitted it cannot be altered. (River, 2022)

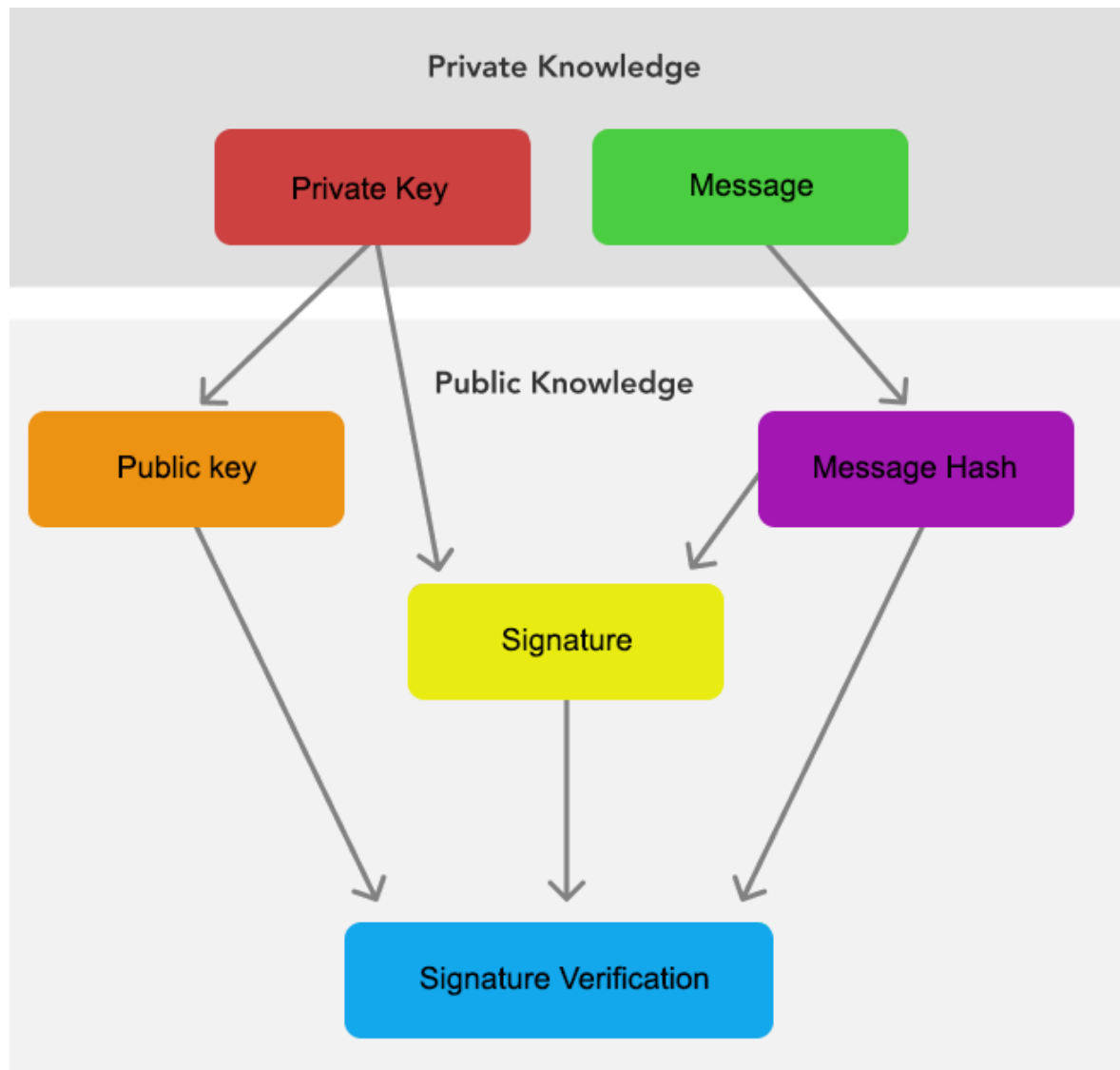
Asymmetric cryptography

In cryptocurrencies asymmetric cryptography (public key cryptography) is used. It is a cryptographic method in which different keys are used to encrypt and decrypt so that anyone with access to the public key can use it to encrypt a message, but only the private key holder can decipher the message. In addition to the possible secrecy of communication, asymmetric cryptography is also used for digital signatures to prove the author of its data. (Stalling, 2017)

Digital signatures

Digital signatures connect approval of data being signed to a specific identity of the signer. Bitcoin protocol allows pieces of bitcoin to be sent to a public key, such that only a valid signature from the corresponding private key can unlock. This signature is then sent to the blockchain so that any miner can verify that the signature, public key, and message match. Thanks to this process, miners can't produce valid signatures for other people and steal their bitcoins. (CoinDesk, 2021)

Figure 3: Digital Signature Structure



own processing according to (River)

1.5 Getting into cryptocurrencies

There are many ways to enter the cryptocurrency space, some don't even require any initial capital which is great because anyone can enter.

Buying

Buying is the standard way to get your hands on some cryptocurrencies. To buy cryptocurrencies, you need a debit/credit card or a bank account from which you can send money onto an exchange where you can swap your fiat currency for a crypto one, the other way around is by a peer-to-peer transaction between two people, these transaction can be risky because there is a lot of scammers in the crypto space, to avoid the huge amount of scams some exchanges have a marketplace for P2P¹ transactions where they take the role of the third party, that makes sure both the buyer and the seller gets their crypto or fiat. Some people prefer this method because buying crypto can be a bit tricky for newcomers. (Forbes, 2022), (Binance, 2022)

To store your bought cryptocurrencies, you can either use a digital or hardware wallet or you can keep it on the exchange if you plan to trade with it. If you want to keep your cryptocurrencies as safe as they can be, the hardware wallet is the way to go. It works as a USB drive which you insert into your PC and to which you can move your funds from an exchange or a digital wallet. A digital wallet works as a browser extension, desktop application or a mobile phone application and is suitable for moving funds to different blockchains and is relatively safe, but it can still get hacked. The real only safe place where this cant happen is the hardware wallet where the hacker cannot access the funds unless he knows the 12 to 24 random words which are called the recovery seed and form the master access key through which you can access all cryptocurrencies associated with it. When you buy a cryptocurrency on an exchange it is still stored by it until you withdraw it to a wallet. Therefore, when an exchange would get hacked your cryptocurrencies could get stolen, this situation has happened a few times, but the exchanges returned the stolen money from its own reserves back to the users who got robbed to save their credibility. (Forbes, 2020)

Mining

Mining is a competitive process that is based on verifying and adding transactions to a block which is then added to the blockchain in PoW based cryptocurrencies. The miners are reward in transaction fees for maintaining the blockchain and in coins they mine for adding a new block to the blockchain. To become a miner a graphics card or a mining machine is required therefore the initial capital is quite big because high tier graphics card and mining machines are currently hard and expensive to get because there are not enough microchips. (FreemanLaw, ©2022)

Staking

In Proof-of-Stake consensus, blockchains transactions must be validated before being added to the blockchain and staking is how it is done.

To stake a cryptocurrency you firstly need to buy it, there are many PoS cryptocurrencies so you can choose from ones like Cardano, Polkadot, Solana and more. Some cryptocurrencies can be staked directly on the exchange after you buy them making staking user friendly but

¹ peer-to-peer

other require withdrawing from the exchange onto a software wallet and depositing in a staking pool. Staking pool consists of a group of users who join forces by combining their funds together for a bigger chance of being chosen to be the validator and get rewarded. (Fool, 2022)

Every time a block is added to the blockchain, new coins are minted and distributed as staking rewards to the validators. Usually, the rewards are the same cryptocurrency as the one being staked but it doesn't have to be always that way.

By staking your cryptocurrency you can earn passive income by earning interest. Using protocols like Anchor you can have as much as 20% APY². (Anchor, ©2022) There is no equipment needed and by staking you help to keep the blockchain running and increasing its efficiency. The only downside is that in certain staking pools there is a lockup period before you can withdraw your funds meaning you can withdraw your cryptocurrency only after a certain period of time which can complicate things when you want to sell the asset instantly. (Fool, 2022)

Yield Farming

Yield farming is another way to put your crypto assets to work utilizing the decentralized finance protocols like Aave or Compound. Yield farming is sometimes called liquidity mining even though it doesn't have anything in common with the PoW mining. It works similarly to staking where you earn interest on your cryptocurrency for locking it up in a liquidity pool, except here is no lockup period to it and you can instantly withdraw the funds. The difference between staking and farming is that yield farming requires two different assets which you deposit in the liquidity pool and is considered riskier, therefore the rewards are higher whereas staking requires only one asset, and the rewards are smaller. (BusinessInsider, 2021)

By depositing assets into the liquidity pools, yield farmers are rewarded with transaction fees and an annualized model which shows how much earnings they can get in percentages if they lockup their funds for a year. The percentage metric of the annualized model is either in APR³ or APY, the difference between them is that APY includes compounded interest, which involves depositing the rewards received back into the liquidity pool. (Forkast, 2021)

Yield farming has its risks, one of them are rug pulls which is a situation where the developer abandons the projects and runs off with investors funds. Another one is the smart contract risk; users have to deposit their funds into protocols smart contracts to earn the yield, if that smart contract is poorly coded, it can crash and all the funds within it are irreversibly lost. At last, there is impermanent loss, since yield farming requires two different assets price of each one could rise or fall, creating unrealized gains or losses which can become permanent with the withdrawal of funds from the liquidity pool. When there is huge volatility in the market keeping your funds on a wallet instead of putting them into a pool might sometimes have better results than staking them in a liquidity pool however when there is huge

² annual percentage yield

³ annual percentage rate

volatility. the liquidity miners are compensated by earning more from the transaction fees which they receive as a reward. (BusinessInsider, 2021)

Play to earn

Play to earn blockchain video games are new trend in the gaming industry and another way to get into cryptocurrencies if you enjoy gaming. The concept allows player to player transactions of items and true ownership of them thanks to non fungible tokens. Some of these games require an initial investment to play but many don't which is perfect because this way you can get cryptocurrency exposure for free. By playing these games and completing quests, winning battles against other players or doing other kinds of challenges you can earn a cryptocurrency which you can then swap in a in-game exchange to other cryptocurrencies like ETH, BTC or other tokens supported by the exchange and withdraw them to your software wallet through which you can deposit it on a regular crypto exchange and swap the cryptocurrencies earned into desired fiat currency. (FinanceFeeds, 2022)

„True potential of play-to-earn games to put food on the tables of families in developing nations. “(Andrius Miron, 2021) Players in a play to earn game called Axie Infinity can earn over \$40 a day just by completing daily quests, that much money in developing nations around the world can mean months wages of work and since the prices of food skyrocketed many people don't even make enough money to feed their families from their regular jobs and this could help them do it. (Beincrypto, 2021)

Variety of these games is truly endless ranging from chivalry times, space exploration in spaceships, card collecting games to horse races in which players have their own stable and breed horses on which they can bet while they compete with other players horses, players can even lend their horses for races for some interest. The options of how players can use in game items and currencies feel almost like the ones they would have in real world, which brings virtual reality even closer to real life.

Cryptocurrency ETF and companies connected to crypto

ETFs and companies connected to crypto are the best way to get into cryptocurrencies for those who don't desire direct exposure but are interested in the technology and space but don't have the ability or knowledge to go through the process of buying cryptocurrency on an exchange.

ETF is an exchange-traded fund, in our case it consists of cryptocurrencies. ETF usually tracks a particular index, sector, or other assets. In our case it is tracking the price of Bitcoin, Ethereum or other cryptocurrencies, it can even track more cryptocurrencies at once. The cryptocurrency ETF trades the same way the stocks do, it can be bought or sold on daily basis as it mimics the derivatives or spot prices. The advantage of it is the indirect exposure to crypto and the increased security level because the ETF providers are the ones responsible for keeping the cryptocurrencies secure so a possible hack wouldn't affect the investor, but that comes along with some fees. (Investopedia, 2022)

There are two types of cryptocurrency ETFs:

1. The ones that are backed by physical cryptocurrencies, where the investment company makes the purchase of the cryptocurrencies and investors buy them as shares and indirectly own them.
2. The ones that are synthetic and track the cryptocurrency derivatives prices like futures contracts. The most known synthetic Bitcoin ETF is traded on the Chicago Mercantile Exchange (CME).

Another way is to get cryptocurrency exposure is to invest in companies involved in the crypto space and technology that are focused on blockchain, crypto mining, or money transaction. Namely Coinbase which is a cryptocurrency exchange that recently went public, MicroStrategy which is an enterprise business intelligence application software vendor and the largest company holder of Bitcoin, owning 121,044 bitcoins, Tesla which is a car and technology-based company holding around 48,000 bitcoins or Hut 8 Mining and Riot Blockchain which are a cryptocurrency mining companies. When it comes to performance, these stocks are very correlated to the prices, trends and overall health of Bitcoin and the rest of the cryptocurrency markets. (Quartz, 2022)

There is also the possibility to get crypto exposure through trust funds. Trust funds can be described as companies that handle investments for other investors who give them their money. The biggest crypto related ones are Grayscale and Osprey. However, the required initial capital to use these ranges between \$25,000 to \$50,000 which is not a beginner friendly amount. Furthermore, there is a risk of dislocation between the performance of the trust fund and cryptocurrency prices. (MoneySmart, 2022)

1.6 Metaverse

Metaverse is an immersive virtual world that incorporates blockchain technology with cryptocurrencies bringing huge social and financial potencial. Using blockchain as the infrastructure gives the game developers an opportunity to implement real economy into a virtual world by having virtual items exchangeable between players for real economic value beyond the metaverse. By combining changing environments of VR, the popularity of video games, inactivity of social media and cryptocurrencies as payment system the metaverse is bound to be a key element in the future of blockchain gaming. (Gemini. ©2022), (Decrypt. 2022)

In metaverse worlds players can build and monetize their own virtual experiences and creations. Through the play-to-earn game model players are rewarded with cryptocurrency for playing the game and completing challenges and quests which they can later use to buy items, cosmetics or upgrade their character or even exchange it for fiat money through an exchange. The opportunities in metaverse worlds are so big that a player can make enough money to cover his costs of living just by playing the game. (Gemini. ©2022), (Decrypt. 2022)

Virtual reality is much closer to our reality than ever before, J.P. Morgan a big United States bank has opened a virtual lounge called Onyx in the metaverse project Decentraland shopping mall where people can see a floating portrait of the bank's CEO and stated that it

is looking for opportunities in the metaverse. (Coindesk, 2022) Electronics giant Samsung bought land in Decentraland on top of which they created an entire city where they held the Unpacked event where they showcased their latest phones and tech, the event also included some minigames which they created for the attendees to enjoy. (Trustnodes, ©2022)

Key part of every metaverse are NFTs⁴ which represent ownership of in-game items which can range from things like real estate, clothes, tools, art to spaceships. These items represented by the tokens can then be exchanged or sold between players using a virtual marketplace. (OpenSea, ©2022)

1.7 NFT

NFT is an abbreviation for non-fungible token. NFTs are tokens that represent ownership of a unique item a one out of one piece on the blockchain. NFTs offer creators tokenize and therefore monetize their art, collectibles, even real estate. An NFT can only have one owner, this is secured by the blockchain so there is no possibility of copy pasting it. (Ethereum, ©2022) Many large video gaming companies like Ubisoft view them as another type of monetization for their in-game content and therefore are looking to implement them into their future games. Not only video game developers but also artists view NFTs as another place to present and sell their artwork. A digital artist called Beeple made an art piece called Everydays: The First 5000 Days which was later sold for 38,525 ETH the USD equivalent at that time was \$69,346,250 making it the biggest NFT sale of all time. (Nftnow, 2022)

In 2017 NFTs first made an appearance in the form of CryptoKitties, it was a game centered around breeding collectible funny looking cats. It looked like a childish game that was ignored by many, but it brought something new to the cryptocurrency space, it opened a gateway to the world of tradable items in the digital world. Previously digital items lived on company servers such as the Steam marketplace but now they exist on public blockchains owned by no single party which wasn't possible before and they can easily be traded between users. (OpenSea, ©2022)

NFTs are the current trend in the cryptocurrency space. The largest NFT marketplace is on the Ethereum blockchain and it called OpenSea, its daily trading volume is between \$80-200M and its monthly trading volume spiked up to \$5,7B during January 2022. (Dune, 2022) On OpenSea you can buy and sell digital items like horses, real estate, cosmetics, playing cards, gifs, paintings and such from all the games backed by the Ethereum blockchain. The cost of these can vary from few dollars to hundreds of thousands of dollars, all the transactions fees and payments are settled in ETH. (OpenSea, ©2022)

⁴ non-fungible tokens

1.8 Cryptocurrencies real life use cases

Store of value

Bitcoin is a deflationary currency meaning there is only a limited amount of it and no extra can be created while fiat money can be printed when needed decreasing its overall purchasing power therefore its future value is expected to decrease. The largest asset by market cap is gold and that is because of its rarity, need in engineering and limited quantity. By those specifics it is considered a store of value asset by many meaning that it is expected to increase in value over time because of what it stands for. Many people look at Bitcoin the same way and call it the digital gold thanks to its similar specifics to gold like the limited amount, difficulty in mining, the fact that there cannot be any extra Bitcoin created after the last one is mined and its overall performance by increasing its value by thousands of percent since its creation. (Ledger, 2021)

Payments

Through banks sending a payment to a different country can take many days or even a week. Banks also don't work on weekends so when someone wants to send money back to his friend for paying for him at the Friday party, their friend will not receive the money until Monday. However, using cryptocurrencies this can be settled within minutes and at any time of the week because crypto is working 24/7. Banks also use the money clients put in their saving accounts to invest, meaning they play with their client's money, for small interest. This doesn't happen in the cryptocurrency world either, people can earn much higher interest on their coins while having full control of them. (Ledger, 2021)

Asset tokenization

Tokenization is a process of converting the ownership of anything from the real world into the cryptocurrency one using a token. Through tokenizations illiquid assets like real estate can be split into many tokens adding more liquidity to an illiquid asset making it transparent and easier to buy or sell. An expensive mansion could be split into tokens making it possible for people to buy parts of it and then receive rent depending on the percentage of the property tokens they own. (Ledger, 2021)

Storage

Using the blockchain anyone can rent out their free data storage space and earn passive income on it. Protocols like Filecoin offer this service and people who want to rent out some storage use the platform's native cryptocurrency to pay for it. (Ledger, 2021)

Lending and borrowing

Borrowing money from a bank is a complicated process that requires a lot of time and paperwork. In cryptocurrency world there are protocols that are made to make things like this much easier and faster. Lending protocols like Aave or Compound offer digital loans using smart contracts, users can borrow money using their cryptocurrencies as collateral within minutes and no irrelevant questions asked and paperwork needed because

everything is written down in the blockchain. These protocols offer much more than loans, user can also deposit their cryptocurrencies and earn interest on them by lending them to other users. (Ledger, 2021)

2 Selected cryptocurrencies

This chapter presents four cryptocurrencies that were precisely chosen due to their different real-life use cases. Their functionality, creation, purpose, and application will be explained in detail and since the thesis is mainly about technical analysis you will be presented also with some data like the price development charts.

2.1 Bitcoin

Bitcoin was invented in 2008 with the publication of a paper titled Bitcoin: A Peer-to-Peer Electronic Cash System written by author with aliases Satoshi Nakamoto. Nakamoto created a decentralized electronic cash system. Due to the system being decentralized it doesn't require any central authority for currency issuance or settlement and validation of transactions. all of this is done by the blockchain and its miners.

- Ticker: BTC
- Market Capitalization: \$770B⁵
- Foundation: 2008
- Creator: Satoshi Nakamoto
- Current price: \$40,608⁶

⁵ market capitalization to date 16.3.2022, available at: <https://www.coingecko.com>

⁶ current price to date 16.3.2022, available at: <https://www.coingecko.com>

In the chart below we can see the price development of Bitcoin since 28th April 2013 till 16th March 2022.

Figure 4: Bitcoin price development



own processing according to (Coinagecko)

The key invention was making the distributed computation system (we now know as Proof-of-Work algorithm), do an election every 10 minutes, allowing the decentralized network to come into a conclusion about the state of its transactions. This solved the issue of double spending⁷ which was the biggest weakness of previous digital currencies. To prevent it, all transaction had to be cleared through a central clearing house before, this job is now done by the miners. (Antonopoulos, 2017)

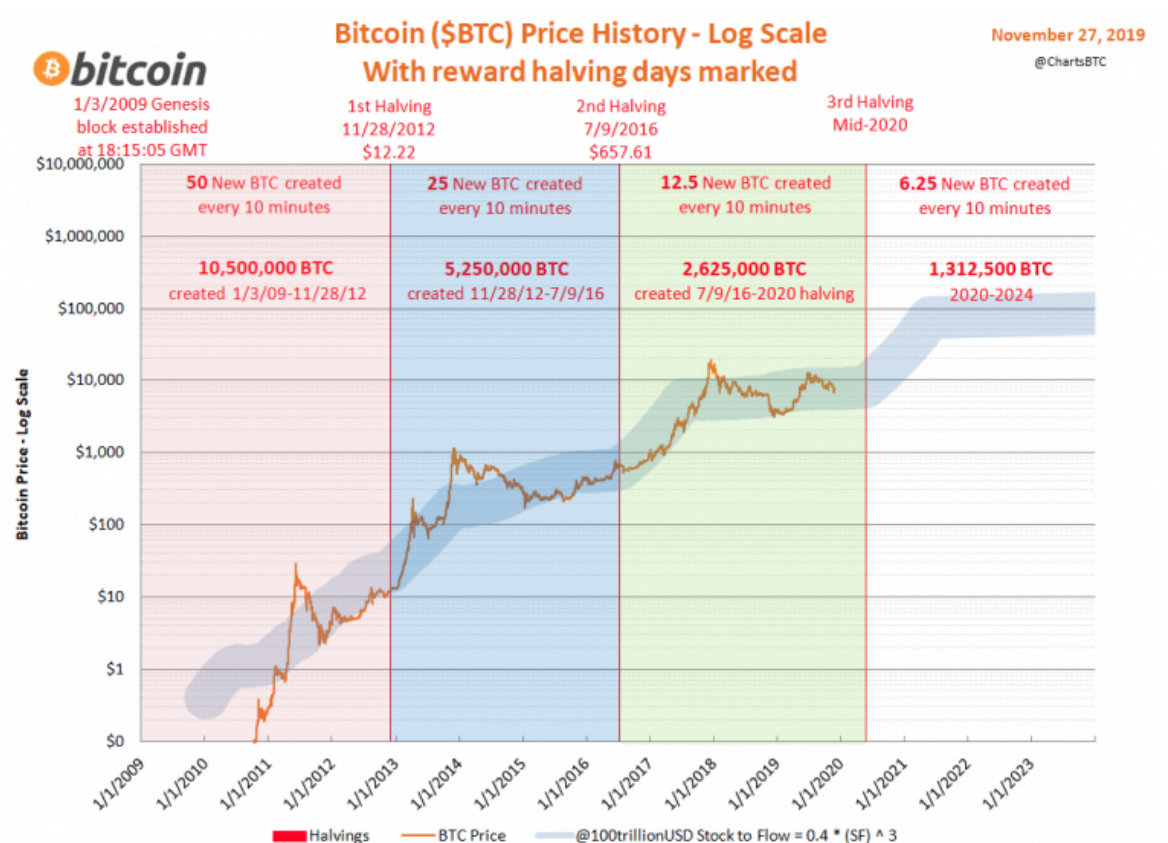
Unlike most electronic payment networks as credit/debit cards, Paypal and bank transfers Bitcoin transactions are irreversible. This introduces a very high risk for people unfamiliar with the system. If you send your Bitcoin to a different wallet address than desired its just gone, and no one can return it back to you. (Antonopoulos, 2017)

⁷ situation where a single currency could be spent twice

The Proof-of-Work consensus is what keeps Bitcoin running, this algorithm is maintained by the miners who provide computing power to secure the network by verifying transactions and mining blocks that group the transaction history. The goal of a miner is to be the one who generates first the hash that is below the last block's hash, the winner gets to add the latest block of transaction to Bitcoin's blockchain and is rewarded with some Bitcoin. Other income of miners are the transaction fees that users pay when they send transactions. There is a limited amount of Bitcoin that can be mined and its 21,000,000 therefore its considerer as a deflationary⁸ currency because of that some call Bitcoin the digital gold. (Fool, 2022)

An event called halving happens to Bitcoin, the idea of it is to keep the maximum supply of Bitcoin fixed, in contrast with the fiat currencies. This event causes the mining difficulty to increase by half, therefore double the amount of computing and energy that is required to mine a new block. Historical data shows that after halvings the price of Bitcoin tends to spike up. (Lavie, BusinessInsider 2021)

Figure 5: Logarithmic chart with Bitcoin halvings



(@ChartsBTC on Twitter,2019)

⁸ there is only a limited amount of it

2.2 Altcoins

Altcoin is a term used for all other cryptocurrencies than Bitcoin. Altcoins share some characteristics with Bitcoin but are different in other ways like in what consensus mechanisms they use and what capabilities they offer.

2.2.1 Ethereum

Ethereum is the second largest cryptocurrency after Bitcoin. It was created in 2015 and uses the same consensus mechanism as Bitcoin. Its creators are trying to upgrade it from the heavy energy consuming Proof of Work to the more sustainable Proof of Stake, the upgrade from PoW to PoS is on its way and should arrive in few years. Another difference between Bitcoin and Ethereum is that Ethereum has an unlimited supply whereas Bitcoin's supply is locked to 21,000,000 (Ethereum, ©2022)

- Ticker: ETH
- Market Capitalization: \$323B⁹
- Foundation: 2015
- Creator: Vitalik Buterin, Gavin Wood, Charles Hoskinson, Anthony Di Iorio, Joseph Lubin
- Current price: \$2683¹⁰

In the chart below we can see the price development of Ethereum since 7th August 2015 till 16th March 2022.

⁹ market capitalization to date 16.3.2022, available at: <https://www.coingecko.com>

¹⁰ current price to date 16.3.2022, available at: <https://www.coingecko.com>

Figure 6: Ethereum price development



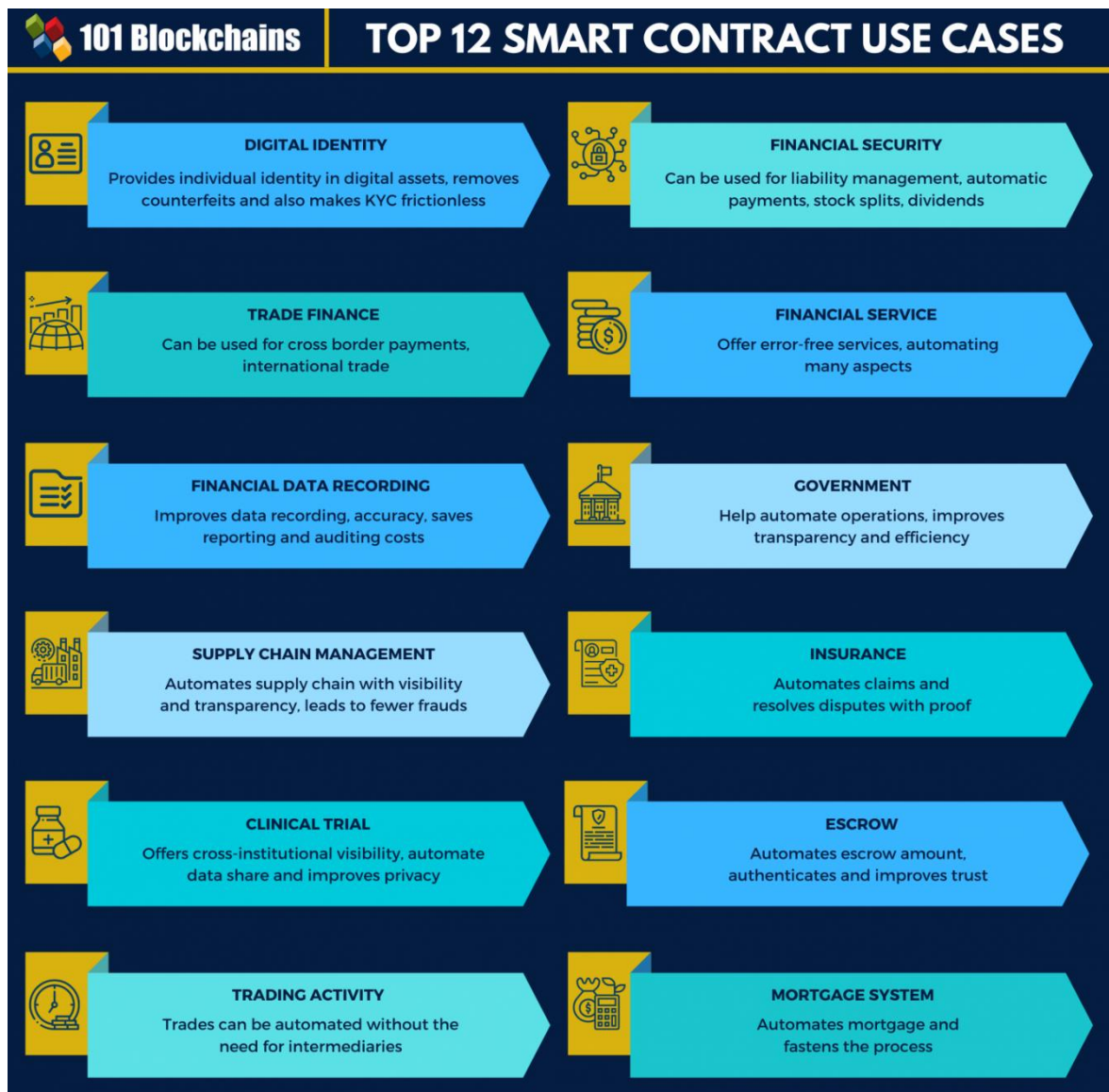
own processing according to (Coingecko)

The idea behind Ethereum was to create an alternative protocol to Bitcoin which would serve to build decentralized applications. It is the first blockchain to have its own programming language, that allows anyone to write smart contracts and decentralized applications where they can implement their own rules for ownership, transaction formats and state transition functions. (Ethereum, ©2022)

Smart contracts are self-executing contracts that facilitate, verify, and enforce transactions on the blockchain, and are a key innovation in Ethereum blockchain. For example, if a person was to buy an NFT on OpenSea which is a Ethereum NFT marketplace, the smart contract would save the agreement between the buyer and a seller by writing it into lines of code and passing the transaction onto the blockchain which would confirm the change of ownership and the seller would receive his money. (Investopedia, ©2021)

The picture bellow presents the possible applications of smart contracts

Figure 7: Smart contract use cases



(Diego Geroni, © 101 Blockchains 2021)

Ethereum has a wide range of use cases ranging from applications, finance, gaming, identity management or supply management. The biggest part of Ethers network are the financial applications which provide users with more options to what they can do with their cryptocurrencies. Theses application belong to the decentralized finance category and are used for peer-to-peer borrowing and lending, making interest on your cryptocurrency holdings or as a decentralized exchange. The most popular ones are Aave, Compound and Curve. (Ethereum, ©2022)

Ethereum also ensures secure information sharing using cryptographic methods which is key for transferring important data like medical records or identity. Additionally the blockchain-based ledger can help with supply chain and logistics management. Businesses and customers can track a products journey and know that what arrived at their house or storage is what they desired and that it wasn't replaced by anything. Everything from goods to organic food can be tracked and traced using the Ethereum network. (Gemini, 2021)

2.2.2 Polkadot

Polkadot is currently the 10th largest cryptocurrency by market cap according to Coingecko. It was created by one of the Ethereum founders.

- Ticker: DOT
- Market Capitalization: \$20,2B¹¹
- Foundation: 2016
- Creator: Gavin Wood
- Current price: \$18.40¹²

In the chart below we can see the price development of Polkadot since 19th August 2020 till 16th March 2022.

Figure 8: Polkadot price development



own processing according to (Coingecko)

¹¹ market capitalization to date 16.3.2022, available at: <https://www.coingecko.com>

¹² current price to date 16.3.2022, available at: <https://www.coingecko.com>

Polkadot is a one-of-a-kind Proof-of-Stake blockchain that is built to connect and secure other blockchains, whether they are public or private. It enables internet where independent blockchains can exchange information under common security guarantees. (Polkadot.network, ©2022)

„Polkadot enables scalability by allowing specilzed blockchains to communicate with each other in a secure, trust-free environment.” (Polkadot.network, ©2022)

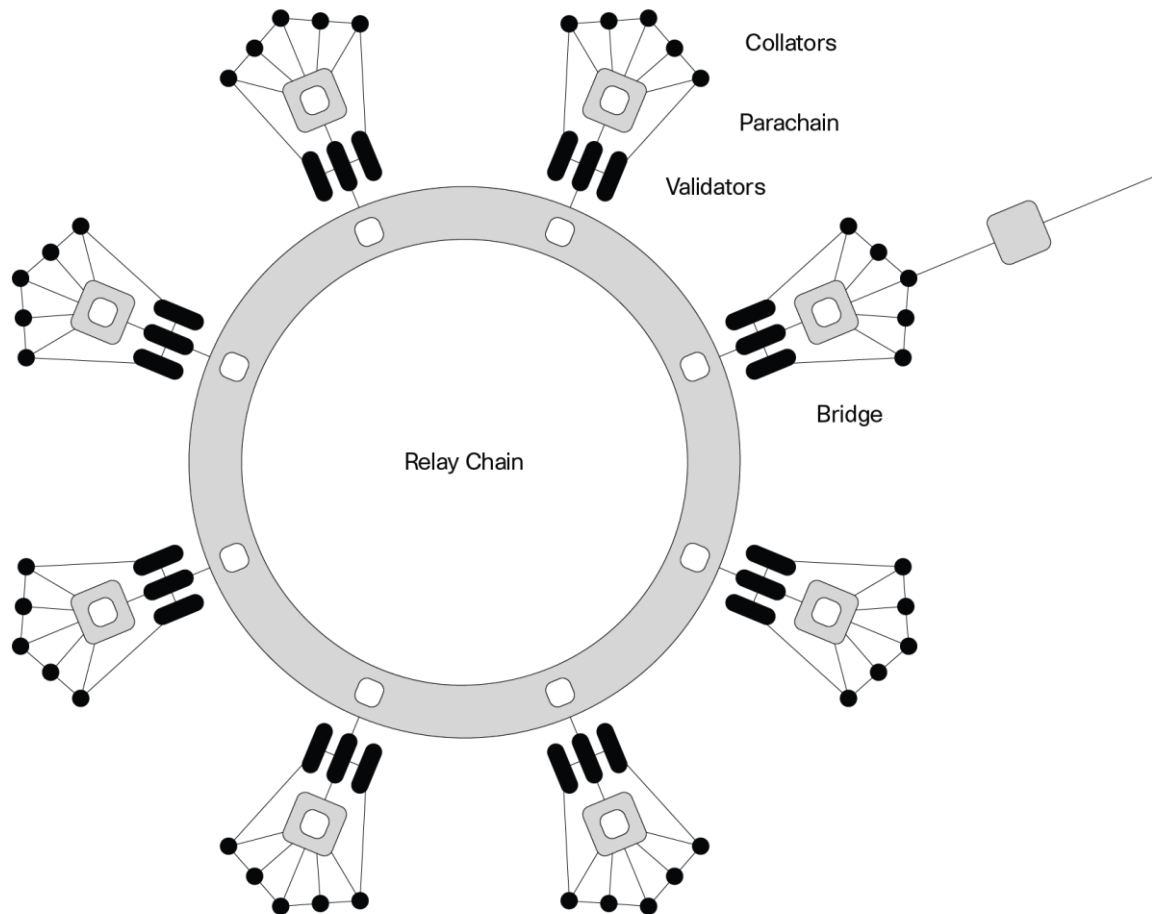
Polkadot network main focus is interoperability¹³ and scalability¹⁴. It uses a sharded model where shards called parachains are connected to one main core which is called the Relay Chain using a bridge. Parachains can be considered smaller blockchains that construct and propose blocks to validators on the Relay Chain which works as the parent chain. There the blocks undergo validity checks before being added to the finalized chain. As the Relay Chain is the one providing security guarantees to all the parachains, collators therefore have no security responsibilities and thus do not require a strong incentive system. This is how the network manages to stay up to date with the amounts of transactions that take place. (Polkadot.network, ©2022)

The following picture presents the Relay Chain which works as a brain of the Polkadot network throughout which parachains can communicate between each other using the Cross-Consensus Messaging Forma (XCM) to send messages of any type. The Relay Chain provides the security and validation of all the messages that are sent between the parachains creating the trust-free environment which can operate faster than any other separate blockchains communicating between each other. (Polkadot.network, ©2022)

¹³ the ability of different blockchain networks to exchange and leverage data between one another and to move unique types of digital assets between the networks

¹⁴ measure of a system's ability to increase or decrease in performance and cost in response to changes in application and system processing demands

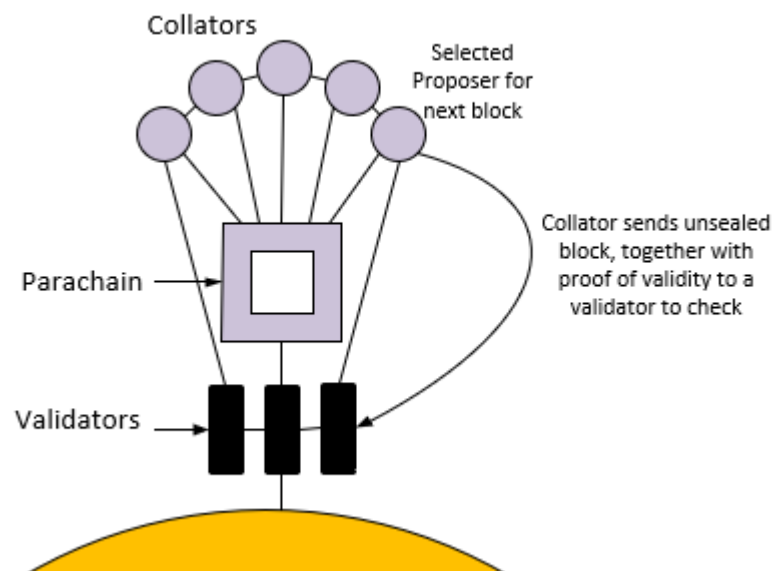
Figure 9: Polkadot network scheme



(Polkadot.network ©2022)

Collators work the same way as Proof-of-Work miners on the blockchain. They maintain parachains by collecting transactions made by the users. The collected data is used to produce state transition proofs that are then checked by the validators on the Relay Chain. The difference compared to PoW miners is that Collators don't validate blocks and they are not responsible for security, their only job is to create and send the transition proofs to the Relay Chain where validators verify them. Collators are the key element of the XMC which serves as an easy messaging system between two parachains. (Polkadot.network, ©2022)

Figure 10: Collators



(Polkadot.network ©2022)

Parachain auction is the only way to enter the Polkadot network, the auction winner can lease a parachain spot for up to 96 weeks with the option to renew, this system helps in having fresh and active blockchains on the network. Parachain auctions are held on the Polkadot network where companies present their blockchain and explain why they should be given the parachain slot. In order to participate in the auction users and companies must lock up their DOT tokens which they can then use to bid for their favored blockchain. Companies can also fund their campaign with a crowdloan, this allows them to accept contributions from DOT holders, in return the holders are rewarded in various ways like airdrops¹⁵ or some privileges on the blockchain. Close to the end of an auction, a snapshot of bids for each blockchain is taken at random time in case anyone would try to cheat and bid more at the end of the auction. The highest bid one wins the spot, and all DOT tokens they were funded by remain locked for the duration of the parachain lease, the remaining DOT tokens are returned to their owner's wallets. (Polkadot.network, ©2022)

Native token of Polkadot Network is called DOT, DOTs can be used for transaction fees, staking, governance, acquisition of a parachain by participating in an auction and more functionalities. (Polkadot.network, ©2022)

Use cases of the DOT token:

- 1 DOT – minimal amount required to participate on the Polkadot network
- 5 DOT – participation in crowdloans for parachain slots
- 20 DOT – on-chain identity and voting right
- 120 DOT – right to nominate validators and earn staking rewards

¹⁵ airdrops involve crypto projects sending free tokens to their communities as a symbol of trust

2.2.3 Decentraland

Decentraland is currently the 36th largest cryptocurrency and is one of the most popular decentralized virtual reality projects on the Ethereum blockchain. Players can join the world through their PC, mobile phone or a VR headset.

- Ticker: MANA
- Market Capitalization: \$3,5B¹⁶
- Foundation: 2015
- Creator: Ariel Meilich, Esteban Ordano
- Current price: \$2.34¹⁷

In the chart below we can see the price development of Decentraland since 28th October 2017 till 16th March 2022.

Figure 11: Decentraland price development



own processing according to (Coinagecko)

¹⁶ market capitalization to date 16.3.2022, available at: <https://www.coingecko.com>

¹⁷ current price to date 16.3.2022, available at: <https://www.coingecko.com>

Decentraland is a metaverse project, it is a virtual reality game created on the blockchain where users can monetize their content. It is divided into 90,601 parcels of land which is represented by an NFT. Users can buy the land on which they can build whatever they desire and later rent or sell it to other players, make money for advertising on their land or house, or even sell various products on the marketplace. The idea behind Decentraland is to bring videogames to the blockchain where people can make connections and create their own perfect world. (diit, 2020)

Decentraland's native token is MANA, which is an ERC-20¹⁸ token, it is used as a digital currency in the virtual world. It has many use cases like buying clothes and cosmetics for your character, renting or buying real estate, visiting a casino where you can play poker or blackjack for it or even buying tickets to virtual concerts where artists like DeadMau5 performed, the opportunities are endless. (diit, 2020)

¹⁸ Ethereum blockchain based token

3 Trading cryptocurrency and traditional markets

In this chapter I would like to talk about the differences and similarities of cryptocurrency and traditional markets in various aspects. I will mention the pros and cons of each, how to approach them and what you need to know if you were interested in investing or trading them. Like what tools to use to track them, how to categorize them, where and when you can buy them, and how you can define strength of an asset.

3.1 Differences and similarities in trading crypto and other markets

Trading a cryptocurrency market is not any different than trading any other market, there are few differences

3.1.1 Charting software

Charting software serves the purpose of making a trading plan by utilizing the tools of technical analysis it offers.

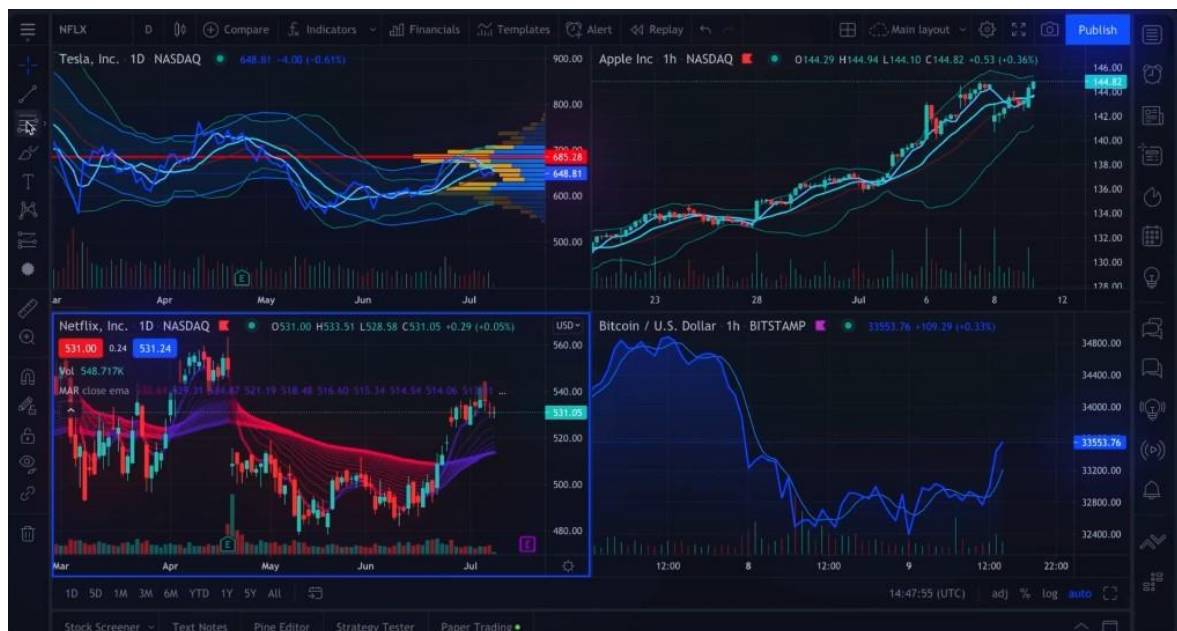
Trading View is the most popular charting software, its not only a website with its own dedicated desktop application it also works as a social media network used by over 30M+ traders. You can view other traders' analysis and ideas, comment on them or share them. It contains built in chart indicators, various technical analysis tools like fibonacci, trendlines, horizontals, multiple window function so you can track up to 8 charts at the same time, strategy tester where you can develop your own strategy and test it on the chart. Alerts which you can set at desired price points where you want to get notified. Pine editor where you can code your own technical indicators which you can then publish to everyone else who uses Trading View or make it private and put a subscription fee to it. Paper trading function where you can create a fake trading account and put your trading skills in practice without the need to lose money. When you are ready to trade with real money you can connect your broker to Trading View and trade from the chart but for that function there are better platform and not every broker is supported. Since Trading View also works as a social network there are data feeds of all the global news so traders can easily keep up with what is going on and position themselves in the right direction before others realize that something happened. (Tradingview, 2022)

Trading view offers charts for everything tradeable ranging from cryptocurrencies, stocks, currencies, commodities, bonds, futures to indices all in one place making it the perfect tool for planning trades or perhaps even trading in some cases. (Tradingview, 2022)

For cryptocurrencies all data feed is streamed in real time directly from the exchanges and there is no fee for viewing it. For stock exchange markets the data feed is also streamed in real time from the exchanges, but there is a certain fee for viewing it live. The fee depends on what market you want to look into ranging from 2 USD up to 110 USD for the ICE US Futures - US financial and commodities. The data feed can be viewed even without paying but it's delayed by around 10 to 15 minutes in most cases, some markets data feeds are delayed until the trading day ends making it untradeable without paying the fee. (Tradingview, 2022) In trading time is key and not having the real time data can be the difference between making or losing money.

Below you can take a look at how Trading View looks in action.

Figure 12: Trading View preview



(Tradingview, ©2022)

In conclusion Trading View works the same for cryptocurrency or traditional markets, all charts on Trading View can be viewed for free, the cryptocurrency ones have a live data feed which is free to view whereas some of the stock exchange ones have a small fee for that feature. This makes the cryptocurrency market friendlier since there are no initial costs.

3.1.2 Trading software

Trading software is used to get a deeper understanding and look into what is going on in the chart, some trading softwares are made only for cryptocurrencies other support only traditional markets, there are few that support both but all in all they all offer the same features with only miniscular differences. For this thesis I will present ATAS which is an abbreviation for Advanced Technical Analysis Software which works for both cryptocurrency and traditional markets.

ATAS comes as a desktop app where you create an account trade with to which you connect your exchanges via API keys. It offers complete customization of workspaces which can be loaded upon logging in, different chart types and indicators that can be changed function and color wise to the smallest detail. (ATAS, 2022)

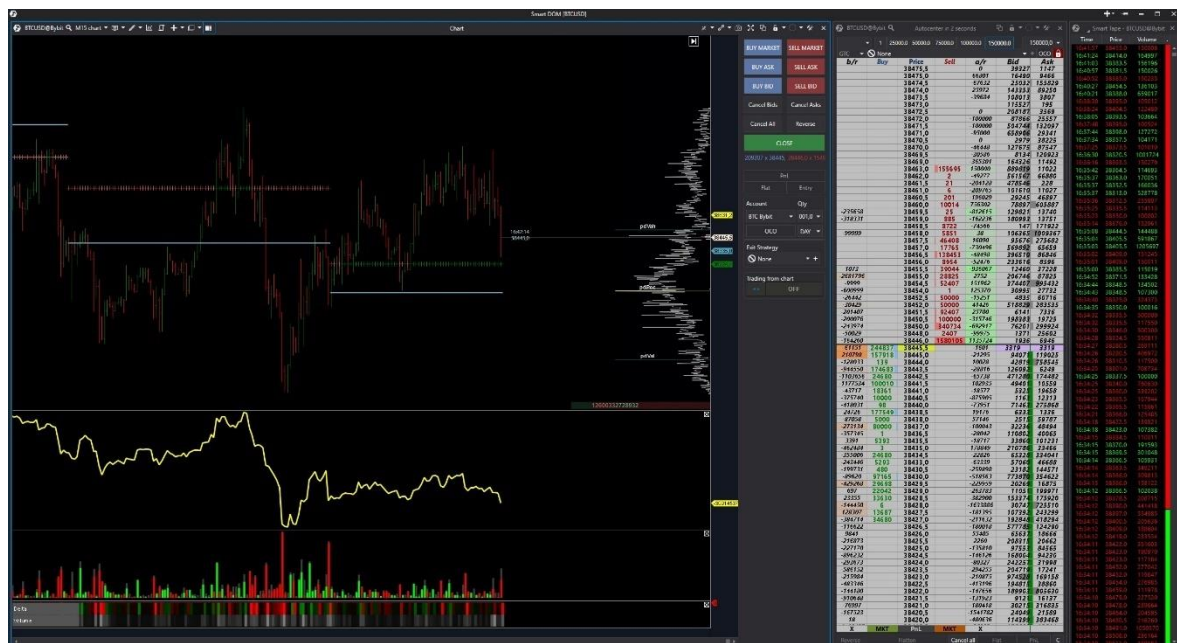
ATAS can be used for trading traditional and cryptocurrency markets, and it offers the same features for both. The only difference is again the pricing. ATAS offers a 14-day free trial for any market and after that period is free if used for cryptocurrencies only but paid for when trading traditional markets. The subscription is paid monthly, and it's valued at 69 EUR. (ATAS, 2022)

ATAS offers features like:

- Different types of charts - volume, time or the traditional candlestick based depending on what you want to trade
- Smart DOM which is used for spotting high frequency algorithms that the market makers use to fool the retail traders meaning we can see big orders appear and moments later be removed from the order book.
- Smart Tape which shows all market orders in full detail
- Bid/Ask Tape which allows us to see the volume of trades, thus trades behaviour can be analyzed when approaching an important price level.
- Cumulative delta (CVD) which
- Volume statistics which display how much volume a candle has and whether the volume is positive or negative
- Open Interest which shows if positions are opening or closing
- Trading from chart function which lets you trade directly from the chart with no delays, you can even program hotkeys to enter positions even faster

In the next example we can see ATAS and its futures - Smart DOM in grey, Smart Tape on the right of the Smart DOM, on the left side we can see volume delta displayed as the yellow line, below are the volume and delta statistics in green, red, and grey. Above is the price chart displayed using bars which are an equivalent to the candlesticks.

Figure 13: ATAS preview



own processing according to (ATAS)

In conclusion we can say that the same trading software that is used for cryptocurrencies can be used for traditional markets. It offers the same features for both but is cheaper for cryptocurrency use since there is no monthly subscription fee.

3.1.3 Correlation

After the big cryptocurrency boom in 2017, institutions spotted the cryptocurrency market as a new way to derisk from the traditional financial markets since it behaved on its own and still wasn't big so the risk to reward ratios was on their side because they could accumulate large amounts of assets for a relatively cheap price and so the adoption began. (IMFBlogs, 2022)

Before the 2020 pandemic crypto assets weren't still showing much correlation with major indices like the S&P 500. This changed after the crisis responses from the banks in early 2020 where both crypto and us stocks surged amid positive global financial conditions and greater risk appetite to make more money. Due to that, the risk diversification benefit of cryptocurrencies declined almost completely because cryptocurrencies started to act almost as any other financial market. (IMFBlogs, 2022)

In the figure below we can see how the price of Bitcoin and the S&P 500 behaved in the last years and the rapid change after the early 2020. (IMFBlogs, 2022) This correlation is still ongoing, therefore it's always a good idea to check what the S&P 500 is doing before entering the cryptocurrency market.

Figure 14: Stocks and Bitcoin correlation

Stronger correlation

Bitcoin and U.S. stocks have moved together more closely during the pandemic.

Bitcoin price and S&P 500 index



Correlation between Bitcoin returns and S&P 500 index



Source: CryptoCompare, Yahoo Finance, and authors' calculations.
Note: Bottom panel shows rolling 60-day correlation coefficient.

IMF

(IMFBlog, 2022)

In conclusion we can say that trading cryptocurrencies is now more than ever correlated to what the equity markets are doing. Therefore, a trader should always look at what is happening in the equity markets before trading cryptocurrencies to understand the overall situation and get the bigger picture. Although the markets are correlated it doesn't mean that it always has to be that way, at certain times the S&P 500 can behave in contrary to what Bitcoin does.

3.1.4 Automated trading

„Automated trading is a method of participating in financial markets by using a programme that executes pre-set rules for entering and exiting trades.“ (ig, © 2022) Meaning instead of trading on your own a bot is used to do the work for you. This type of trading is used because it removes any emotions from trading a person has which could lead to a bad decision and since trading is buying and selling emotions, removing them from the entire process helps significantly. Also, a bot can work 24 hours a day 7 days a week and doesn't need as much care as a person does, this reduces the screen time needed because the bot trades until stopped while the trader can just check from time to time and focus on different things.

To start automatically trading a platform where a bot can be created is needed, preferably a one which can be connected directly to an exchange, or a broker using an API and offers the possibility of backtesting to see how the bot performs before it is put in practice. (MetaTrader4, © 2022)

To work properly the bot needs a set of rules based on which it will execute orders and there are various approaches. The bot can either trade by using a combination of mathematical equations like Sin, Cos or it can be an algorithmic one which follows an indicator like the MACD, RSI or moving averages and sells and buys depending on how the indicator behaves for example if the 50-day moving average crosses the 100-day moving average, do something. (Ig, © 2022) It can also be set to trade a grid meaning a defined price range in which it will trade. It will then place sell orders when the price goes up and buy orders when the price goes down. Next up is the DCA bot, DCA is an abbreviation for dollar-cost-averaging, this is a style of accumulating an asset that you plan to hold for a long time. This bot automatically purchases a pre-defined amount of an asset at a specific time of day and repeats this process, it serves for averaging down on traders' entry price of an asset. (Kucoin, 2021) The most complex bots use volume analysis, order flow, and monitor the order book but those are out of reach for the casual traders due to the fact that coding a bot of that sort requires months of testing and an entire development team. (ATAS, ©2022)

There are two ways to create a bot, with a programming background a trader can code his own bot. This option is more complex and therefore harder. The second option is using a rule-based bot that works by the trader choosing an indicator like an RSI and setting values at which the bot should execute an order. (Trality, © 2022) This bot is easier to set up but also less rewarding because other traders can copy its settings much easier than if coded on your own and when it comes to automated trading the more people the strategy is used by the less efficient it is. Therefore, the majority of bots are coded due to that and also because most platforms only support the coded ones since the rule-based version is relatively new on the market.

Trading environment is also very important because a bot that worked yesterday and had great results doesn't need to work the same way today. The market can be more volatile, go only in one direction or any other day to day change can happen which can affect it. That's why the bot needs to be adjusted every now and then to keep up with the market or more bots for different types of price action are needed. (Investopedia, 2022)

Trading traditional and cryptocurrency markets using an automated bot is a bit different, the same strategies can apply to both markets, but the big difference are the settings of the bots, due to the fact that cryptocurrency markets are more volatile compared to the traditional ones the values of the indicators that track the price need to be adjusted and turned down a bit. Also, when you want to use a bot some cryptocurrency exchanges offer the grid and DCA versions directly on their website, which none of the stock exchanges do for retail traders therefore a software like MetaTrader4 or ATAS has to be used.

In conclusion there are many types of trading bots used for automated trading. The same strategies work for both cryptocurrency and traditional markets the only differences are in the parameters of indicators that are used by the bots. Also the current state of market matters a lot, because bots will usually perform better in a sideways range than a trending market.

3.1.5 Trading hours

Trading hours are one of the biggest differences in cryptocurrency and traditional markets. Stock exchanges do not work on the weekends and state holidays and are opened for trading only few hours a day. Whereas cryptocurrency exchanges are up and running 24 hours a day, 7 days a week, which makes crypto tradeable at any time.

Exchange trading hours in UTC+1 time (Prague):

Tokio Stock Exchange session: 01:00 – 07:00

London Stock Exchange session: 09:00 – 17:30

New York Stock Exchange session: 15:30 – 22:00

Close session: 22:00 - 01:00 – no trading

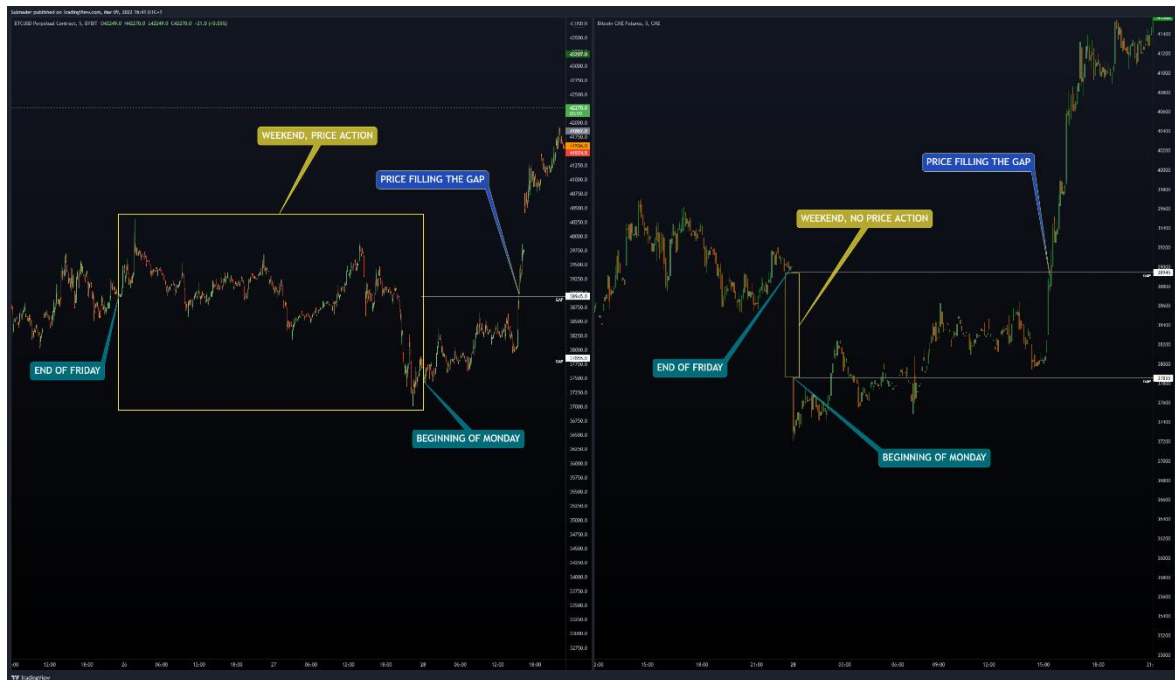
(Signaltrade, 2022)

Since the stock market opens and closes at certain time a gap in price can appear when the opening price differs from its closing price. This gap usually gets filled as it acts as an inefficiency on the chart and there are many trading strategies involving it. (Mytradingskills, ©2022)

Although crypto exchanges don't close and there are no gaps on the cryptocurrency charts, they still affect the price of Bitcoin. Bitcoin is traded on the CME futures exchange 24 hours a day 5 days a week starting Monday and ending Friday. Since there is no trading on the stock exchange on weekends there is a gap formed every Monday. The price on the stock exchange then opens on Monday at the same price point as the crypto exchanges price, which is affected by the weekend trading and the gap is formed. Even though this gap doesn't appear on the crypto exchange charts it still affects them. When the stock exchange opens on Monday, the price is driven to fill the gap on the stock exchange chart and the crypto exchanges follow. (CMEGroup, © 2022)

In the figure below you can see how the gaps are formed. On the left you can see the crypto exchange chart which trades during the weekend and on the right side the stock exchange with the gap in price since it doesn't trade on the weekends.

Figure 15: Gap creation



own processing according to (TradingView)

In conclusion trading hours are very different when it come to cryptocurrencies and traditional markets. Traditional markets are closed at some time of the day whereas crypto runs 24/7. Due to that we can see gaps in the price in traditional markets and in stock market traded Bitcoin whereas in the crypto exchange charts we don't see any gaps, but price is still influenced by them.

3.1.6 Trading styles

There are 4 ways to approach trading. Each style uses different timeframes and tests different capabilities of a person. The cryptocurrency and traditional market differ in how long each one of the approaches lasts. One is focused on the quick intraday moves where the trader goes in and out of the market rapidly, the second one focuses on bigger moves during the day which can take even hours to happen, the third one focuses on moves during a long period of time and the fourth one can last up to a lifetime to develop.

Scalping is the most advanced trading style which works the same way in the cryptocurrency and traditional markets. During the day scalpers take dozens of trades which they hold for seconds to minutes, they aim for small moves to get in and out fast. They use the 30 second to 5-minute charts to enter and monitor their trades. Scalpers tend to trade the most at the beginnings and ends of sessions which is the point in time with increased volatility and trading volume. (Capitalindex, © 2022)

Day trading is also a short-term trading style where the trader aims for larger moves and taking less trades. These moves can take minutes to hours until fully developed so the preferred timeframe for day trading is the 5 to 15-minute chart for entering and the 1 to 4-hour to get the confirmation. Day traders close their positions at either a profit or a loss at the end of the day to avoid overnight fees and any risk of a large overnight move. Trading traditional markets this way works the same as trading the cryptocurrency ones the only difference is that in the cryptocurrency market you are the one who defines when it's the end of the trading day since the markets run 24/7. (Capitalindex, © 2022)

Swing trading is a long-term trading style in which cryptocurrency and traditional markets differ in how long the position is held, for cryptocurrency the typical duration is between days to weeks, in traditional markets it ranges from weeks to months and some positions can even be held for over a year. (Capitalindex, © 2022)

Investing is a very different approach to the markets compared to previous trading styles. It is a long term one, we are talking years to a lifetime. Investing means allocating capital in an asset and only taking small profits while keeping the position running for a long time. This approach is preferred by people who want their money to work for them but are not interested in paying much attention to it at. Investing is putting money into something you believe in. If you believed in cryptocurrencies early on and invested in Bitcoin at the beginning of January 2015 and sold the November 2021 all time high you could have made around 45 000% on your investment, in comparison if you invested at the same time in Apple, you would have made only 600%.

In the figure you can see the measured performance of Apple and Bitcoin from the beginning of 2015 until March 2022.

Figure 16: Apple and Bitcoin performance



own processing according to (TradingView)

In conclusion trading styles of cryptocurrency and traditional markets are almost the same with only a small difference being the time of how long the position is held for in the higher term timeframe trading.

3.1.7 Leverage

Leverage is a powerful tool through which a trader can increase his position size to greater amount than he directly owns by borrowing capital from a broker or exchange to maximize the potential return of the investment. The returns can be maximized but so do the risks of downside if the market was to turn the other way than expected. Investors can lever their assets using futures, options, or margin accounts.

Cryptocurrency exchanges offer up to 200x leverage for the high liquid markets like Bitcoin or Ethereum, but most markets can be traded only with up to 50x leverage and a certain maximum position size, whereas brokers offer various leverages for different markets. The highest achievable leverage is 1000x and it's used for trading currency pairs. (PrimeXBT, ©2022)

Although high leverage is very risky to use and requires a lot of knowledge to be used well exchanges and brokers tend to offer this tool to the newcoming traders to rip them of their money because of that many regulatory laws have been issued to protect the investors. Previously there were no limits to leverage but Regulation T changed that. It was established by the US Federal Reserve System to provide rules which limited the margin trading. Traders borrowing limit was set to no more than 50% of the securities purchase price meaning traders had to deposit 50% of the desired position size as the margin when opening a leveraged trade. In India the borrowing limit was already raised to 100%. (Investopedia, 2020)

Margin is a special type of leverage used in trading. It involves the trader using his assets as a collateral to increase buying power by borrowing money from an exchange or broker for an interest rate. Therefore, margin can be used to create leverage, for instance trader having \$1,000 worth of USD on his account and wanting to buy \$20,000 worth of Bitcoin or any other stock or cryptocurrency a trader would be using 5x leverage therefore having a 1:20 margin, meaning the broker or exchange would lend him \$20,000 for his \$1,000. (Investopedia, 2022)

This formula is used to calculate potencial gains or losess when using leverage.

$$Profit\ or\ Loss = (Initial\ Margin) \times (\% \text{ price movement}) \times (Leverage)$$

(1)

Playing with more money than a trader has involves the risk of liquidation. Liquidation is a forced event where a trader's position is automatically closed at market price by the exchange or broker because he doesn't meet the margin requirements of the contract meaning the price of an asset has reached a certain level at which the money he deposited as a security to borrow isnt enough to handle the loss. Let's use the same example as above a trader buys \$20,000 worth of Bitcoin only having \$1,000 USD as margin, the position is

using 20x leverage and will be liquidated if the price of Bitcoin asset moves 5% against his position ($100/20 = 5$). (Coindesk, 2021) Liquidations are one of the ways exchanges and brokers make money.

This formula is used to keep track of how much a market need to move against a position to get it liquidated.

$$Liquidation \% = \frac{100}{Leverage}$$

(2)

In conclusion the type of leverage used in trading is called margin and it works the same way in both cryptocurrency and traditional markets the only difference is the multiply amout which in crypto is much lower than in traditional markets to protect the investors due to the natural volatility of cryptocurrency markets.

3.1.8 Market cycles

There are two types of market cycles, the bull run where price go up only and bear market where the exact opposite happens. These cycles are tightly connecte to human emotions and what is happening in the world.

Bull run is market where prices rise, the trend is upwards, economies are growing, and general attitude is positive. (Investopedia, 2021)

Bear market is when economies are receding, the trend is downwards, most assets decline in value and the general attitude is negative due to some crises or risks like wars or increase in unemployment. (Investopedia, 2021)

Market cycles are referred to as periods between two high term timeframes lows or highs in the market. In the cryptocurrency market that is led by Bitcoin the cycle last approximately 4 years. The halving event which reduces mining speed and happens every 4 years is believed to be the middle of the cycle. In traditional markets the cycle takes much more time. (Cryptopotato, 2021)

Cryptocurrency market cycles have a specific money flow. The end game play of most cryptocurrency investors is to accumulate Bitcoin since it's the largest and best performing asset. The figure below describes how to behave in the cycles to maximaze profits.

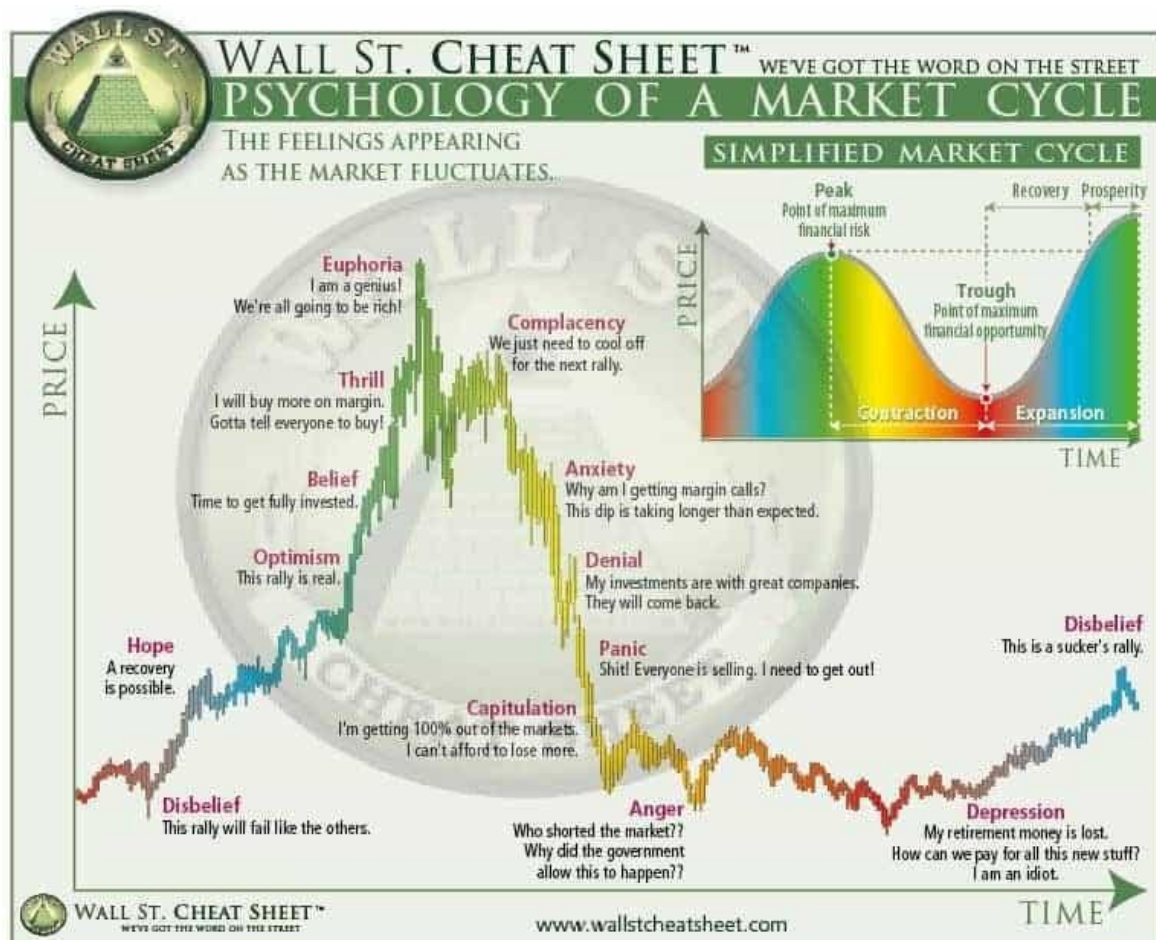
Figure 17: Flow of the crowd during a cryptocurrency cycle



(@SecretsOfCrypto on Twitter,2021)

Emotions play a big role in all markets. They drive the markets much more than any fundamentals. The following figure describes the emotions investors experience as the market fluctuates. These stages repeat over time and form the market cycle, nothing in trading is precise so some of these phases can be faster, slower, or even skipped. The key takeaway is that nothing goes up for ever and neither does it go to zero. The markets usually recover even from the biggest downside because we as humanity advance forward. (Cryptopotato, 2021)

Figure 18: Psychology of a market cycle



(Karen Bennett ©Wall Street Cheat Sheet 2020)

"Be fearful when others are greedy and greedy when others are fearful." - Warren Buffett
an investor icon once said, this perfectly resonates with the figure above.

Market cycles are important for investors but not for traders, since in trading money can be made if prices rise or go down. Traders don't think about bear or bull market, they are more focused on whether the current trend is up or down to determine if long or short positions are more favorable.

In conclusion market cycles are built on human emotions and are similar in the cryptocurrency and traditional markets. The difference is the time they last which in the cryptocurrency space is much shorter than in the traditional markets where the cycles can last for many years. The end goal of cryptocurrency investors is to multiply their bitcoins using altcoins during the market cycles.

3.1.9 Volatility

The major difference between cryptocurrency and traditional markets is the volatility. While a big daily movement of NASDAQ is considered to be 1-2%, Bitcoin, which is considered the most stable cryptocurrency and the base asset, can fluctuate more than 20% daily. Cryptocurrencies are like stocks on steroids, the main reason for that is their newness that boosts the volatility. Some altcoins can even increase 300% in price during a day but these sharp movements tend to pull back almost completely, so the traders who don't take their profit end up being break even or even in a losing position. (Cryptopotato, 2021)

When the bull market is on it is normal that some altcoins rise over thousands of percent in few weeks. However, when the bear market hits, many of those retrace even 95% from their all time high and some remain there for the rest of their existence. making the cryptocurrency markets ruthless and unforgiving.

Traditional markets are stable compared to the cryptocurrency ones. Movements of few percent a day are considered as large. However, even traditional markets can be volatile at times and either rise or fall tens of percent during a day but that is usually related to news events or what is happening in the world.

In conclusion traditional markets tend to be more stable but at certain times can be as volatile as the cryptocurrency ones which are very volatile on daily basis.

3.1.10 Defining strength of an asset

Defining strength of an asset is a key factor to observe before entering it especially when looking at it from a long-term investment perspective. (Investopedia, 2021)

To define the strength of an asset we must compare it to something and for that there are indexes which work as a benchmark to which the asset is compared. Indices are trackers of performance of a group of assets. They can measure performance of a group of stocks, interest rates or inflation. (Investopedia, 2021) The biggest part of indices like Nasdaq and S&P consists of tech companies which is perhaps the reason why cryptocurrencies are so heavily correlated to them because they are a tech based digital asset.

The well-known indices are:

- S&P 500 - consists of 500 most widely traded companies in USA
- Nasdaq 100 - consists of 100 most widely traded companies in USA
- Dow 30 - consists of 30 most widely traded companies in USA

The goal of the asset is to outperform the indices. If the asset doesn't outperform them, there is no reason to invest in it because you can invest in the index and have better gains. You cannot invest directly in the index, but you can buy index funds, which track the indices. (Investopedia, 2021)

In the figure below we can see a representation of indices and what view each of them represents.

Figure 19: Indices



(Sabrina Jiang © Investopedia 2020)

Defining a strength of a cryptocurrency works the same way. The only difference is that there is no index to compare it with, instead we compare smaller cryptocurrencies to the biggest one – Bitcoin which works as our index, so the goal in the cryptocurrency market is to outperform Bitcoin.

Bitcoin has been the best performing cryptocurrency, so it hasn't been outperformed in the long-term but since it's the biggest cryptocurrency its also the slowest moving one and that's why traders often switch their Bitcoin into Altcoins for the short-term to outperform Bitcoin and then convert it back for the sake of making more Bitcoin because Altcoins tend to be more volatile in the short-term.

In conclusion we use indices to track the performance of traditional markets and Bitcoin to track the performance of Altcoins. This principle is the same for traditional and cryptocurrency markets and the end goal of this is to determine whether to invest in that market.

4 Cryptocurrency exchanges and brokers

In this chapter I will briefly talk about the history of exchanges, present some cryptocurrency exchanges and brokers, go through services they offer, what are the fees for using them, create a trader profile for each. Then I will compare the presented cryptocurrency exchanges between each other and the broker in the services offered and summarize the results.

4.1 History of exchanges

A stock exchange is a physical or digital place where investors can buy and sell stocks, bonds or shares. The price of each asset is driven by supply and demand. The more people want to buy, the higher the price goes. Less demand, and the price decreases.

The first stock exchange appeared in 17th century Amsterdam where the Dutch East India Company was the first and for a long time only stock to be traded. Currently the biggest exchange in the world is the New York Stock Exchange (NYSE) which was founded in 1792. (Intercontinental Exchange Inc., ©2022) The first cryptocurrency exchange called Bitcoinmarket went live in March 2010, it worked more as a marketplace than an exchange and the initial price of 1 Bitcoin was around \$0.003. The first proper exchange was BitStamp which was founded in 2011 and is still operating. (Medium, 2020) Currently the biggest cryptocurrency exchange is Binance which was founded in 2017, its trading volume peaked at \$76 billion in 2021. (BusinessofApps, 2022)

4.2 Binance

Binance is the world's largest cryptocurrency exchange with access to hundreds of cryptocurrencies, over 1600 markets, daily trading volume around \$75B, and advanced features for investors with low trading fees. (Coinmarketcap, ©2022) However, it is not available to U.S. customers, American citizens will be forced to use Binance US, which has less functionalities compared to the regular one like high leverage. (Binance, ©2022)

Binance allows deposits of crypto and fiat currencies with no minimum size, fiat currencies can be either deposited instantly using a credit card for a 1,8% fee or by a bank transfer which usually takes a day or two to process. Leverage wise it offers up to 125x for the highest liquid markets. (Binance, ©2022)

Trading wise Binance includes TradingView and its own charting interface and makes it easy to buy or sell assets at specific price and amount, includes most fiat currency to cryptocurrency trading pairs and offers a huge variety of futures contracts but lacks behind other exchanges in features like placing and moving orders directly in the chart, order flow,

possibility of having open both short and long positions of an asset in futures contracts and the fact that the minimum order size has to be greater than \$10. (Binance, ©2022)

Binance offers much more than trading futures, options, spot, and margin. It has its own NFT marketplace where people can buy and sell their digital art pieces, its own credit card which you can top up with various cryptocurrencies and pay with anywhere across the world while earning interest on your purchases, a peer-to-peer marketplace where people can directly buy cryptocurrencies from others, launchpool where traders can get exposure to cryptocurrencies before their listings¹⁹ on exchanges and the possibility of staking and yield farming to earn passive income on your coins directly on the exchange without the need of withdrawal and deposit elsewhere. (Binance, ©2022)

Binance has its flaws, one of them is its technical support, its impossible to resolve a time sensitive issue fast because the waiting queues to get in contact can take up to few hours at times of large traffic. Another serious one is that even though it is the biggest exchange with over 28 billion traders it lags and its UI²⁰ glitches during huge spikes in prices, making them untradable due to the number of traders entering positions and liquidations happening all at the same time, also being the largest cryptocurrency exchange doesn't help either in case of any crypto related regulations because it will be the target number one. (Binance, ©2022)

In conclusion Binance has its flaws and might look a bit user unfriendly at first but after few hours it feels natural. It is still the best solution for traders with large scale portfolios because of its trading volume even though it glitches at times and the variety of cryptocurrencies and features designed to maximize profits is tremendous.

4.3 Bybit

Bybit is primarily a cryptocurrency derivatives exchange with a wide range of advanced trading tools and options, daily trading volume around \$10,5B and almost 300 markets. (Coinmarketcap, ©2022) It has top-notch security and a no-down-time commitment, but due to its derivatives nature it is not available in the United States. (Bybit, ©2022)

Unfortunately, Bybit allows deposits only in cryptocurrencies and doesn't support fiat currencies at all. The only way to get your fiat onto Bybit is either directly buying a cryptocurrency - BTC, ETH or USDT from Bybit using a debit/credit card or through the Peer-to-Peer marketplace which compared to Binance doesn't charge any fees. Leverage wise it offers up to 100x for the highest liquid markets.

Trading wise, Bybit is almost flawless. It is hands down one of or maybe even the best exchange with no downtime or glitches even at times of huge spikes in price and volatility. It offers its own and Trading View interface for charting, possibility of placing orders

¹⁹ time when an asset is opened for trading on an exchange

²⁰ user interface

directly on the chart and easily moving them, fast and swift orderflow meaning it executes orders instantly upon trader's mouse click and at shown price which is needed for a precise entry unlike Binance where the orders are often filled few dollars off the shown price. The trading fees are also low compared to other exchanges. (Bybit, ©2022)

Bybit's main focus are the futures markets, but recently spot trading was enabled too. It offers features similar to what Binance offers. An NFT and P2P marketplaces, launchpool where traders can get exposure to cryptocurrencies before their listings, staking and yield farming to earn passive income on your assets. Other than that, it has its own risk-free paper trading²¹ testnet that is available for anyone who would like to try trading without losing money, educational resources for technical analysis and seminars and great technical support which is live 24/7, easy to get and solves problems right on. (Bybit, ©2022)

Even Bybit has its flaws but there are not many. The major one is the lack of option to sell your cryptocurrencies into fiat and withdraw it to your bank account, to do that a trader needs to withdraw their cryptocurrencies from bybit to another exchange which offers that. There are also not that many assets to trade compared to Binance, but the team is working fast and adding new ones that are requested by traders almost weekly. Getting the desired position size can also be an issue since Bybit only offers traders to type in the desired order value as number of coins, not dollars which can complicate things. (Bybit, ©2022)

In conclusion Bybit is an exchange for non-US experienced traders only since it is derivatives based. It offers the top-tier technical tools and indicators, functions, security, learning materials and is probably the best exchange orderflow wise.

4.4 KuCoin

KuCoin is the 5th largest spot cryptocurrency exchange. Same as Binance and Bybit it doesn't it isn't available for the United States due to the possibility of using high leverage. Its daily trading volume is around \$5B and it offers more than 1100 markets. (Coinmarketcap, ©2022) (Kucoin, ©2022)

Same as Bybit, Kucoin only supports cryptocurrency trading. To buy cryptocurrencies and the options are again limited to BTC, ETH and USDT you can either use your debit/credit card, bank account or Revolut and PayPal depending on the fiat currency users are paying with. Other possibility is to buy through the Peer-to-Peer marketplace where there are no fees. The fact that Kucoin doesn't support fiat currencies can be a downside but on the other hand it allows the exchange to skirt some regulations and provide a more anonymous service to users. (KuCoin, ©2022)

Kucoin trading wise offers the same as Binance, derivatives, spot and margin trading through its own and Trading Views interfaces that can be used for charting, sadly it offers even the same flaws as the inability to place orders directly on the chart. It even includes

²¹ trading without real money

the few flaws Bybit has like the accurate order sizing. Leverage wise it allows up to 100x for the high liquid markets like Ethereum or Bitcoin but derivatives aren't the reason to be here. (KuCoin, ©2022)

The huge upside of Kucoin is the variety of trading pairs. Kucoin focuses mainly on spot trading and because of that it is the exchange where many companies list their cryptocurrencies first giving the users the ability to buy in early for cheaper prices because once it is listed on a major exchange like Binance, the price tends to spike up in the immediate short-term therefore the Kucoin users are given the opportunity to sell their holdings at much higher prices than they bought to the newcoming crowd. (KuCoin, ©2022)

Kucoin offers same services as Bybit and Binance like staking, yield farming and other ways to earn passive income on your holdings. However, it also offers lending where a user can lend his holdings to others for an interest, futures brawls where users bet on whether an asset's price will increase or decrease in the immediate short-term, lottery where users can buy tickets for a chance of winning some cryptocurrency and at last the feature that no other exchange has. Five trading bots where each one offers different strategies like DCA or grid trading. The bots are fully integrated in the exchange trading terminal and come with learning material making them relatively easy to set up. (KuCoin, ©2022)

In conclusion Kucoin is not for inexperienced traders and has many flaws even though it is the 5th largest cryptocurrency exchange. Although it has flaws, they are well compensated for by the potential it offers for users to be ahead of others when it comes to new cryptocurrency listings and low market cap altcoins which can offer returns in hundreds of percent. Additionally, it offers features like trading bots and mini games which no other exchange does in such manner.

4.5 eToro

eToro is the biggest retail broker. It works as a social broker platform where traders can post their analysis, ideas and plans and trade them at the same time. Unlike the other cryptocurrency exchange eToro is available globally and even supports Czech language. It is easy to use for beginners, well secured and offers low fees for traditional markets. Deposits can be made in any fiat currencies, even in the Czech crown by a wire transfer, credit/debit card or electronic wallets like Revolut or PayPal. (eToro, ©2022)

Investing wise eToro is great, it has a simple buy button where you enter the desired buy amount and just press buy. Trading wise it is much worse than all the cryptocurrency exchanges, the user interface is not transparent and friendly. eToro has their own special charting software called proCharts but there is not much difference from the regular one they offer. ProCharts offers some indicators, and technical analysis tools but compared to Trading Views features they are very poor. Highest leverage eToro offers is 30x and it is used for currency pair trading, 10x for most commodities it offers, 5x for CFD stocks and 2x on cryptocurrency CFDs. (eToro, ©2022)

The great benefit of eToro is that since it is a broker it can offer trading of stocks, commodities, currency pairs, ETFs, indices and as well as cryptocurrencies. When trading cryptocurrencies not only traders can speculate using CFDs, but they can directly buy the assets and move them to their software wallets from eToro if desired. A unique feature of eToro is copy trading, users can view statistics and success of other traders and by one click copy the trades they make using a bot, it is simply and easy to setup as most other functions. eToro also offers a demo account to explore trading without the need to put in your money. Same as cryptocurrency exchanges eToro offers staking but with worse terms than the cryptocurrency exchanges. (eToro, ©2022)

Although eToro offers the feature to buy cryptocurrencies, it is more efficient to buy cryptocurrencies directly on crypto exchanges than on eToro because of big spreads²² and the fees for trading cryptocurrencies are up to 5 times higher and the fact that once you buy and withdraw a cryptocurrency it cannot be deposited back. Sadly, a live chat with the technical support is only available for users with over \$5000 on their account, for those who don't meet this requirement there is only email and ticket support available. (eToro, ©2022)

Overall eToro is made for beginners from all around the world that are looking to start their investing journey. The interface is made to look easy to use but does not offer the best technical analysis tools or indicators but it has its benefits in the large variety of assets to trade and being minimalistic and friendly but that comes with the cost of not having better technical tools.

4.6 Exchange tokens

Exchange token is a cryptocurrency created by an exchange which it either hands out to its users as a token of gratitude for using it or it can be bought as any other cryptocurrency. Accumulating a certain amount of it gives privileges to the owner in form of perks, discounts, and other benefits on the platform. Not every exchange has its own token though.

The most common perk exchanges offer is a trading fee discount on orders. If a trader has a certain amount of exchange tokens his fees to execute orders decrease, or if his trading volume for the exchange token pair is at certain volume he is rewarded with lower fees for the other pairs. The other way around is that traders can pay their trading fees at a discounted price using the exchange tokens but that is not very lucrative. (Gemini. 2022)

Some exchange tokens are about governance and give their owners the right to vote on how the platform will move forward and what changes will or won't be done since cryptocurrencies are about decentralization this is as far it can get. (Gemini. 2022)

Exchange tokens can be used to increase trading activity on certain assets, the exchange can reward traders for reaching a specific trading volume in certain time with them. This gives the trader additional prize for trading the asset and brings more activity and money onto

²² gap between buy and sell prices

the exchange. This can work the same way with staking, exchanges can offer their tokens on top of the regular rewards to users for staking their coins at the exchange. (Gemini. 2022)

The most opportunity exchange tokens offer are the launchpads. Launchpads are like startup incubators for new cryptocurrency projects. Exchanges help to raise funding for new projects through offering users the possibility to invest their exchange tokens into them and in return get some cryptocurrency of the incubated project depending on how much they invested. The received cryptocurrency can then be instantly sold when the projects are listed on the exchange because once these projects are listed the price tends skyrocket and the early investors can secure nice profits. (Jumpstart, 2022)

Binance's exchange token has the name Binance coin and ticker BNB, it can be used to participate in launchpads, get a discount on trading fees, receive a cashback in BNB for paying using the Binance credit card and earn flexible percentage yields by depositing it on the BNB chain or pay for various real-life services like VPN and travel expenses. (Binance, ©2022)

Bybit's exchange token has the name Bit and ticker BIT, it can be used to invest in launchpad projects, earn yield by providing liquidity or vote on what happens next with the platform. (Bybit, ©2022)

Kucoin's exchange token has the name Kucoin token and ticker KCS, it can be used to get discount on trading fees, get a higher VIP level, get a daily bonus which comes from 50% of Kucoin's daily trading fee revenue if holding over than 6 KCS, use decentralized payment methods, pay in metaverse games, pay travel expenses, or used as collateral to borrow USD. (KuCoin, ©2022)

eToro doesn't have any kind of exchange token or any other currency that could be utilized to get discounts or any advantages on the platform.

4.7 Trading fees

Trading fees are the transaction costs charged to the trader upon the execution of his order by the exchange. There are two types of trading fees, a taker and maker one each of them is connected to a different order type.

Taker fee removes liquidity from order books and is filled immediately, taker fee is paid when traders use market orders, to get in action instantly. (Bybit, ©2022)

Maker fee provides liquidity to order books and is not filled immediately, maker fee is paid when traders use limit orders, these orders are set at predefined price level and are executed once the price reaches them. (Bybit, ©2022)

All cryptocurrency platforms use a tier-based system of fees (VIP levels) except for eToro which doesn't offer any discount on fees and doesn't differ between a market or a limit order either. The more you trade the less fees you pay. Some exchanges offer using their

exchange token as payment for the fees to get a discount on them. For this comparison the lowest and the highest tiered fees were used.

In the table below we can see the non-VIP level tier and the values of trading volume and exchange token owned it is defined by. The non-VIP tier is where most traders begin to trade on either of the exchanges mentioned. To get to a higher VIP level a trader either needs to trade over a certain amount of money each month on the futures contracts or spot or own a certain amount of the exchange token, but this option is only available with Binance and Kucoin, Bybit doesn't offer such option.

Table 1: non-VIP level

Non-VIP	BINANCE		BYBIT		KUCOIN	
	SPOT	FUTURES	SPOT	FUTURES	SPOT	FUTURES
30-day trading volume	<1M	<15M	<1M	<10M	<50 BTC	<100 BTC
Minimum exchange token owned	<25 BNB		not supported		<1 000 KCS	

The friendliest option to advance to the next level is owning 25 BNB which according to today's BNB price on Coinmarketcap would cost around \$11,000 which is not even close to being beginner friendly. Therefore, beginning traders will be stuck in the non-VIP tier for quite some time.

In the table below we can see the requirements for the highest tier available, this tier is achievable only by the hedge funds or few of the biggest traders with large capital. Only buying 5,500 worth of BNB would cost around \$24M according to Coinmarketcap, which is a lot of money to allocate only in one asset and not an amount many people have at their disposal.

Table 2: Maximum VIP level requirements

Max VIP	BINANCE		BYBIT		KUCOIN	
	SPOT	FUTURES	SPOT	FUTURES	SPOT	FUTURES
30-day trading volume	>5B	>25B	>60M	>1.5% of total derivatives market	>80 000 BTC	>160 000 BTC
Minimum exchange token owned	>5 500 BNB		not supported		>150 000 KCS	

In the table below we can see the discounts on trading fees that are available if exchange tokens are used to pay for them. We can see that Binance offers the best discounts that can be applied even to futures contracts. Using exchange token to pay for trading fees is an option, but it is not frequently used because if the price of the exchange token would go up

and the trader would be spending it on fees, he would be reducing his potential profit and the exchange tokens have much better use cases than that.

Table 3: Exchange token fee discount

Exchange token trading fee discount	BINANCE		BYBIT		KUCOIN	
	SPOT	FUTURES	SPOT	FUTURES	SPOT	FUTURES
	25%	10%	not supported		20%	not supported

In the tables below we can see the taker and maker fee for spot and futures contracts on the three mentioned cryptocurrency exchanges and eToro. The first row shows non-VIP tier which conditions we mentioned in the table n.2, and the second row presents the fees for the highest VIP tier offered.

Table 4: non-VIP and maximum VIP level fee structure

	BINANCE				BYBIT			
	SPOT		FUTURES		SPOT		FUTURES	
	Taker Fee	Maker Fee	Taker Fee	Maker Fee	Taker Fee	Maker Fee	Taker Fee	Maker Fee
Non-VIP	0.100%	0.100%	0.040%	0.020%	0.100%	0.100%	0.060%	0.010%
Maximum VIP	0.040%	0.020%	0.017%	0.000%	0.020%	0.000%	0.030%	0.000%

	KUCOIN				ETORO			
	SPOT		FUTURES		CRYPTO		ETFS, STOCKS...	
	Taker Fee	Maker Fee	Taker Fee	Maker Fee	Taker Fee	Maker Fee	Taker Fee	Maker Fee
Non-VIP	0.100%	0.100%	0.060%	0.020%	1%	1%	No fee	
Maximum VIP	0.025%	-0.005%	0.030%	-0.015%				

As we can see the fees for trading spot as a non-VIP are the same everywhere, only if we were to pay for fees using exchange tokens there would be a difference, Binance would then be the definitive winner with its 25% discount. The worst place to trade cryptocurrencies is eToro where the fees are ten times higher. Therefore, cryptocurrencies shouldn't be traded on eToro because the fees are sky high compared to crypto exchanges so day trading there would be highly inefficient.

Where we can see some differences in the non-VIP tier are the futures contracts, Binance offers the lowest fees which can be even smaller by using BNB to pay, it is also the only exchange that offers this discount when trading futures and since non-VIP traders don't have that much money to play with they need to join the market makers and that is best done by market orders because it's hard to fill a limit order when the price is moving fast in one direction. Kucoin and Bybit offer the same terms, in which case Bybit would be the second-best option because of its security and order flow.

When it comes to trading spot as the maximum VIP tier, Bybit offers the best terms when using market orders, but Kucoin comes as the winner when it comes to limit orders because traders get a small rebate²³ when using them and buying spot is mostly done by using limit orders.

For the highest VIP tier futures contract trading the best option is Kucoin. It offers even higher rebates than when trading spot and since a high VIP level trader probably has a lot of money, he needs a lot of liquidity to fill his large orders therefore he has to enter using limit orders because if he used a market order, he would push the price too fast and wouldn't fill the order size he wants. On the other hand, when using market orders Binance is the winner.

4.8 Security

Security plays a big role when it comes to investing and especially when it concerns digital assets. Let's first take a look at how safely the exchanges store the traders' cryptocurrencies and then compare the security measures they offer to protect their customers and themselves due to the fact that they are financial service providers and they have to keep track about their customers using some kind of identification process.

Identification and limits

The type of identification process banks, exchanges and brokers use is KYC in other words know-your-customer. It is a requirement imposed by regulators on financial service providers. Service providers have an obligation to identify their customers, keep identification records and pass them on to public authorities. KYC usually has the obligation to scan an identity card (ID card or passport), which is often accompanied by a request for a selfie photo of the document holder and other documents, such as proof of residence or bank statement. The key fact is that the institutions implementing the KYC are obliged to keep all documents and share them with the state authorities upon request.

Due to Binance being the biggest cryptocurrency exchange and first target of the regulators their KYC policy is very strict and without sending all required documents users can't trade at all, also to increase maximum deposit amount users, must include some kind of proof of address like an internet bill or lease agreement where their address is written. (Binance, ©2022)

Since Bybit and Kucoin don't offer trading pairs of a cryptocurrency and a fiat currency their KYC doesn't have to be that strict. To trade, deposit and withdraw up to 2 BTC on Bybit and 5 BTC on Kucoin a day the KYC isn't required at all. Many traders abuse these exchanges to evade paying taxes since the account they have is not registered to anyone, they can stay anonymous. (Bybit, ©2022) (Kucoin, ©2022)

²³ a partial refund of a payment

eToro is a broker that primarily offers non crypto assets therefore the KYC is required even before the users can start doing something on the platform, but unlike Binance the proof of address and facial recognition is not required.

Security of exchanges and brokers

Security is the most important thing traders should care about when it comes to exchanges due the fact that there have been many hacks over the last few years.

Binance protects its customers by keeping most of the funds in an offline cold storage, storing 10% of trading fees in a secure place that would be used to recover any losses to traders in case of a hack and using address whitelisting which enables withdrawing assets to only predefined addresses after inserting an email and two-factor authentication codes. (Binance, ©2022)

Bybit offers an even better security measures keeping 100% of customers funds offline in a cold storage and individually review every single withdrawal request manually to avoid unauthorized withdrawals while also having the withdrawal whitelisting possibility, two-factor authentication, and possible login device management. It also participates in a bug bounty program which rewards ethical hackers to report loopholes they find in the system. Bybit also does background checks on all its employees. (Bybit, ©2022)

Funds on Kucoin are stored in a cold storage but in an unknown percentage. It has a partnership with Onchain Custodian to secure the cryptocurrencies located on the exchange including possible hacks and such. Same as Bybit it rewards ethical hackers for finding bugs and loopholes in its systems. Kucoin offers the same security measures as Bybit and Binance - address whitelisting, two-factor authentication and a unique one which is a trading passcode that users have to write down when placing the first trade after logging in. (Kucoin, ©2022)

eToro doesn't offer such personalization when it comes to security measures nonetheless it is a fully regulated broker with investor protection that offers additional private insurance up to \$1,000,000 in cash, securities and CFDs on assets, therefore it is a safe place to use and store your assets. (eToro, ©2022)

In the table below we can see all the security measures the exchanges and broker offer. We can see that eToro offers much less security measures compared to the cryptocurrency exchanges and has even bigger withdrawal and inactivity fees if a user doesn't trade for over a year. Therefore, using the cryptocurrency exchanges is favorable if a person wants to trade cryptocurrencies due to the lower fees and higher security standard they offer, the only upside eToro has against them is the possibility to withdraw all funds at once.

Table 5: Security measures

Security	BINANCE	BYBIT	KUCCOIN	eToro
Email adress verification	Required	Required	Required	Required
SMS verification	Available	Available	Available	Available
Password	Required	Required	Required	Required
Two-factor authentication	Available	Available	Available	Available
Security key	Available	Not available	Not available	Not available
Trading password	Not available	Not available	Required	Not available
Withdrawal whitelist	Available	Available	Available	Not available
Anti-Phishing code	Available	Available	Available	Not available
Login safety phrase	Not available	Not available	Available	Not available
Trusted devices	Available	Available	Not available	Not available
Identification – KYC				
ID/Passport/Driver's licence	Required	Optional	Optional	Required
Personal information (name, birth date etc.)	Required	Optional	Optional	Required
Address	Required	Optional	Optional	Required
Proof of address (internet bill etc.)	Optional	Optional	Optional	Not available
Facial recognition	Required	Optional	Optional	Not available
Daily withdrawal limit				
Without KYC	\$0	2 BTC	5 BTC	\$0
With KYC	\$96 000 000	up to 100 BTC	up to 3000 btc	All
Deposit / Withdrawal fees				
Deposit fee	\$0	\$0	\$0	\$0
Withdrawal fees	\$1,65	\$1	\$1	\$5
Inactivity fee	\$0	\$0	\$0	\$10

5 Technical analysis and trading cryptocurrencies

In this chapter I will present various approaches to trading, what technical analysis is and how it is done, what knowledge is needed for it and get into specifics about what you need to know about a chart like how candlesticks work, what are horizontal levels, fibonacci theorem or initial balance. I will also present some indicators and oscillators which can add another layer of understanding of what is happening in the chart and how a trader should approach it.

5.1 Technical analysis

Technical analysis has its roots in 17th century and refers to the study of past price action and supply and demand as guidance in forecasting future prices because the historical price developments of assets are likely to be repeated in the future. It doesn't take any fundamentals in consideration because all data needed is the volume and price and that is all in the chart. (Rockefeller, 2011)

The goal of technical analysis is to find the place and time where the best setup for a trade with a good risk to reward ratio could appear. Nevertheless, it is a game of probabilities where nothing is certain, so even the best-looking setups can end up going wrong therefore a good risk management is needed to be profitable over a long period of time. (Rockefeller, 2011)

There are no real differences between cryptocurrency, stocks, Forex, futures, commodities, indices, or any other instruments whose price is affected by supply and demand when it comes to technical analysis. The same tools and strategies can be applied on all in the same way and on any time frame. The only small difference is that cryptocurrency markets tend to be more volatile as mentioned before.

The biggest issue of technical analysis is that it is very subjective so two traders can look at the same chart and come out with different ideas.

5.1.1 Charts

There are many types of trading charts but the main three are the candlestick, line, and bar ones. Each of these have their own specifics and are used on different occasions.

Line chart

Line chart is the simplest one that that can be seen when charts are presented in television and other media because it is easy to understand and has no distractions to it. The line chart looks like a line that is drawn from on close of price to the next one. (xtb, ©2022)

It is used to get a general idea of what is happening in the chart and find chart patterns because it is much easier to do so when the chart is just a clear line, therefore it is good for beginners. (xtb, ©2022)

Candlestick chart

Candlestick charts present the open, high, low, and close of price. Each time period is represented as a candle that has a body which shows the open and close of the price and shadows that are vertical lines coming out of the body which represent the maximal and minimal price of that time period. The longer the body is the more intense is the buying or selling pressure. This chart type is used by the professional traders. (xtb, ©2022)

Bar chart

Same as the candlestick charts the bar charts provide the complete open, high, low, and close price package. The difference between the candlestick and bar chart is that the candle body is presented as a vertical line (bar) and the opens and closes of price as horizontal lines that connect all the bars together creating the chart. This chart type works as an alternative to the candlestick chart, the advantage of it is being visually more minimalistic. (xtb, ©2022)

5.1.2 Candlesticks

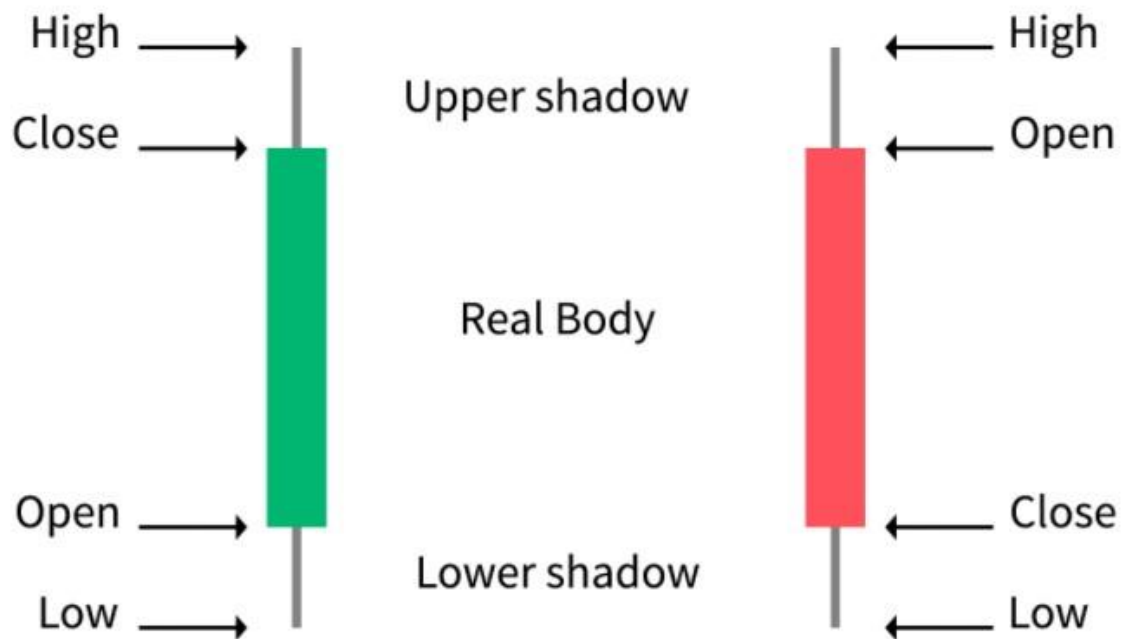
Candlestick charts have their origins in Japan where a man named Homma discovered that there is a link between supply and demand of rice and that the markets are heavily influenced by people's emotions. (Achelis, 2014)

What candlesticks do is display that emotion visually by the size and color of them. Traders use candlesticks to read the emotions and forecast short-term price action. (Rockefeller, 2011)

A candlestick is formed of an open, high, low, and close of price. It has a body and shadows that come out of the body on both sides. The body represents the price between the open and close of the candle whereas the shadows represent the high and low of the candle. Each candle is printed during a certain time period which is chosen by the trader. For short-term trading traders use the 1-minute up to 1-hour candlesticks and for long-term the 4-hour to 1-month candles are used. (Achelis, 2014)

In the figure below we can see the description of the parts of a candlestick.

Figure 20: Anatomy of a candlestick



(Sam Oulmet, ©Coindesk 2021)

5.1.3 Candlestick patterns

Candlestick patterns are a technical analysis tool which is used to graphically interpret movements of the price during a day on the candlestick chart. Candlestick patterns are something every trader should have in memory but trading by only using candlesticks can be risky.

Doji

Doji is a neutral pattern made of only one candlestick. It happens the opening and closing prices of an asset are equal or have small differences. It looks like a cross or a plus sign, with a small or no body. This is rare candlestick pattern which shows uncertainty on the market where neither the bulls nor bears want to act. (Rockefeller, 2011)

The doji pattern can have two more variations beside the neutral one. A dragonfly doji which is bullish and has the body at the top of the candle and indicates that strong upside is incoming and gravestone doji which is bearish and has the small body at bottom signaling a sell-off is bound to happen. (Rockefeller, 2011)

Bullish and Bearish Engulfing

The bullish and bearish engulfing pattern is formed by two candlesticks. In the bullish case the first candle has a short red body that is completely engulfed by a larger green candle. The shadows don't count so the bigger candle can have a smaller shadow than the small one and still be considered engulfed. In the bearish case it is the exact opposite the first candle has a short green body that is completely engulfed by a larger red candle. Usually, these two

candlestick patterns signify a possible trend reversal especially when it if they happen on the higher timeframes.

Three white soldiers and three black crows

The three white soldiers and the three black crows are trend reversal patterns that consist of three consecutive candlesticks of the same color preferably with long bodies that go in one direction where each candle opens below the previous one. White soldiers are the bullish version, black crows the bearish one. Both signal a probable continuation of the trend they start.

Three-line strike

The Three Line Strike Candlestick Pattern is formed by four candlesticks and can be found during both upwards and downwards trend. The pattern is spotted by the formation of three candlesticks during a one-way trend where the fourth candle goes the opposite way. (Achelis, 2014)

In most cases, the pattern indicates a trend reversal. There are two types of three-line strike pattern a bearish and bullish one. In the bullish pattern, the first three candles are going up with the trend and the fourth one goes down. The bearish pattern leads to the formation of three bearish candles during a downtrend which then reverses to the upside with the green candle erasing the previous red candles. (Achelis, 2014)

5.1.4 Trend analysis

Trend analysis is one of the many techniques of technical analysis. It tries to predict the future price of an asset based on its recent trend data utilizing technical tools such as trend lines and support and resistance levels. The historic data is mixed with the recent when forecasting the long-term direction. (Rockefeller, 2011)

A trend is a direction where the market goes during a period of time, the overall trend of an asset is usually determined on the daily and higher time frames. There are three trend types in the market uptrend, downtrend, and no trend. (Rockefeller, 2011), (Griffis and Epstein, 2009)

- Uptrend is when the market is bullish, and prices are rising making new higher highs and higher lows.
- Downtrend is when the market is bearish, and prices are decreasing making lower highs and lower lows.
- No trend is a situation where price goes sideways and is contained within a range where it bounces up from the bottom to the top and falls from the top to the bottom. Until it is pushed outside of the range and starts trending again.

(Rockefeller, 2011),

Trends are split into three different time frames. When all time frames are trending the same way, the market is united and further push to that side should be expected. (Griffis and Epstein, 2009)

- Short-term trend which is defined by the market structure on the 1-minute up to 1-hour time frame.
- Mid-term trend which is defined by the market structure on the 4-hour to 1-day time frame.
- Long-term trend which is defined by the market structure on the 1-week and higher time frames.

(Griffis and Epstein, 2009)

Trend lines

Trend lines are straight lines that connect two price points on a chart and are drawn underneath or above pivot points in price and work as support and resistance lines that the trend should hold if it is supposed to continue. When touched by price they should give a reaction. Drawing them is very subjective but at least two points are needed. In an uptrend the trend lines should connect the higher lows and in a downtrend the lower highs. Once a trend line is broken traders should expect the trend to either go into sideways consolidation phase or down. (Griffis and Epstein, 2009) (Rockefeller, 2011)

There are few rules about trend lines. The higher the time frame the more reliable the trend line is. The more times the price touches a trend line the more likelihood is that it will go through it on the next time. (Griffis and Epstein, 2009)

In the following figure we can see a drawn trend line by connecting the swing lows. It then successfully holds the trend three times until it is broken and backtested, flipping support into resistance after which prices drop down.

Figure 21: Trend line example



own processing according to (TradingView)

5.1.5 Horizontals

Horizontals are another tool of the technical analysis which works on the basics of supply and demand. As their name says they are horizontal lines. They are used to display areas of support where price stops when falling and resistance where price stops when rising and work the same way as trend line but are not angled and typically drawn along swing highs and lows or better along a series of them where each high or low is at or close to the horizontal line. When price reaches them, traders can expect the price to either get a bounce or drop whether it is rising to the resistance or dropping to the support. (Achelis, 2014), (Griffis and Epstein, 2009)

There are few rules to about horizontals. The more times a resistance or support gets tested the more likely it is to break and also the higher the time frame they are marked in, the stronger they are. If a resistance or support is broken the traders can expect the price to continue that way and should look for new horizontal levels to trade of. Support and resistance levels can flip into being the opposite once they are broken and backtested, a backtest is when price comes back to the broken level from the other side, touches it and bounces switching a resistance into a support or gets rejected switching a support into a resistance. If a strong support or resistance is broken, it is then expected to be a strong resistance or support when price come back to it. (Achelis, 2014), (Griffis and Epstein, 2009)

In the next figure we can see the price first dropping down and then forming a range creating the support and resistance horizontal levels between which it trades until the resistance is broken which leads to further continuation to the upside.

Figure 22: Horizontal's example



own processing according to (TradingView)

5.1.6 Fibonacci theorem

The Fibonacci sequence has been here for a long time. It is an infinite series of numbers that begins as follows: 1,1,2,3,5,8,13,21,34. etc., the next number is always the sum of the two preceding. The mathematical relationship was known already in ancient Greece but the first person to describe it was Leonardo Pisano an Italian mathematician. (Boroden, 2008)

Dividing any two successive numbers of the Fibonacci sequence gives the value of the golden ratio represented by Greek letter *phi* $\phi \approx 1,618$. It is known as the perfect ratio and is created by dividing the line into two parts so that the ratio of the larger part to the smaller part will be the same as the ratio of the whole line to the larger part. The rule of the golden ratio is found everywhere in nature, in the structure of leaves, in snail shells, and even elephant tusks. The golden ratio is often encountered in design and painting too a great example is the Mona Lisa or Vitruvian Man by Leonardo da Vinci. (Boroden, 2008)

Based on the Fibonacci sequence, specific ratios were created and are now used as technical analysis tools for trading. There are several tools that use the Fibonacci sequence, the most important ones are the Fibonacci Retracements, Fibonacci Extensions and Fibonacci Fans and Fibonacci Time. Combining all of them gives a great outlook. (Boroden, 2008)

Fibonacci retracements

After a significant price move either up or down, prices will often retrace a significant portion if not all of the original move. As price retraces, support and resistance levels are often found near or at the Fibonacci Retracement levels which display how much has the retraced from the original move. (Achelis, 2014)

The strongest level in trading is considered to be the golden Fibonacci ratio of 61.8%, which represents the inverse value of ϕ , the golden ratio. Other important levels are 38.2%, 23.6%, 50.0% or 78.6% which also serve as levels of support and resistance. (Boroden, 2008)

The main extensions of the Fibonacci sequence are levels 161.8%, 261.8% and 423.6%. Like the other levels, these show where the trend is likely to reverse therefore they are a good take profit areas while trading. (Boroden, 2008)

Fibonacci Retracement is used to identify potential horizontal support and resistance levels. When using it, it is important to stretch the Fibonacci from a swing low to a swing high when looking for support and from swing high to swing low when looking for resistance, Fibonacci should always be pulled from left to right side because that is how the market moves. (Boroden, 2008)

It is an excellent tool because it works on all timeframes the same way. The higher the timeframe the stronger the Fibonacci levels are. If a trader needs to confirm the conclusions of his technical analysis with other tools, the Fibonacci sequence ones are a good choice. (Boroden, 2008)

Strong Fibonacci levels such as 61.8%, 38.2%, 23.6%, 50.0% or 78.6% can be used in trading as points to enter or exit the trade. Some traders use a strategy where they set buy limit, sell stop or take profit orders according to Fibonacci levels. Fibonacci is easy to use, reliably displays support and resistance levels and how strong they are. (Boroden, 2008)

In the figure below we can see the 61.8% retracement holding the price twice before a move up.

Figure 23: Fibonacci retracement example



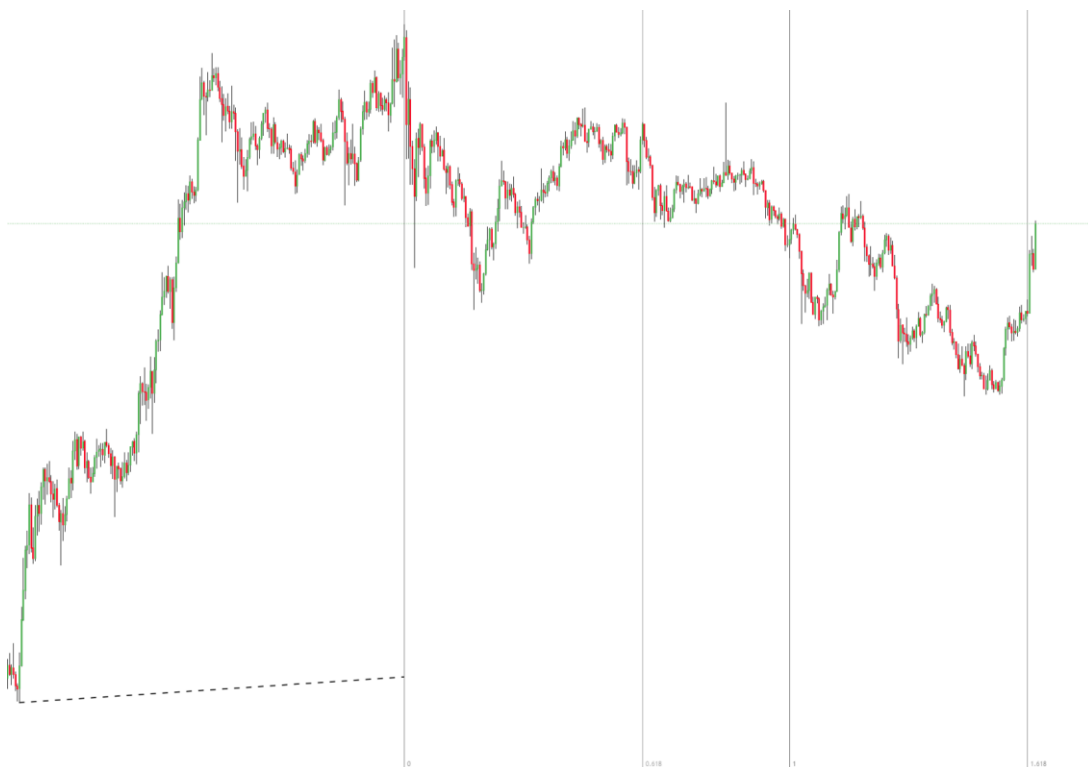
own processing according to (TradingView)

Fibonacci time

Fibonacci time is a tool based on time, and it is supposed to show exact time period where a major swing high or low could happen. It works the same way as the retracements, it is pulled from a swing high to a swing low or the other way around. The difference is that it shows vertical lines instead of horizontal since it isn't used to measure price but time and the best levels to use are the 100%, 161.8%, 200%, 261.8%, traders should expect a strong move to any side once these time periods come. (Boroden, 2008)

In the figure below we can see that close to the Fibonacci time zones a significant move happens.

Figure 24: Fibonacci time example



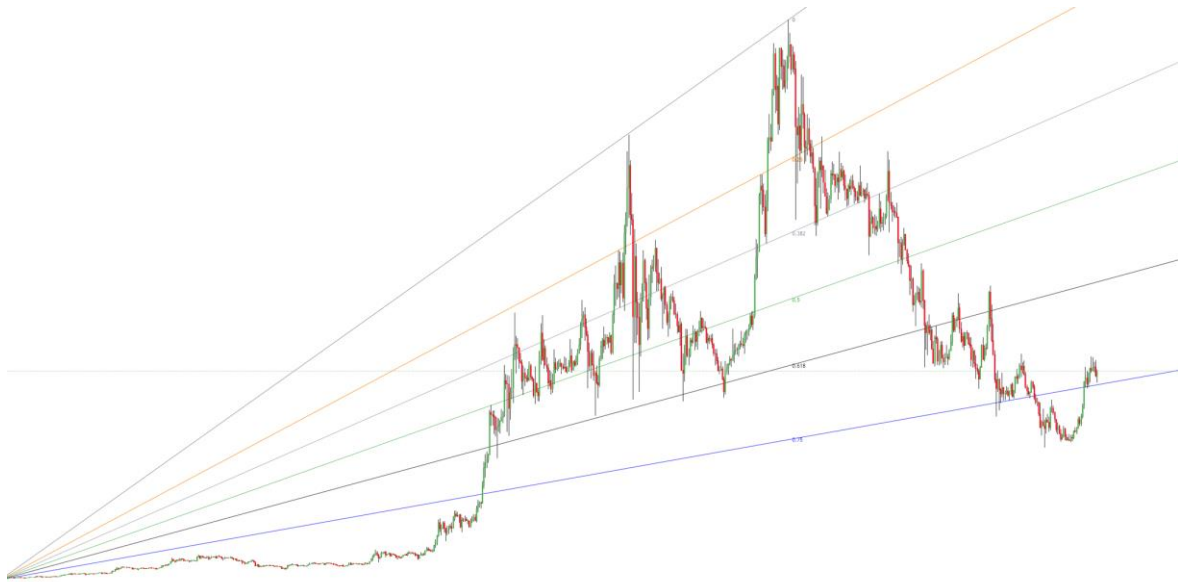
own processing according to (TradingView)

Fibonacci fan

Fibonacci fan is tool based on both time and price therefore, it is very powerful. Same as the retracements and time, it is pulled from a swing high to a swing low or the other way around. The fan looks like a set of sequential trendlines drawn starting from the beginning of the pull, these trendlines then act as resistance and support levels. The recommended levels to use are the 61.8%, 38.2%, 25.0%, 50.0% or 75.0%, once reached they should give a reaction. (Boroden, 2008)

In the following figure we can see how the 61.8% Fibonacci fan level switches from being a strong support into a strong resistance.

Figure 25: Fibonacci fan example



own processing according to (TradingView)

5.1.7 Initial balance

Initial balance is a price range that is formed during the first hour of trading, this happens once a stock exchange is opened for trading, in cryptocurrency markets this range is starting to form from 00:00 UTC time (Prague 02:00). The look of this range can often foretell what will happen during the rest of the trading session. Therefore, understanding the initial balance day types is key to a successful day of trading. (Dalton, Jones, and Dalton, 1999)

The day types of initial balance:

- Primary day - which is a regular day where the initial balance is wide and mostly contains the extremes for the day. In such day's traders would look to short the high and long the low of the initial balance.
- Variation of a Primary day where the initial balance is narrow, the early session is less volatile, and price stays inside it. Later in the day price is pushed outside the initial balance to its range extensions which are equal to 50% and 100% of the initial balance range and traders then look to take the trade from the extensions back into the initial balance range.
- Double distribution day - where the initial balance forms a narrow base in the early part of the trading session reflecting low participation in the market, in other words, a price area where two-sided trading is taking place while the price is pushed sideways. The narrower the range gets, the easier it is to overwhelm it and once that happens price sharply moves in one direction where it forms another range in which it stays for the rest of the day. Traders look to position themselves on both sides of the market and then join side which gets the sharp move that is usually backed with high volume.
- Balanced day - where there is no trend only sideways price action early in the session, this day is similar to a double distribution day because price forms a narrow

base, the difference is that the price stays contained within the base, and the range is never extended. These days usually develop before the release of some news events or reports. They show that market has come to a balance and is waiting on further information to decide its next direction.

(Dalton, Jones, and Dalton, 1999)

Other than that, there are two more types of initial balances in today's markets that are most important for trading guidance:

- Rotational day - where the initial balance base is so narrow that it fits inside the previous day initial balance where it moves sideways for some time until the price gets sharply pushed up to take out the previous day's high and then goes straight down to take out previous day's low doing the so-called full rotation. Trading the rotation trader usually wants to fade ²⁴the first move and join the second one.
- Trend day - where the market starts to trend off from the daily open and doesn't stop creating no proper initial balance. When such days happen, traders should look to join the trend because attempting to apply the initial balance concept would likely place them on the wrong side of the market.

(Dalton, Jones, and Dalton, 1999)

5.1.8 Indicators and oscillators

Moving averages

One of the most popular tools of technical analysis are moving averages. The moving averages are a trend monitoring lagging²⁵ indicator. They are used to track a trend, which means they are either rising or falling in line with the market. Financial asset prices can fluctuate and change rapidly and moving averages help to smooth out the price movements and thus facilitate the identification of trends. In addition, moving averages are used not only to determine the direction of the trend, but also to enter a trade. (Investopedia, 2022)

Most moving averages are based on closing prices. For example, the ten-day simple moving average is the sum of the closing prices of the last ten days divided by ten. As its name suggests, the moving average is the average of those moves. There are many types of moving averages, the common ones are the simple moving average and exponential moving average. There are many strategies using the the moving averages and often traders use multiple ones at once to get a better view of the market over a longer period of time. Common time periods are 50, 100 and 200 for the long-term trend confirmation and 5, 10, 20 for the short-term. (Griffis and Epstein, 2009)

The biggest drawback of moving averages is the fact that they use data from the past, therefore they spot the recation when it already happened and because of that they are used

²⁴ take an opposite position into a high-momentum trend

²⁵ it is based on past prices

more as a confirmation tool rather than a one to base trades upon by the more experienced traders. (Griffis and Epstein, 2009)

Simple moving average – SMA

The Simple Moving Average is a type of moving average that is calculated by adding the price of an instrument over a number of time periods and then dividing the sum by the number of time periods. It basically shows the average price of a given time period, with equal weighting given to the price of each period. The commonly used combination is formed of the 20, 50, 100 and 200-day. (Investopedia, 2022)

$$SMA = \frac{A_1 + A_2 + \dots + A_n}{n} \quad (3)$$

Where:

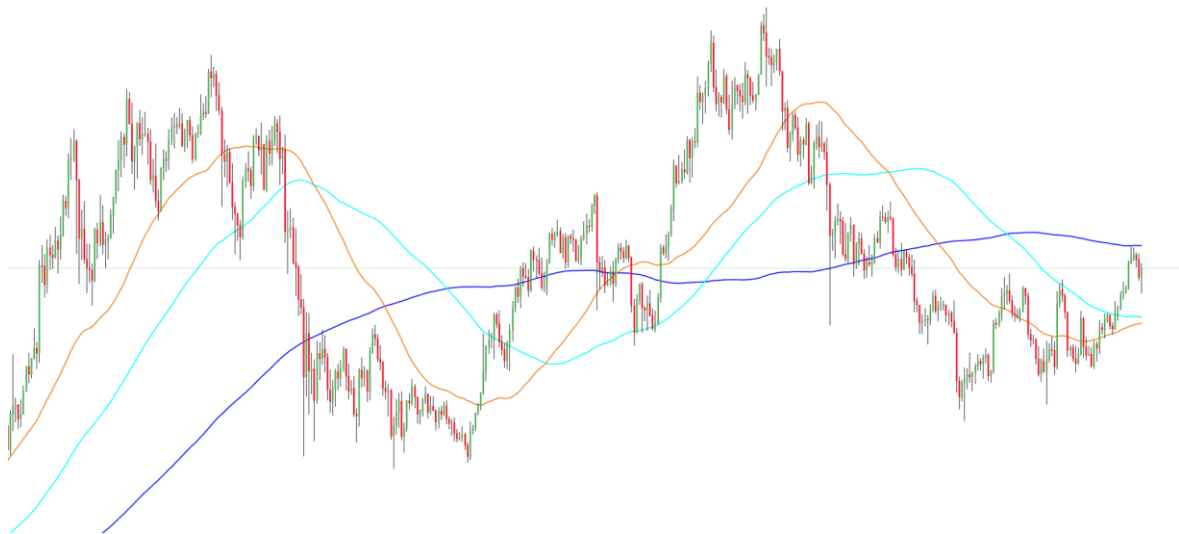
A_n = the price of an asset at period n

n = the number of total periods

A popular use case of SMAs is the golden and death crosses. This situation happens when the 50-day SMA moves below the 200-day SMA, it is known as the death cross and it is considered a bearish signal, indicating possible further downside. The cross of the 50-day SMA above the 200-day SMA is a bullish signal that indicates a possible upside movement. (Investopedia, 2022)

In the figure below we can see the 50 SMA in orange, 100 SMA in teal and 200 SMA in blue on a daily Bitcoin chart.

Figure 26: 50/100/200 SMA



own processing according to (TradingView)

Exponential moving average - EMA

Exponential moving average is another type of a moving average. The major difference between the simple and exponential one is the sensitivity of how each one reacts to the data its calculated by, the EMA gives higher value to more recent data, while the SMA gives equal weight to all the values. Same as the SMA, EMA is used to smooth out price fluctuations and since EMA is based upon more recent data it responds quicker to price changes which is the reason why many traders prefer it over the SMA because new data is considered more relevant. (Griffis and Epstein, 2009)

The EMA can be calculated in many ways. One way of calculating it is by computing the SMA, weighting multiplier k and the current EMA. (Griffis and Epstein, 2009)

$$EMA = Price_{today} * k + EMA_{yesterday} * (1 - k)$$

(4)

Where:

n = the number of total periods

$k = \frac{2}{n+1}$ = the weighting multiplier, which determines the sensitivity

$Price_{today}$ = the current closing price

$EMA_{yesterday}$ = the previous EMA value

EMA_{today} = the current EMA value

Short-term traders favor the 12 and 26-day EMA while the 50 and 200-day is popular among the long-term traders. We can say that the same time periods for EMA and SMA are used for the long-term trading on the other hand short-term trading periods are distinct. (Investopedia, 2022)

In the figure below we can see the 50 EMA in orange, 100 EMA in teal and 200 EMA in blue on a daily Bitcoin chart.

Figure 27: 50/100/200 EMA



own processing according to (TradingView)

Relative strength index – RSI

Relative strength index is a momentum indicator, it determines the strength or weakness of an asset by measuring the rise or fall of its price. It is an oscillator that follows price with a range between 0 and 100. It can be utilized to spot overbought or oversold conditions of an asset, identify support and resistance levels using trend analysis same as on a chart and spot divergences in prices which can indicate possible reversals and more. (Investopedia, 2022), (Achelis, 2014)

When RSI was introduced by J. Wells, the recommended time period was the 14-day. Since then, the 9 and 25-day periods also risen in popularity. The fewer days used to calculate the RSI, the more volatile the indicator. (Achelis, 2014)

It is calculated by the formula below, which takes into account positive and negative exchange rates changes for a specified period of time. This ratio then forms the relative strength of the asset.

$$RSI = 100 - \left[\frac{100}{1 + \frac{\text{Average gain}}{\text{Average loss}}} \right]$$

(5)

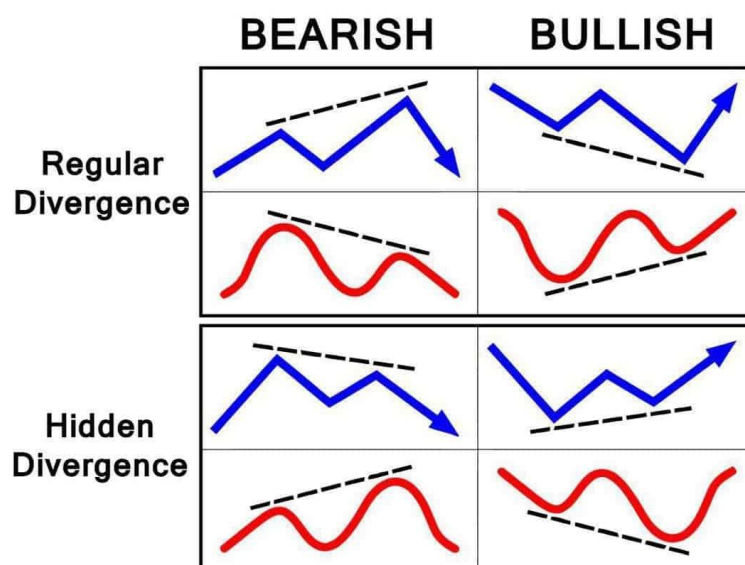
Where:

Average gain = an average of uptrend price change

Average loss = an average of downtrend price change

Divergences occur when the price makes a new high or low that is not confirmed by a new high or low on the RSI showing exhaustions or continuation of a trend. There are two types of divergences regular and hidden, each one has its specifics and meanings described visually in the figure below. Hidden divergences spot trend continuation whereas regular spot trend reversal. (Achelis, 2014)

Figure 28: Divergences cheat sheet

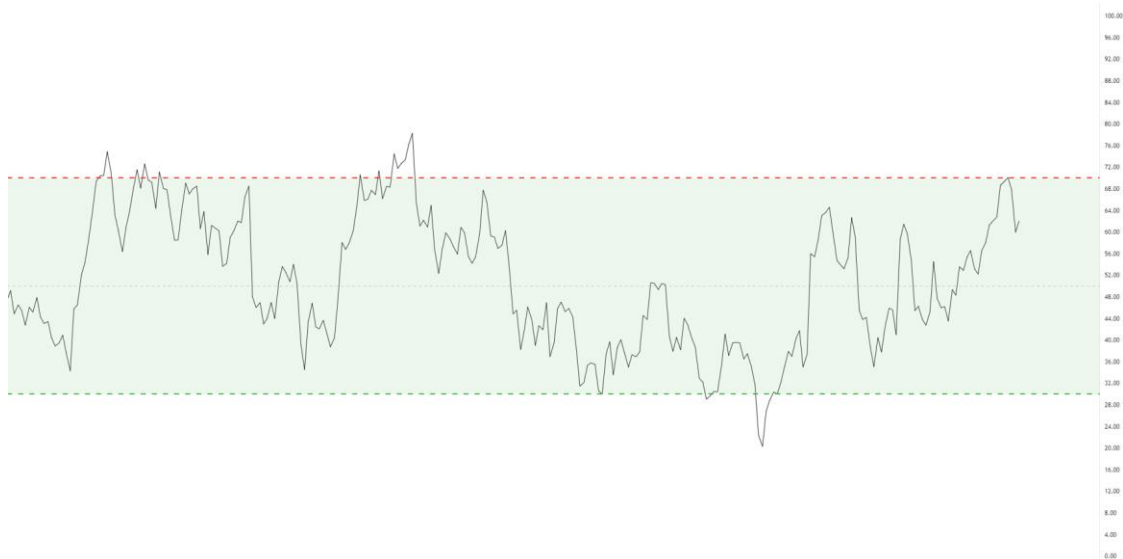


(@Abhishekkar_ on Twitter, 2019)

- Regular bullish divergence happens when **price** makes a lower low and **indicator** makes a higher low creating the divergence, this signals a possible trend reversal to the upside.
- Hidden bullish divergence happens when **price** makes a higher low and **indicator** makes a lower low creating the divergence, this signals that the uptrend should continue.
- Regular bearish divergence happens when **price** makes a higher high and **indicator** makes a lower high creating the divergence, this signals a possible trend reversal to the downside.
- Hidden bearish divergence happens when **price** makes a lower high and **indicator** makes a higher high creating the divergence, this signals that the downtrend should continue.

Overbought and oversold conditions happen when the RSI reaches either 70 for overbought or 30 for oversold. In the figure below we can see the 70 represented as the red dashed line and 30 as the green dashed line. The RSI usually bottoms out below 30 or tops out above 70, it is common that it forms three tops/bottoms before reversing to the other side. (Achelis, 2014)

Figure 29: Overbought and oversold RSI



own processing according to (TradingView)

Overbought condition means that an asset is heavily pushed upwards with no retracements by constant buying, suggesting that a correction is bound to happen. An overbought assets price is considered above fair value meaning it is overvalued. (Yahoo.finance, 2014)

Oversold condition means that an asset is heavily pushed to the downside with no proper retracements by constant selling. An oversold assets price is considered below fair value meaning it is undervalued. (Yahoo.finance, 2014)

Volume weight average profile - VWAP

VWAP averages the closing prices of candles during a given time periods. The most common are session which in traditional markets is the time from open to close of trading day and in cryptocurrency markets the entire day, weekly which resets every Monday, monthly which resets every first day of the month and yearly which resets every first day of a year. (Speedtrader, ©2022)

It is calculated by the formula below which combines the sum of price and volume.

$$WVAP = \frac{\Sigma Price * Volume}{\Sigma Volume} \quad (6)$$

VWAP is represented as a line that works as support and resistance (below the level expect resistance, above the level expect support) it is very impactful because it is calculated by incorporating price and volume statistics unlike moving averages. It can be used on all time frames, from a very high time frame for swing traders, to a very low time frame (1 min). Previous session, weekly, monthly, and yearly VWAP closes act as support or resistance lines too, when reached they usually give a reaction and there are multiple trading strategies built on top them. (Speedtrader, ©2022)

Average true range - ATR

Average true range indicator belongs to the family of oscillators. It is visualized as a curve that reaches values from 0 to infinity and used to determine the degree of volatility in the markets for chosen time period. (Achelis, 2014)

ATR is used for detecting a volatile and therefore riskier periods in the markets. When ATR values are high a trend might be close to an end, on the other hand when ATR values are low a huge move can be unfolding soon. It can also be used to determine stoplosses, the larger the ATR the larger the SL because the market is more volatile so more impulsive moves can happen. (Rockefeller, 2011)

If the market does not move, the ATR curve sits around values close to 0. On the contrary, when announcing important news, reports or any other fundamental events the market is usually set in instability and nervousness and volatility enters the market setting the ATR in motion. (Rockefeller, 2011)

There are many ways to calculate the ATR using various moving averages. One way is to calculate the true range which is the greatest value out of these three below, then the value is inserted into the second formula which includes the desired time period and from that we get the final result. (Investopedia, 2021)

$$TR = Max[(H - L), Abs(H - C_p), Abs(L - C_p)] \quad (7)$$

$$ATR = \left(\frac{1}{n}\right) * \sum_{(i=1)}^{(n)} TR_i$$

(8)

Where:

TR = True range

Max = maximum value

$H - L$ = high – low

$Abs(H - C_p)$ = absolute value of (high – previous close)

$Abs(L - C_p)$ = absolute value of (low – previous close)

n = the number of total periods

TR_i = the greatest of the values above

Open interest – OI

Open interest is the sum of opened contracts that traders in the market are holding at the end of every trading day, counting both the long and short position. It is more of a data feed than an indicator, it's used to measure the total level of activity of a market and helps with determining whether a move is strong or weak. Combining open interest with volume is even more powerful, tracking these two data feeds can lead to a transparent view of whether money is entering or leaving the market. (EconomicTimes, ©2022), (Achelis, 2014)

Open interest has four rules:

- Prices moving up in an uptrend and total open interest increasing shows that new money is coming into the market which is considered bullish
- If prices are rising and open interest is declining the movement up is likely caused by shorts closing. This signals that money is leaving rather than entering the market. Once shorts stop closing the trend is likely to reverse to the downside.
- While price is in a downtrend and open interest is rising this indicates that new money is coming into the market in form of aggressive shorts selling which is considered bearish.
- Situation where total open interest is declining along with declining price is considered bullish because it shows that there are no aggressive short positions opening which indicates that the downtrend is weak.

In the table below we can see the four rules of open interest visualized for better understanding. Red background color means prices are going down and green that they are rising.

Table 6: Open interest rules

Price	Volume	Open Interest	Market
Rising	Increasing	Increasing	Bullish (strong uptrend)
Rising	Decreasing	Decreasing	Bearish (weak uptrend)
Declining	Increasing	Increasing	Bearish (strong downtrend)
Declining	Decreasing	Decreasing	Bullish (weak downtrend)

5.2 Trading

Trading isn't a get rich quick scheme, it works the same way as any other craftsmanship, traders must study many hours, test their hypothesis, and practice their craft to improve before they can become masters which can take years.

Trading is different from investing in the way that wherever the price goes doesn't actually matter because the market can be traded both ways, therefore traders can make money not only when the prices rise entering long positions but also when they drop down entering short position. (Griffis and Epstein, 2009)

To trade a trader needs a plan and set of rules upon which he reacts. The plan consists of scanning the market, choosing an asset, applying technical analysis to it to find a right place for entry, take profit and stop loss and creating a scenario with a set of rules upon which a short or long position is entered. (Griffis and Epstein, 2009)

Short or Long

Going short or long means entering the market either on the bid (buy) side or the ask (sell) side. When entering a long position, the trader is speculating that prices will rise, and he will be able to sell his assets at a higher price whereas entering a short position the trader thinks the prices will decrease therefore he sells his assets in order to re-buy them for a cheaper price. (Rockefeller, 2011)

5.2.1 Trading tools

Trading tools are used by the traders to control and position themselves in the market and respond to its behaviour.

Take profit – TP

Take profit is a tool used to pre-set a limit order at a price point where the trader wants to exit his position in profit. Once the price point is reached it is automatically executed. It is common that traders set multiple take profits on a single position at important price points on the chart to secure at least some profits if the trade was to go their way only partially. (Griffis and Epstein, 2009)

Stop loss - SL

Stop loss is a tool that is pre-set by the trader used to minimize losses, once hit it executes a market order which exits the position. It is a price point defined by the trader at which he thinks his idea is no longer valid and therefore he wants to exit his position. Professional traders instead of using the pre-set stop losses use manual ones meaning they exit the trade by themselves once it is invalidated, this approach is not recommended for beginners, because they usually don't have the guts to quit a losing trade and end up losing much more than if they had used the pre-set one. (Griffis and Epstein, 2009)

Since cryptocurrency markets run 24/7 it is a good habit to place a stop loss when a trader goes to sleep because there is nothing more psychic damaging than waking up to a huge losing trade.

5.2.2 Risk and money management

Money and risk management are key aspects of long-term profitability, they are a predefined plan with a set of rules that the trader defines and must obey at any times. The rules can define things like after losing three trades in a row I will go outside because my head isn't thinking straight, if I make over 3% of my entire portfolio a day I stop to trade for the rest of the day or only take the most probable trades, I will not take trades that have a risk to reward ratio smaller than 2:1. These rules are not only about trading itself but also about traders feelings so when a trader feels bad he shouldn't trade because he is more inclined to involve his feelings into trading which can lead to a bad decision therefore losing money. If the trader doesn't obey any of his predefined rules, he should stop trading for the day. (Rockefeller, 2011), (Griffis and Epstein, 2009)

A huge part of money and risk management is the position sizing where the trader defines how much he is willing to lose if the market behaves differently than he expected. With the proper management even a low winrate can be profitable if right trades are taken. The goal of trading and investing is to win and minimize losses while being overall profitable. (Rockefeller, 2011), (Griffis and Epstein, 2009)

5.2.3 Position sizing

Position sizing refers to a process where the trader determines how much money he allocates to a trade and how much he is willing to lose if his idea doesn't play out. Position size is subjective and traders account size and risk tolerance must be taken in consideration

when determining the right size. Position sizing should be adjusted to market conditions, at times of uncertainty traders should decrease their account risk. (Rockefeller, 2011)

Account risk

Account risk is the amount of money usually described in percentages that a trader is willing to lose of his entire portfolio in a single trade. In a single trade, trader should not expose himself to a loss greater than 1-2% of his entire portfolio, beginners should not use more than 1% to not lose all their money before they learn how to properly trade. When swing trading a larger percentage (up to 5%) of portfolio can be allocated into a trade since the trader is looking for a higher-term time frame play which can take many days even weeks to develop. When scalping, no more than 1% of the entire portfolio should be lost if a trade goes the other way. (Rockefeller, 2011), (Investopedia, 2022)

Trade risk

After deciding how much money a trader is willing to lose on a single trade if his idea is wrong, he proceeds to develop a trading plan which consists of an entry, take profit and a stop loss. The trade risk is the distance between the entry and the stop loss prices, which is usually defined in percentages. (Rockefeller, 2011), (Investopedia, 2022)

Position size

When the account risk and trade risk values are solved, the trader can proceed to calculate the position size that he will be entering the trade with which is defined by them. This whole process can be solved using the formula below. (Rockefeller, 2011), (Investopedia, 2022)

$$Position\ size = \frac{Portfolio\ size\ (USD) * Portfolio\ risk\ \%}{Stop\ loss\ \%}$$

(9)

5.2.4 Journaling

Journaling is a very important part of trading and being profitable long-term. Trading journal works as a trader's diary, which includes his past trades, thoughts about the market, even emotions, basically anything he might find helpful to improve his trading. (Griffis and Epstein, 2009)

Traders use the journal to find patterns in their past trading. Using basic statistical software like Excel they can quickly find where they performed poor or well, which setups work the best for them and look to repeat those in the future, not only this helps them to trade better but it also boosts their confidence. (Griffis and Epstein, 2009)

There are many versions of trading journals available on the internet including some that are linked to a trader's exchange account and automatically save every trade he takes using an API but for beginners a simple Excel spreadsheet is more than enough. (Griffis and Epstein, 2009)

6 Practical part

In this chapter I will be presenting a few trading strategies that are based on different tools and indicators of technical analysis. Each strategy has its own predefined rules for entry, take profit and stop loss upon which a long or short position will be opened. The strategies will be tested on all the previously mentioned cryptocurrencies in the thesis.

The goal of this chapter is to put the presented trading strategies in use and backtest them to verify if they are profitable or not based on the historic data for each cryptocurrency and compare their results.

6.1 Backtesting

Backtesting is the process of testing a trading strategy on historical data of various assets to find whether it is profitable or not. Depending on the results the strategy can then be optimized by changing its parameters or setting a different stop loss or take profit to improve its results. The idea behind backtesting is to let the trader know whether the strategy is or isn't worth trading. (Rockefeller, 2011)

There are two ways to backtesting, traders can either do it manually by writing all the entries and exits by hand into a statistical software like Excel in which the results can be easily displayed through various charts but the con of this is that writing down everything manually takes a lot of time. The second option is automatical backtesting which is much faster than writing everything down but requires programming knowledge making it less accessible to everyone also if there is an error in the code the strategy might not work at all or display wrong results which can lead to poor decisions. It is done by creating a script which when executed goes through all the chart data and if programmed conditions are met a long or short position is placed. (Rockefeller, 2011)

For this thesis both types of backtesting were put in use the automatical one for testing the RSI + MA strategy using pine script language and backtesting terminal in TradingView and the manual one using an Excel spreadsheet to test the the Initial balance backtest and VWAP close strategies.

These statistics from the backtest are used to determine whether a strategy is worth using:

- PnL²⁶
- Winrate in %
- Profitability in %
- Maximum draw down

²⁶ profit and loss

- Duration of the backtest
- Equity curve

Since backtesting works with historical data only it has its risks because no one can foretell the future and the markets evolve every day therefore a tested and optimized profitable strategy can still turn into a losing one. (Rockefeller, 2011)

6.2 Trading strategies

Trading strategies used in this thesis are all based on the knowledge, technical analysis tools and indicators presented in the theoretical part and here they will be put in practice. The out-sample backtesting method was used to test the profitability of each strategy. The in-sample consisted of data from the time between 01.01.2021 to 31.03.2022 and the out-sample consisted of data from the time between 01.04.2022 to 30.04.2022.

Out-sample backtesting is when a backtest is divided into two parts: in-sample and out-sample. The in-sample test is where the rules, signals, and parameters are set. The out-sample is where the set rules, signals and parameters are tested on new data. The whole point of backtesting is to forecast the future performance of a strategy that worked in the past by applying it to new data to see whether it still performs well.

The initial capital invested in each strategy tested was \$10,000. The starting number of contracts bought for each cryptocurrency had the equivalent value of around \$10,000 to use the entire capital from the beginning. The number of contracts bought each trade then remained the same in order to make the final results comparable between the tested cryptocurrencies. The values in the tables representing profit, average profit per trade, etc., represent values in USD although the dollar sign isn't indicated in the tables.

All tested strategies always open only one trade, either a long or a short position, which is closed by the set take profit or stop loss. In case of the RSI + MA strategy the take profit means a switch from a long to a short position or the opposite and trades are entered using a market order for this strategy. The VWAP close strategy uses a limit order for entry and has a set take profit and stop loss depending on the average of the last 24 hours of volatility.

To find the highest profitable strategy, the parameters of each, the stop losses and the take profits are optimized.

All the charts and data with indicators were downloaded from tradingview.com server and modified for the purpose of this work. All the values below can be found in the tables in the In-sample.xlsx and Out-sample.xlsx files that are part of this work.

6.3 RSI + MA strategy

The first strategy that will be tested combines the relative strength index and two moving averages (SMA) one with a short period (fast one) and one with a long period (slow one) of calculation. The difference between the regular SMA cross strategy that is often used and this one is that the values for the moving averages are calculated from the RSI instead of the assets price directly. The time frame the strategy uses is the 30minute, therefore it is considered a day to swing trade strategy.

A position is opened at upon the cross of the moving averages if the fast one crosses the slow one from above it is a buy signal, and a long position is opened at market price. If the fast one crosses the slow from below it is a sell signal, and the long position is exited and a short position is opened. This strategy uses only market orders and only one position at a time can be opened.

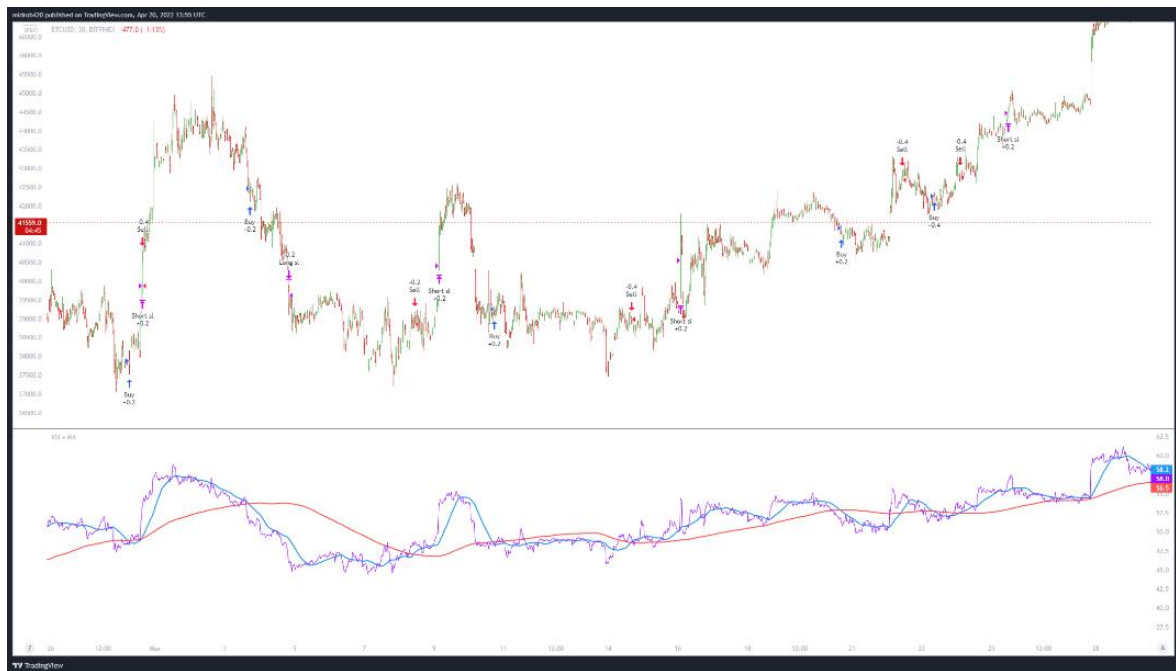
Since this strategy trades both sides of the market and switches between long and short positions there is no predefined take profit for it because it automatically switches from a long into a short position at market once the moving averages cross. The only thing required is a stop loss for both the long and short side to cut the bad trades.

The strategy can be optimized by changing the input parameters from which the relative strength index and moving averages are calculated and the values of stop losses. The input parameters may vary between the cryptocurrencies tested in order to find the best result for each one because they behave differently.

The initial capital for this strategy was \$10,000 for each cryptocurrency. To make the results comparable between all four cryptocurrencies a constant number of contracts was used for each trade, the number of contracts for each cryptocurrency was calculated by dividing the initial capital by the average daily candle close price of the entire backtesting period. The strategy was tested in the period of 01.01.2021 to 31.03.2022. The values in the tables that record profit, average profit, etc., represent values in USD, but the USD designation is not directly in the table.

In the upper half of the following figure, we can see the chart of Bitcoin including the trade signals and stop losses marked with red, blue and purple arrows (red=short signal, blue=long signal, purple=stop loss) and in the bottom half the fast-moving average in blue, the slow moving average in red and the RSI in purple.

Figure 30: RSI+MA strategy detail



own processing according to (TradingView)

6.3.1 Bitcoin

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$46,143.

Since this strategy uses only market orders an exchange with low taker fees would be the ideal place to test it, therefore Binance was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.04%. Those 0.04% are accounted for in the backtesting.

In the following figure we can see the strategy in action. Two long and one short signals are displayed in the upper half of the chart by the red and green text bubbles. On the bottom half we can see multiple crossovers of the fast- and slow-moving averages marked by the red and green circles depending on whether the cross is bullish therefore price should rise (green) or bearish therefore price should drop (red) these crosses create the buy and sell signal arrows upon which the trader reacts and enters either a long or short position depending on the signal direction.

Figure 31: RSI+MA strategy on Bitcoin



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Relative strength index length: 137
- RSI oversold / overbought value: 30 / 70
- Fast moving average length: 29
- Slow moving average length: 221
- Long position stop loss: 6.7%
- Short position stop loss: 3.8%

The following table presents the results of the tested strategy on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 7: RSI+MA Bitcoin in-sample results

BITCOIN	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	13989,27	139,89%	8190,97	81,91%	5798,30	57,98%
Gross Profit	45388,49	453,88%	24902,27	249,02%	20486,22	204,86%
Gross Loss	31399,23	313,99%	16711,31	167,11%	14687,92	146,88%
Max Drawdown	2195,20	15,51%				
Profit Factor	1,45		1,49		1,40	
Commission Paid	1785,63		896,88		888,75	
Total Closed Trades	219		110		109	
Number Winning Trades	144		77		67	
Number Losing Trades	75		33		42	
Percent Profitable	65,75%		70,00%		61,47%	

Avg Trade	63,88	0,64%	74,46	0,74%	53,20	0,53%
Avg Winning Trade	315,20	3,15%	323,41	3,23%	305,76	3,06%
Avg Losing Trade	418,66	4,19%	506,40	5,06%	349,71	3,50%
Largest Winning Trade	1273,09	12,84%	1089,84	12,84%	1273,09	10,17%
Largest Losing Trade	933,97	6,78%	933,97	6,78%	547,73	3,88%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$45,388 and gross loss of \$31,399 creating a profit factor of 1,45 and an overall profit of \$13,989 which equals to 139.89% of the initial capital. Of that \$13,989 profit \$8,191 was from long positions which equals to 81.91% and \$5,798 from short positions which equals to 57.98% showing that overall long positions made more money. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$2195.

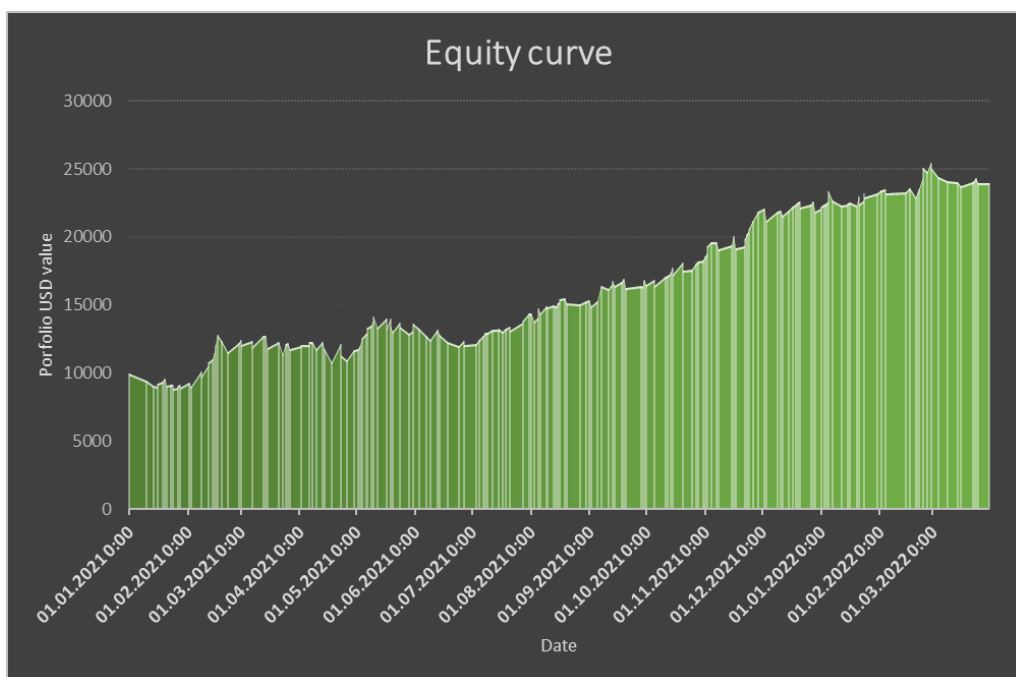
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions \$1,786.

The strategy executed 219 trades in total from which 110 were long positions and 109 short. From the total, 144 were profitable – 77 longs and 67 shorts, this led to an overall profitability of 65.75%, 70% for the side of long positions and 61.47% for the shorts so long positions were more successful than short ones.

The average profit of a trade was nearly \$64. The average profit of a winning one was \$315 and \$418 for the losing one, this metric shows that when losses came, they were bigger than wins but thanks to the overall trade profitability being almost 66% the winning trades simply outweighed the losing ones making the strategy profitable

The largest winning trade that led to a gain of \$1,273 was a short position whereas the largest losing position that led to a loss of \$934 was a long position, the reason for this was probably the tighter stoploss on short positions and higher on the long ones.

Figure 32: RSI+MA Bitcoin in-sample equity curve



The figure above shows the development of the equity curve where we can see a slight drawdown at the beginning which is continued by two rises and time of a sideways trend, which is then continued by a steady uptrend until the very end where it is currently going sideways again. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$23,989 showing a rise of almost 140%.

6.3.2 Ethereum

The initial capital for trading Ethereum was \$10,000 and for each trade 3,57 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$2,803.

Since this strategy uses only market orders an exchange with low taker fees would be the ideal place to test it, therefore Binance was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.04%. Those 0.04% are accounted for in the backtesting.

In the following figure we can see the strategy in action. Three long signals, three short signals and one stop loss are displayed in the upper half of the chart by the red, green and purple text bubbles. On the bottom half we can see multiple crossovers of the fast- and slow-moving averages marked by the red and green circles depending on whether the cross is bullish therefore price should rise (green) or bearish therefore price should drop (red) these crosses create the buy and sell signal arrows upon which the trader reacts and enters either a long or short position depending on the signal direction.

Figure 33: RSI+MA on Ethereum



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Relative strength index length: 160
- RSI oversold / overbought value: 30 / 70
- Fast moving average length: 28
- Slow moving average length: 215
- Long position stop loss: 5.7%
- Short position stop loss: 5.9%

The following table presents the results of the tested strategy on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 8: RSI+MA Ethereum in-sample results

ETHEREUM	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	15985,26	159,85%	12073,33	120,73%	3911,93	39,12%
Gross Profit	56309,8	563,10%	32144,1	321,44%	24165,71	241,66%
Gross Loss	40324,54	403,25%	20070,77	200,71%	20253,78	202,54%
Max Drawdown	2294,14	12,85%				
Profit Factor	1,396		1,602		1,193	
Commission Paid	1673,82		838,79		835,03	
Total Closed Trades	210		105		105	
Number Winning Trades	132		67		65	
Number Losing Trades	78		38		40	
Percent Profitable	62,86%		63,81%		61,90%	

Avg Trade	76,12	0,76%	114,98	1,15%	37,26	0,37%
Avg Winning Trade	426,59	4,27%	479,76	4,80%	371,78	3,72%
Avg Losing Trade	516,98	5,17%	528,18	5,28%	506,34	5,06%
Largest Winning Trade	1824,22	15,43%	1824,22	15,43%	1525,53	15,25%
Largest Losing Trade	977,44	5,98%	977,44	5,78%	924,18	5,98%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$56,310 and gross loss of \$40,325 creating a profit factor of 1,396 and an overall profit of \$15,985 which equals to 159.85% of the initial capital. Of that \$15,985 profit \$12,073 was from long positions which equals to 120.73% and \$3,912 from short positions which equals to 39.12% showing that overall long positions made much more money and shorts were almost break-even gross profit wise. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$2,294.

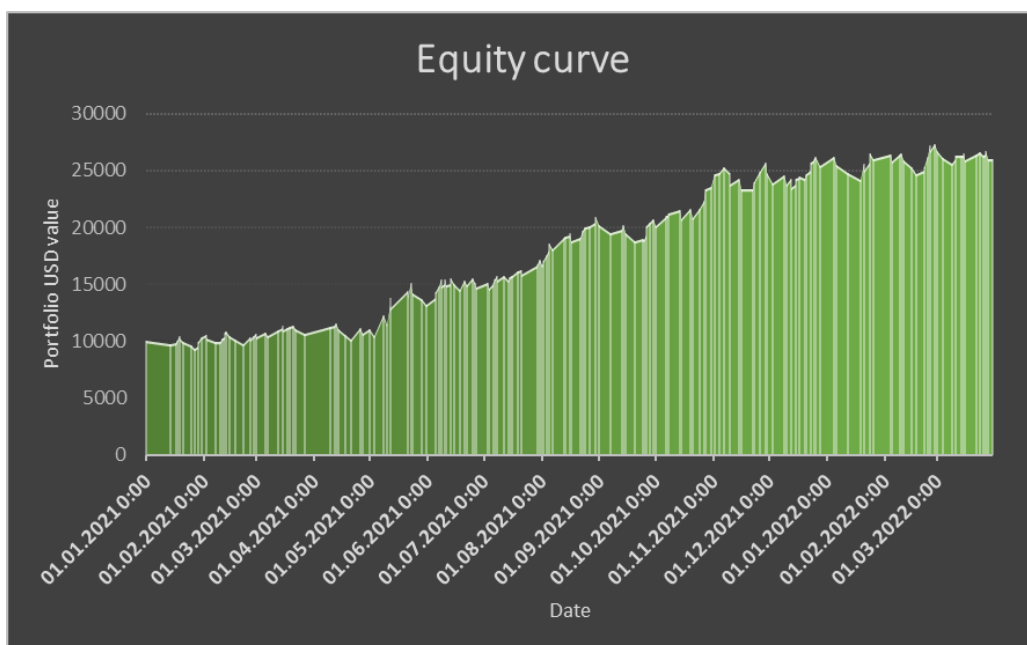
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions \$1,674.

The strategy executed 210 trades in total from which 105 were long positions and 105 short. From the total, 132 were profitable – 67 longs and 65 shorts, this led to an overall profitability of 62.86%, 63.81% for the side of long positions and 61.90% for the shorts so both were almost evenly profitable percentage wise.

The average profit of a trade was nearly \$76. The average profit of a winning one was \$427 and \$517 for the losing one, this metric shows that when losses came, they were slightly bigger than wins but thanks to the overall trade profitability being almost 63% more than half of the trades were wins therefore the strategy was profitable because the bigger losses were outweighed by the smaller wins by their quantity.

The largest winning trade that led to a gain of \$1,824 was a long position and the largest losing trade that led to a loss of \$977 was also a long position.

Figure 34: RSI+MA Ethereum in-sample equity curve



The figure above shows the development of the equity curve where we can see a continuous rise from the beginning with only slight pullbacks. It shows that the trader's capital is growing over the time without any greater losses. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$25,985 showing a rise of almost 160%.

6.3.3 Polkadot

The initial capital for trading Ethereum was \$10,000 and for each trade 366 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$27,3.

Since this strategy uses only market orders an exchange with low taker fees would be the ideal place to test it, therefore Binance was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.04%. Those 0.04% are accounted for in the backtesting.

In the following figure we can see the strategy in action. Two long signals, two short signals and one stop loss are displayed in the upper half of the chart by the red, green and purple text bubbles. On the bottom half we can see multiple crossovers of the fast- and slow-moving averages marked by the red and green circles depending on whether the cross is bullish therefore price should rise (green) or bearish therefore price should drop (red) these crosses create the buy and sell signal arrows upon which the trader reacts and enters either a long or short position depending on the signal direction.

Figure 35: RSI+MA on Polkadot



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Relative strength index length: 68
- RSI oversold / overbought value: 30 / 70
- Fast moving average length: 15
- Slow moving average length: 246
- Long position stop loss: 3.2%
- Short position stop loss: 1.2%

The following table presents the results of the tested strategy on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 9: RSI+MA Polkadot in-sample results

POLKADOT	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	26383,48	263,83%	10962,02	109,62%	15421,45	154,21%
Gross Profit	70185,01	701,85%	37577,40	375,77%	32607,62	326,08%
Gross Loss	43801,54	438,02%	26615,37	266,15%	17186,17	171,86%
Max Drawdown	5609,78	33,01%				
Profit Factor	1,60		1,41		1,90	
Commission Paid	2442,68		1222,12		1220,57	
Total Closed Trades	302,00		151,00		151,00	
Number Winning Trades	83,00		53,00		30,00	
Number Losing Trades	219,00		98,00		121,00	
Percent Profitable	27,48%		35,10%		19,87%	

Avg Trade	87,36	0,87%	72,60	0,73%	102,13	1,02%
Avg Winning Trade	845,60	8,46%	709,01	7,09%	1086,92	10,87%
Avg Losing Trade	200,01	2,00%	271,59	2,72%	142,03	1,42%
Largest Winning Trade	5530,32	34,78%	3975,00	34,78%	5530,32	33,68%
Largest Losing Trade	566,24	5,22%	566,24	3,49%	531,74	5,22%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$70,185 and gross loss of \$43,801 creating a profit factor of 1,6 and an overall profit of \$26,383 which equals to 263.83% of the initial capital. Of that \$26,383 profit \$10,962 was from long positions which equals to 109.62% and \$15,421 from short positions which equals to 154.21% showing that overall short positions made almost half more money than longs thanks to the tight stop loss that stopped most of the losing short trades early. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$5,609.

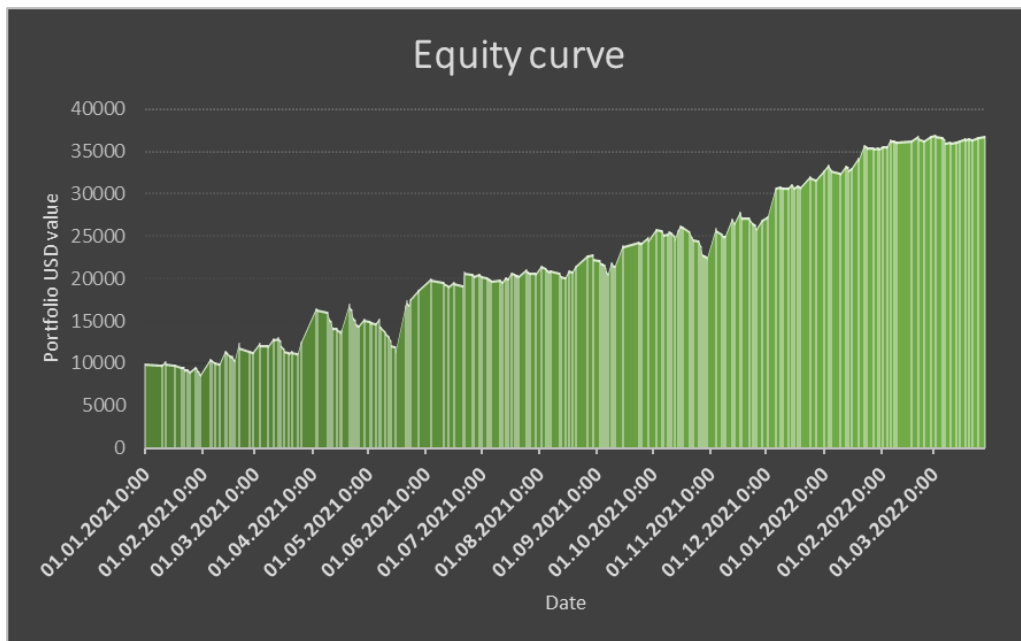
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions \$2,442.

The strategy executed 302 trades in total from which 151 were long positions and 151 short. From the total, 83 were profitable – 53 longs and 30 shorts, this led to an overall profitability of 27.48%, 35.10% for the side of long positions and 19.87% for the shorts. Even though the profitable percentage is quite low the strategy is profitable thanks to the fact that the average winning value of trades is over four times bigger than the average value of the losing ones.

The average profit of a trade was nearly \$87. The average profit of a winning one was \$845 and \$200 for the losing one, this metric shows that when wins came, they were much bigger than losses so even the fact that the overall trade profitability was low didn't make the strategy unprofitable.

The largest winning trade that led to a gain of \$5,530 was a short position whereas the largest losing position that led to a loss of \$566 was a long position.

Figure 36: RSI+MA Polkadot in-sample equity curve



The figure above shows the development of the equity curve where we can see a slight sideways trend in the beginning which then changes into a steady uptrend until the end only with a few drops which indicates a solid performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$36,383 showing a rise of almost 264%.

6.3.4 Decentraland

The initial capital for trading Ethereum was \$10,000 and for each trade 6660 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$1,5.

Since this strategy uses only market orders an exchange with low taker fees would be the ideal place to test it, therefore Binance was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.04%. Those 0.04% are accounted for in the backtesting.

In the following figure we can see the strategy in action. Two long signals, two short signals and one stop loss are displayed in the upper half of the chart by the red, green and purple text bubbles. On the bottom half we can see multiple crossovers of the fast- and slow-moving averages marked by the red and green circles depending on whether the cross is bullish therefore price should rise (green) or bearish therefore price should drop (red) these crosses create the buy and sell signal arrows upon which the trader reacts and enters either a long or short position depending on the signal direction.

Figure 37: RSI+MA on Decentraland



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Relative strength index length: 17
- RSI oversold / overbought value: 30 / 70
- Fast moving average length: 18
- Slow moving average length: 50
- Long position stop loss: 30%
- Short position stop loss: 3.5%

The following table presents the results of the tested strategy on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 10: RSI+MA Decentraland in-sample results

DECENTRALAND	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	62025,72	620,26%	36289,57	362,90%	25736,16	257,36%
Gross Profit	171770,67	1717,71%	98220,82	982,21%	73549,86	735,50%
Gross Loss	109744,95	1097,45%	61931,25	619,31%	47813,7	478,14%
Max Drawdown	9915,73	20,69%				
Profit Factor	1,565		1,586		1,538	
Commission Paid	5173,01		2591,65		2581,36	
Total Closed Trades	665		333		332	
Number Winning Trades	264		148		116	
Number Losing Trades	401		185		216	
Percent Profitable	39,70%		44,44%		34,94%	

Avg Trade	93,27	0,93%	108,98	1,09%	77,52	0,78%
Avg Winning Trade	650,65	6,51%	663,65	6,64%	634,05	6,34%
Avg Losing Trade	273,68	2,74%	334,76	3,35%	221,36	2,21%
Largest Winning Trade	9311,87	76,51%	9311,87	76,51%	6577,62	27,06%
Largest Losing Trade	2963,3	21,44%	2963,3	21,44%	1240,99	6,47%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$171,770 and gross loss of \$109,745 creating a profit factor of 1,565 and an overall profit of \$62,025 which equals to 620.26% of the initial capital. Of that \$62,025 profit \$36,289 was from long positions which equals to 362.90% and \$25,736 from short positions which equals to 257.36% showing that overall long positions made more money. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$9,916.

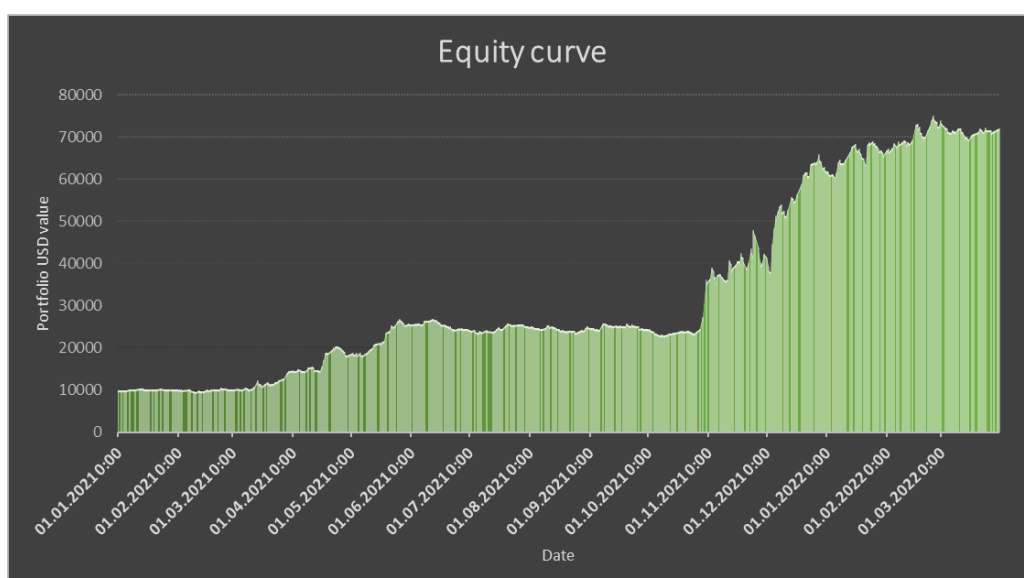
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions \$5,173.

The strategy executed 665 trades in total from which 333 were long positions and 332 short. From the total, 264 were profitable – 148 longs and 116 shorts, this led to an overall profitability of 39.70%, 44.44% for the side of long positions and 34.94% for the shorts. Even though there were more losing trades than winning the strategy is profitable thanks to the fact that the average winning value of trades is over two times bigger than the average value of the losing ones.

The average profit of a trade was nearly \$93. The average profit of a winning one was \$650 and \$273 for the losing one, this metric shows that when wins came, they were double the size of losses so the fact that the overall profitability was 39.70% didn't matter because the size of the winning trades outweighed the losing ones making the strategy profitable

The largest winning and losing trades were both long positions. The winning long led to a gain of \$9,312 and the losing one to a loss of \$2,963, the reason behind the biggest gain and loss being long positions is probably the loose stop loss but with a tighter one the strategy worked worse.

Figure 38: RSI+MA Decentraland in-sample equity curve



The figure above shows the development of the equity curve where we can see a sideways trend at the beginning which is then continued by three small run up phases after which a small downtrend over a few months appears but is then swiftly erased by an extremely fast rise which continues until the end of the testing period. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$72,025 showing a rise of 620%.

6.3.5 In-sample results summary

The following table presents the results of the tested strategy on all mentioned cryptocurrencies for the 455-day period. The first slope of each cryptocurrency presents the combined results of long and short positions in USD and the second one in percentage value.

Table 11: RSI+MA in-sample results comparison

	BITCOIN		ETHEREUM		POLKADOT		DECENTRALAND	
	All \$	All %	All \$	All %	All \$	All %	All \$	All %
Net Profit	13989,27	139,89%	15985,26	159,85%	26383,48	263,83%	62025,72	620,26%
Gross Profit	45388,49	453,88%	56309,8	563,10%	70185,01	701,85%	171770,67	1717,71%
Gross Loss	31399,23	313,99%	40324,54	403,25%	43801,54	438,02%	109744,95	1097,45%
Drawdown	2195,20	15,51%	2294,14	12,85%	5609,78	33,01%	9915,73	20,69%
Profit Factor	1,45		1,396		1,60		1,565	
Commission	1785,63		1673,82		2442,68		5173,01	
Trades	219		210		302		665	
Winning	144		132		83		264	
Number	75		78		219		401	
Profitable	65,75%		62,86%		27,48%		39,70%	
Avg Trade	63,88	0,64%	76,12	0,76%	87,36	0,87%	93,27	0,93%
Avg Winning	315,20	3,15%	426,59	4,27%	845,60	8,46%	650,65	6,51%
Avg Losing	418,66	4,19%	516,98	5,17%	200,01	2,00%	273,68	2,74%
Largest Winning Trade	1273,09	12,84%	1824,22	15,43%	5530,32	34,78%	9311,87	76,51%
Largest Losing Trade	933,97	6,78%	977,44	5,98%	566,24	5,22%	2963,3	21,44%

As mentioned before the initial capital invested in each cryptocurrency tested was \$10,000 and the number of traded contracts for each trade was adjusted to make the results comparable. The figure above shows that money wise the most profitable cryptocurrency for this strategy was Decentraland making \$62,025 which was over double the amount of what the second largest profit amount was. We can also see that the smaller market cap cryptocurrencies which are Polkadot and Decentraland made larger profits than Ethereum and Bitcoin, the reason for that is that both Bitcoin and Ether are much less volatile therefore have lower percentage moves compared to the smaller cap ones that can easily rise 30% in a day. The highest profit factor was achieved on Polkadot which reached 1,60 which was the best ration between the gross profit and loss making it the overall best performing cryptocurrency for this strategy. Even though Decentraland made greater gains it wasn't as efficient as Polkadot was.

There is no surprise that the highest commissions paid for trading was on Decentraland because the strategy opened 665 trades and the commission for each one was 0.04%.

When it comes to the number of trades, the strategy made much more buy and sell signals on Polkadot and Decentraland than on Bitcoin and Ether. The reason for that is again the increased volatility of smaller market cap altcoins because it requires a lot less money to push their price higher or lower than it would on assets like Bitcoin and Ether for the market makers therefore, the strategy produces more signals on them because they rapidly go up and down.

When it comes to profitability Bitcoin and Ether were the best performers both reaching around 65% winrate but the high overall winrate doesn't mean they made the biggest profit,

because even though a strategy has a high winrate it can realize greater losses than gains making it unprofitable overall. So, although the strategy on Polkadot and Decentraland lost much more trades than won, the average profit of a winning trade made two or more times bigger gains compared to the losses making the strategy realize much more profit on Polkadot and Decentraland even though they had a poor winrate compared to Ether and Bitcoin which had a higher winrate, but the average loss of a trade was greater than the average profit.

The biggest winning (\$9,311) but also losing (\$2,964) trade was recorded on Decentraland, which was probably because of the loose stop loss on the side of long positions. Even though the Decentraland recorded the largest loss, its largest winning trade was three times bigger therefore, it performed better than Ether and Bitcoin. The best cryptocurrency in this metric was Polkadot which difference between the biggest gain was almost ten times bigger than its biggest loss.

6.4 RSI + MA strategy out-sample

As mentioned above the out-sample period was between 01.04.2022 to 30.04.2022. The out-sample was made to verify whether the created strategy with its rules, signals and parameters is still reliable even in different market conditions. Same number of contracts and initial capital was used as in the in-sample to bring the out-sample as close as possible to the in-sample and to make the results easily comparable.

The out-sample data was processed the same way as in the in-sample and that is by using Excel with its logical functions and operators and TradingView terminal.

The following picture is the Bitcoin chart on the 4-hour timeframe with the marked out-sample period between the blue lines. During the month the price of Bitcoin was decreasing and with it were the prices of all other cryptocurrencies as well and faster. The 3 other tested altcoins did the exact same as Bitcoin with only fast and sharp movements to the upside which were then swiftly retraced even lower than where they originally began. Therefore, we can say that the market trend for the month was to the downside therefore, long positions were much riskier than short ones.

Figure 39: Bitcoin out-sample price development



own processing according to (TradingView)

Bitcoin

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. The same exchange was used (Binance) to have the same trading fees as before (0.04%) which are accounted for.

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 12: RSI+MA Bitcoin out-sample results

BTCOIN	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	251,27	2,51%	-355,3	-3,55%	606,57	6,07%
Gross Profit	1930,57	19,31%	634,59	6,35%	1295,97	12,96%
Gross Loss	1679,3	16,79%	989,9	9,90%	689,4	6,89%
Max Drawdown	1071,47	10,25%				
Profit Factor	1,15		0,641		1,88	
Commission Paid	102,76		50,88		51,89	
Total Closed Trades	14		7		7	
Number Winning Trades	9		4		5	
Number Losing Trades	5		3		2	
Percent Profitable	64,29%		57,14%		71,43%	
Avg Trade	17,95	0,18	-50,76	-0,51	86,65	0,87
Avg Winning Trade	214,51	2,15	158,65	1,59	259,19	2,59
Avg Losing Trade	335,86	3,36	329,97	3,3	344,7	3,45

Largest Winning Trade	503,89	5,66	250,66	2,87	503,89	5,66
Largest Losing Trade	683,68	6,78	683,68	6,78	349,46	3,88

The table above shows that the strategy was profitable after the 30 days it was running. It made a gross profit of \$1,185 and gross loss of \$1,679 creating a profit factor of 1,15 which is quite low and an overall profit of \$251 which equals to 2.51% of the initial capital. The total profit was entirely made of short positions which brought in \$606, the long positions were not profitable and brought in a loss of \$-355 but that isn't surprising since the market went down the entire month, but it shows a weakness in the strategy, which should be optimized. The maximum drawdown was as high as \$1,071 which is also not the best.

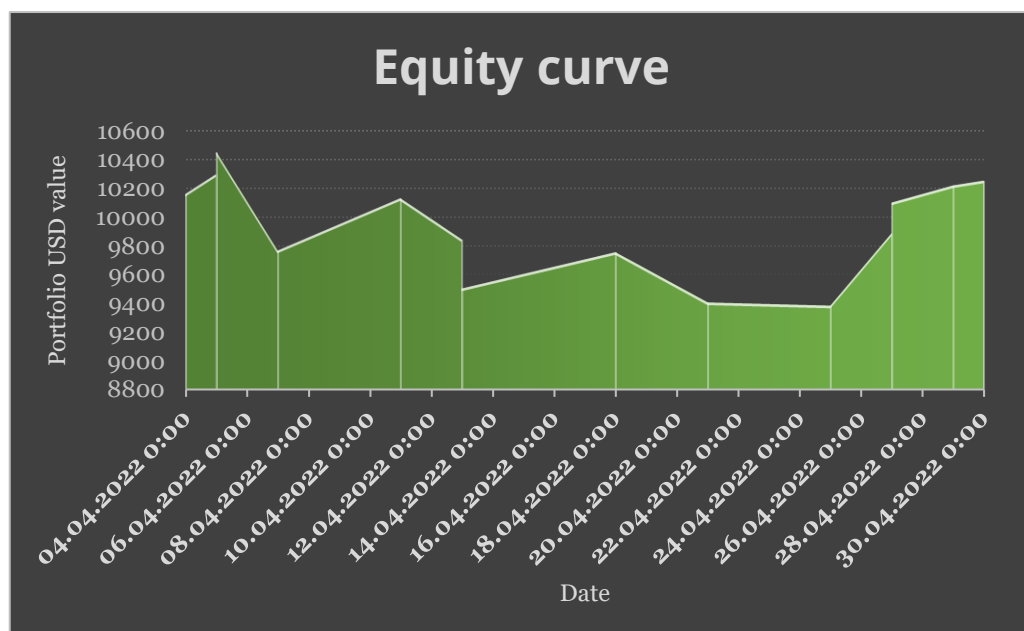
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions was almost \$103.

The strategy executed 14 trades in total from which 7 were long positions and 7 short. From the total, 9 were profitable – 4 longs and 5 shorts, this led to an overall profitability of 64.29%, 57.14% for the side of long positions and 71.43% for the shorts. Even though the strategy made profit, the winrate was high and short positions had very good results the long positions brought in huge losses almost evening out the gains which is something that needs more adjusting.

The average profit of a trade was nearly \$18. The average profit of a winning one was \$214 and \$335 for the losing one, this metric shows that when losses came, they were much bigger than wins but thanks to the high winrate the strategy remained profitable.

The largest winning trade that led to a gain of \$503 was a short position whereas the largest losing position that led to a loss of \$683 was a long position. Proving short positions worked much better than long ones with the help of the market conditions.

Figure 40: RSI+MA Bitcoin out-sample equity curve



The figure above shows the development of the equity curve where we can see a decrease from the beginning until almost the very end where it spiked up this indicates a uncertain performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.04.2021 to 30.04.2022. The initial capital was \$10,000 which over the course of a month has increased to \$10,251 showing a rise of around 2.5%.

Ethereum

The initial capital for trading Ethereum was \$10,000 and for each trade 3,57 contracts were bought or sold. The same exchange was used (Binance) to have the same trading fees as before (0.04%) which are accounted for.

The following table presents the results of the out-sample on Ethereum. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 13: RSI+MA Ethereum out-sample results

ETHEREUM	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	1688,66	16,89%	-759,42	-7,59%	2448,08	24,48%
Gross Profit	3672,38	36,72%	1224,3	12,24%	2448,08	24,48%
Gross Loss	1983,72	19,84%	1983,72	19,84%	0	0,00%
Max Drawdown	1032,54	9,61%				
Profit Factor	1,851		0,617		N/A	
Commission Paid	143,82		50,88		51,89	
Total Closed Trades	16		8		8	
Number Winning Trades	13		5		8	
Number Losing Trades	3		3		0	
Percent Profitable	81,25%		62,50%		100,00%	
Avg Trade	105,54	1,06%	-94,93	-0,95%	306,01	3,06%
Avg Winning Trade	282,49	2,82%	244,86	2,45%	306,01	3,06%
Avg Losing Trade	661,24	6,61%	661,24	6,61%	N/A	N/A
Largest Winning Trade	785,98	7,32%	516,83	4,97%	785,98	7,32%
Largest Losing Trade	717,11	5,78%	717,11	5,78%	N/A	N/A

The table above shows that the strategy was profitable after the 30 days it was running. It made a gross profit of \$3,672 and gross loss of \$1,983 creating a profit factor of 1,851 which is impressive and an overall profit of \$1688 which equals to 16.89% of the initial capital. The total profit was once again made entirely of short positions which brought in \$2,448, the long positions were not profitable and brought in a loss of \$-759 but that isnt surprising since the market went down the entire month, but it shows a weakness in the strategy, which should be optimized but overall, the performace was good. The maximum drawdown was as high as \$1,032 which is also not the best but better than Bitcoin in comparison of the ending net profit.

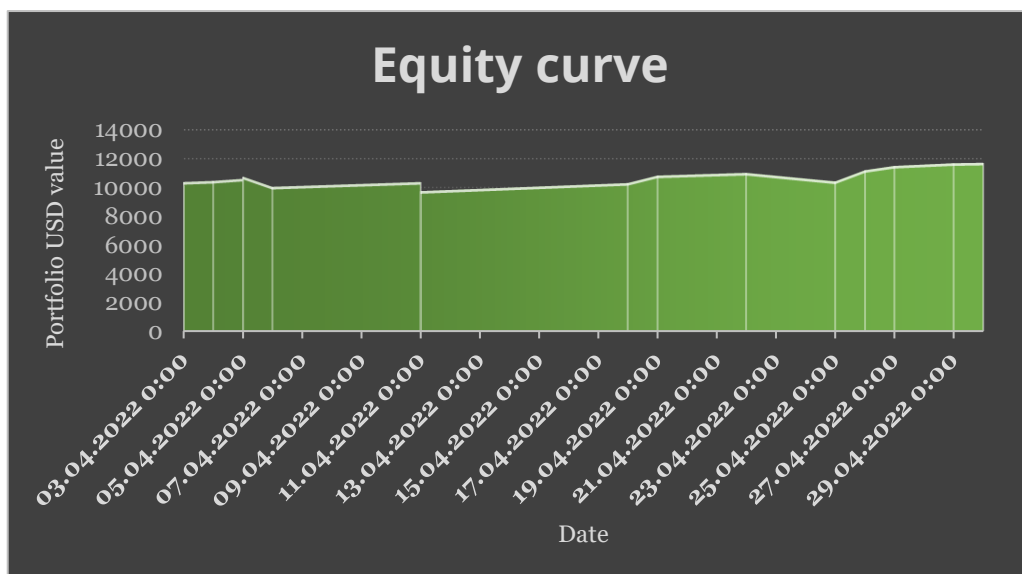
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions was almost \$143.

The strategy executed 16 trades in total from which 8 were long positions and 8 short. From the total, 13 were profitable – 5 longs and 8 shorts, this led to an overall profitability of 81.25% which is very high, 62.50% for the side of long positions and 100% for the shorts which is a very good result. The results on the side of short position were none less than perfect and the overall winrate was very high but only the 3 bad trades on the side of long positions brought a great loss.

The average profit of a trade was nearly \$105. The average profit of a winning one was \$282 and \$661 for the losing one, this metric shows that when losses came, they were greater than wins but thanks to the insanely high winrate the strategy was profitable.

The largest winning trade that led to a gain of \$785 was a short position whereas the largest losing position that led to a loss of \$717 was a long position obviously because shorts had a 100% winrate. This proves short positions worked much better than long ones with the help of the market conditions and good parameters.

Figure 41: RSI+MA Ethereum out-sample equity curve



The figure above shows the development of the equity curve where we can see a slow but steady rise with no sharp declines which indicates a good performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.04.2021 to 30.04.2022. The initial capital was \$10,000 which over the course of a month has increased to \$11,688 showing a rise of around 16.9%.

Polkadot

The initial capital for trading Polkadot was \$10,000 and for each trade 366 contracts were bought or sold. The same exchange was used (Binance) to have the same trading fees as before (0.04%) which are accounted for.

The following table presents the results of the out-sample on Polkadot. The first two slopes present the combined results of long and short positions in USD and percentage value. The

next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 14: RSI+MA Polkadot out-sample results

POLKADOT	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	407,92	4,08%	-781	-7,81%	1188,92	11,89%
Gross Profit	2041,44	20,41%	498,8	4,99%	1542,64	15,43%
Gross Loss	1633,52	16,34%	1279,8	12,80%	353,72	3,54%
Max Drawdown	503,43	4,77%				
Profit Factor	1,25		0,39		4,361	
Commission Paid	111,8		50,88		51,89	
Total Closed Trades	20		10		10	
Number Winning Trades	8		3		5	
Number Losing Trades	12		7		5	
Percent Profitable	40,00%		30,00%		50,00%	
Avg Trade	20,4	0,20%	-78,1	-0,78%	118,89	1,19%
Avg Winning Trade	255,18	2,55%	166,27	1,66%	308,53	3,09%
Avg Losing Trade	136,13	1,36%	182,83	1,83%	70,74	0,71%
Largest Winning Trade	630,58	7,74%	209,33	2,74%	630,58	7,74%
Largest Losing Trade	277,43	3,31%	277,43	3,31%	97,42	1,33%

The table above shows that the strategy was profitable after the 30 days it was running. It made a gross profit of \$2,014 and gross loss of \$1,633 creating a profit factor of 1,25 and an overall profit of \$408 which equals to 4.08% of the initial capital. We can see that the gross profit of longs almost evens out the gross loss of short positions and vice versa. The total profit was once again made entirely of short positions which brought in \$1,188, the long positions were not profitable and brought in a loss of \$-781 but that isn't surprising since the market went down the entire month, but it shows a weakness in the strategy, which should be optimized but the overall performance could be ranked from average to poor. The maximum drawdown was as high as \$503.

The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions was almost \$112.

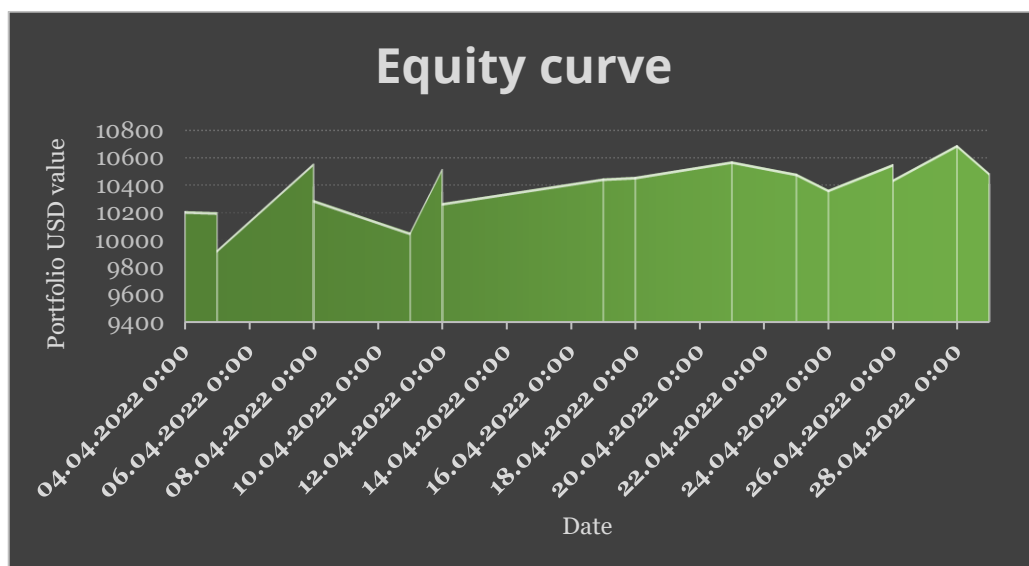
The strategy executed 20 trades in total from which 10 were long positions and 10 short. From the total, 8 were profitable – 3 longs and 5 shorts, this led to an overall profitability of 40%, 30% for the side of long positions and 50% for the shorts. The results on the side of short positions were again better with the help of the market conditions but overall, an average result winrate wise.

The average profit of a trade was nearly \$20. The average profit of a winning one was \$255 and \$136 for the losing one, this metric shows that although there were more losses than gains but when gains came, they were almost double and thanks to that the strategy was overall profitable even though the winrate was below 50%.

The largest winning trade that led to a gain of \$630 was a short position whereas the largest losing position that led to a loss of \$277 was a long position. This proves short positions worked much better than long ones with the help of the market conditions and good parameters.

The following figure shows the development of the equity curve where we can see huge drops and rises in the beginning which are transform into a slow uptrend equity curve wise the performance was uncertain in the beginning but improved after some time indicating an average performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.04.2021 to 30.04.2022. The initial capital was \$10,000 which over the course of a month has increased to \$10,407 showing a rise of around 4.08%.

Figure 42: RSI+MA Polkadot out-sample equity curve



Decentraland

The initial capital for trading Decentraland was \$10,000 and for each trade 6660 contracts were bought or sold. The same exchange was used (Binance) to have the same trading fees as before (0.04%) which are accounted for.

The following table presents the results of the out-sample on Decentraland. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 15: RSI+MA Decentraland out-sample results

DECENTRALAND	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	3735,69	37,36%	-653,64	-6,54%	4389,33	43,89%
Gross Profit	8941,11	89,41%	2559,06	25,59%	6382,05	63,82%
Gross Loss	5205,42	52,05%	3212,7	32,13%	1992,72	19,93%

Max Drawdown	1915,16	13,51%				
Profit Factor	1,718		0,797		3,203	
Commission Paid	463,44		50,88		51,89	
Total Closed Trades	40		20		20	
Number Winning Trades	19		8		11	
Number Losing Trades	21		12		9	
Percent Profitable	47,50%		40,00%		55,00%	
Avg Trade	93,39	0,93%	-32,68	-0,33%	219,47	2,19%
Avg Winning Trade	470,58	4,71%	319,88	3,20%	580,19	5,80%
Avg Losing Trade	247,88	2,48%	267,72	2,68%	221,41	2,21%
Largest Winning Trade	1419,76	8,32%	944,25	6,52%	1419,76	8,32%
Largest Losing Trade	796,02	5,11%	796,02	5,11%	512,47	3,59%

The table above shows that the strategy was profitable after the 30 days it was running. It made a gross profit of \$8,941 and gross loss of \$5,205 creating a profit factor of 1,72 which is very good and an overall profit of \$3,735 which equals to 37.36% of the initial capital and is the best result between the 4 assets. The total profit was entirely made of short positions which brought in \$4389. Although the long positions were not profitable and brought in a loss of \$-653 the gross loss and profit difference of long positions was the best here out of all 4 tested assets. The fact that long positions ended up overall negative isnt surprising since the market went down the entire month, but it shows a weakness in the strategy, which should be optimized. The maximum drawdown was as high as \$1,915.

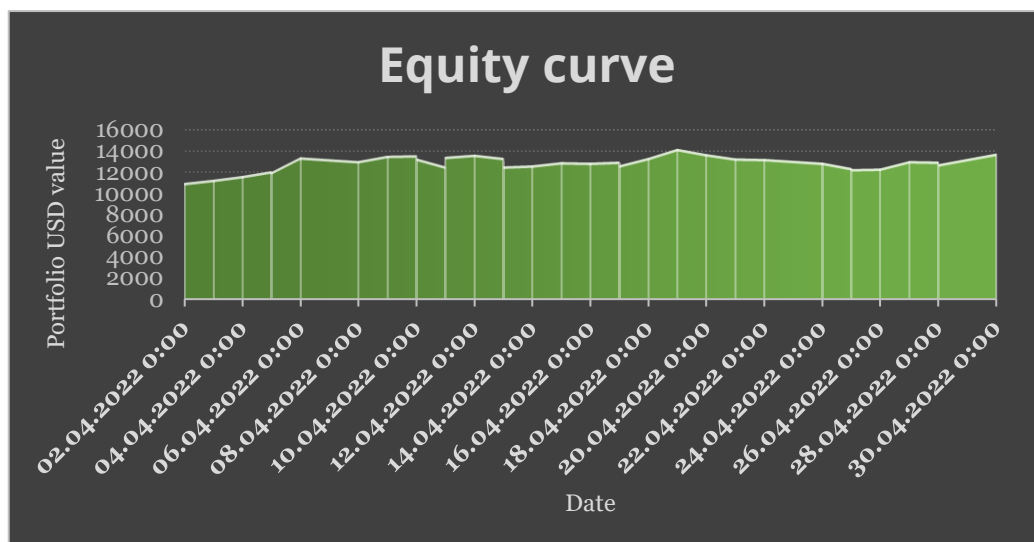
The amount paid to the exchange as trading fees for executing orders at market price was 0.04% for each trade making the total cost of opening all the positions was almost \$463.

The strategy executed 40 trades in total from which 20 were long positions and 20 short. From the total, 19 were profitable – 8 longs and 11 shorts, this led to an overall profitability of 47.50%, 40% for the side of long positions and 55% for the shorts. This shows that winrate is not everything and that even with an average one a great profit can be achieved.

The average profit of a trade was nearly \$93. The average profit of a winning one was \$470 and \$247 for the losing one, this metric shows that when wins came, they were almost double than losses and thanks to the winrate being around 50% the strategy was very profitable.

The largest winning trade that led to a gain of \$1419 was a short position whereas the largest losing position that led to a loss of \$796 was a long position. Proving short positions worked much better than long ones with the help of the market conditions and that the biggest win could compensate for the biggest loss twice.

Figure 43: RSI+MA Decentraland out-sample equity curve



The figure above shows the development of the equity curve where we can see a slow and steady rise with only small decreases which indicates a very good performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.04.2021 to 30.04.2022. The initial capital was \$10,000 which over the course of a month has increased to \$13,735 showing a rise of around 37.36% which is very impressive and the best result of the out-sample.

RSI+MA out-sample results summary

The following table presents the results of the tested strategy on all mentioned cryptocurrencies for the 30-day period. The first slope of each cryptocurrency presents the combined results of long and short positions in USD and the second one in percentage value.

Table 16: RSI+MA out-sample results comparison

	BITCOIN		ETHEREUM		POLKADOT		DECENTRALAND	
	All \$	All %	All \$	All %	All \$	All %	All \$	All %
Net Profit	251,27	2,51%	1688,66	16,89%	407,92	4,08%	3735,69	37,36%
Gross Profit	1930,57	19,31%	3672,38	36,72%	2041,44	20,41%	8941,11	89,41%
Gross Loss	1679,30	16,79%	1983,72	19,84%	1633,52	16,34%	5205,42	52,05%
Drawdown	1071,47	10,25%	1032,54	9,61%	503,43	4,77%	1915,16	13,51%
Profit Factor	1,15		1,851		1,25		1,718	
Commission	102,76		143,82		111,80		463,44	
Trades	14		16		20		40	
Winning	9		13		8		19	
Number	5		3		12		21	
Profitable	64,29%		81,25%		40,00%		47,50%	
Avg Trade	17,95	18,00%	105,54	1,06%	20,40	0,20%	93,39	0,93%
Avg Winning	214,51	215,00%	282,49	2,82%	255,18	2,55%	470,58	4,71%
Avg Losing	335,86	336,00%	661,24	6,61%	136,13	1,36%	247,88	2,48%
Largest Winning Trade	503,89	566,00%	785,98	7,32%	630,58	7,74%	1419,76	8,32%
Largest Losing Trade	683,68	678,00%	717,11	5,78%	277,43	3,31%	796,02	5,11%

The initial capital invested in each cryptocurrency tested was \$10,000 and the number of traded contracts for each trade was the same as in the in-sample to make the results comparable. The figure above shows that money wise the most profitable cryptocurrency for this strategy was again Decentraland making \$3,735 which was over double the amount of what the second largest profit amount was. What changed is that Polkadot didn't perform as good as before and its second spot was taken by Ether which had much better results, overall we can see that the biggest change in the behaviour of the two where Ether performed okay in the in-sample where Polkadot shined but in the out-sample Polkadot stumbled whereas Ether was overall the most efficient asset of the four with a profit factor of 1,851 which is an amazing result.

There is no surprise that the highest commissions paid for trading was again on Decentraland because the strategy opened 40 trades and the commission for each one was 0.04%.

When it comes to the number of trades, they are pretty much the same as in the in-sample, the strategy made much more buy and sell signals on Decentraland than on Polkadot, Bitcoin and Ether where each had similar count. The only real difference was Polkadot which had less trades compared to the many in the in-sample might be because traders simply lost interest in trading the asset hence there was no one to push the price.

When it comes to the overall winrate there were big changes between the best performers. Bitcoin had almost the exact same winrate as before, but the other assets made much better results, the best one was Ethereum which had an 81% winrate which was almost 20% higher than in the in-sample. The results of Polkadot and Decentraland were also about a 10%

better than in the in-sample. Winrate doesn't mean everything so although Ethereum had an insanely high one the average on decentraland was enough to make more profit.

Also, what changed was that the average profit and loss with Ethereum being the worst performer with the average loss being double the win compared to its in-sample where the two were almost the same. Bitcoin and Decentraland behaved the same as in the in-sample, but Polkadot struggled by decreasing its difference between the average win and loss from over four times to just two times.

The biggest profit (\$1.419) but also loss (\$796) was recorded on Decentraland, which was probably because of the loose stop loss on the side of long positions. Even though the Decentraland recorded the largest loss, its largest winning trade was almost two times bigger therefore, it performed better than Ether and Bitcoin. But worse than Polkadot which had over two times bigger winning trade than the losing one therefore, it was the best asset in this metric.

Overall, the RSI + MA strategy proved to be profitable in both the in-sample and the out-sample making it suitable for trading. The two best performing assets were Ethereum for slow but steady gains and Decentraland for traders that are looking for a fast-paced action which can come with great gains but also great losses.

6.5 VWAP close strategy

The second strategy tested combines the VWAPs closing price of the end of the day with average true range indicator (ATR). The idea behind this strategy is that the previous day VWAP closing price should act as a support or resistance level after the price moves away from it a certain distance and returns back. The average true range which shows a range of price of volatility is used to calculate the stop loss and take profit levels for trades based on multiplying the average of the 24hour ATR with a selected number where 1 would equal to 1 time the range of the average of the 24hour ATR, the result of this is then subtracted or added to the entry price level depending on whether it's a long or a short position giving the stop loss or take profit price levels. This way the trading strategy better responds to surges of volatility instead of using fixed percentages for take profits and stop losses which could cause harm.

For example, if we would enter a long position and we would like our stop loss to be equal to 50% of the average 24hour ATR range and take profit to 150% of it, we would have to multiply the 24hour ATR average by 0,5 where 0,5 stands for the 50% and then subtract it to the entry price which would give us our stop loss price level. The same goes for the take profit, where we would multiply the 24hour ATR average by 1,5 (150%) and then add it to the entry price which gives us our take profit price level. For short positions this is done in reverse by adding the result to the entry price we find the stop loss level and by subtracting the take profit.

The time frame the strategy uses is the 15minute, therefore it is considered a scalp to day trade strategy and since it aims to trade only short-term fast price movements such significant gains which came with the first strategy cannot be expected. The strategy resets every day because the daily closing price of the VWAP changes every day, and we always want to trade the newest one because it's the most relevant and recent one and because the older the close the less importance, strength and reaction it usually has. The strategy only uses only limit orders, also only one position at a time can be opened and it trades both short and long positions.

A limit order is set at the previous day VWAP close price level upon the price reaching the required distance from it and a position is entered if the price returns back to it. Then either the take profit or stop loss is hit and the trade is closed for profit or loss. If an opened trade doesn't hit a take profit or a stop loss during the day, it is automatically closed at the end of it. If the required price distance to place the limit order, had been reached but the limit order hasn't been executed during the day, the order is automatically cancelled at the end of it. The limit order is set only after the required distance is reached otherwise there is no trade for the day.

The strategy can be optimized by changing the input parameters from which the stop loss and take profit are calculated. The other thing that can be adjusted for better performance is the distance between the price and the VWAP close that must be reached in order to activate the setup. The stop loss, take profit and distance parameters may vary between the cryptocurrencies tested to find the best result for each one because they behave differently.

The initial capital for this strategy was \$10,000 for each cryptocurrency. To make the results comparable between all four cryptocurrencies a constant number of contracts was used for each trade, the number of contracts for each cryptocurrency was calculated by dividing the initial capital by the average daily candle close price of the entire backtesting period. The strategy was tested in the period of 01.01.2021 to 31.03.2022. The values in the tables that record profit, average profit, etc., represent values in USD, but the USD designation is not directly in the table.

In the upper part of the following figure, we can see the chart of Decentraland. The blue line is the daily VWAP, the red lines mark the daily close of VWAP and in the bottom part we can see a red line which is the ATR. These are the tools that the strategy works with.

Figure 44: VWAP close strategy detail



own processing according to (TradingView)

6.5.1 Bitcoin

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$46,143.

Since this strategy uses only limit orders an exchange with low maker fees would be the ideal place to test it, therefore Bybit was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.01%. Those 0.01% are accounted for in the backtesting.

In the following figure we can see the strategy in action. The required distance is reached by the price getting above the blue rectangle and a long limit order is placed on the red line which is the previous day VWAP close. Where VWAP ended can be seen in the blue circle. When the price reaches the VWAP close a long position is entered which is marked by the

green circle. A take profit is then reached which is marked by the orange arrow. In the bottom part of the picture, we can see the ATR which is used to calculate the stop loss and take profit.

Figure 45: VWAP close strategy on Bitcoin



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Distance filter: 800
- Stop loss: 0,75
- Take profit: 1,15

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 17: VWAP close Bitcoin in-sample results

BTC/USD	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	3717,93	37,18%	1836,15	18,36%	1881,78	18,82%
Gross Profit	9692,29	96,92%	4857,84	48,58%	4834,45	48,34%
Gross Loss	5731,05	57,31%	2885,47	28,85%	2845,58	28,46%
Max Drawdown	361,19	3,61%				
Profit Factor	1,69		1,68		1,70	
Commission Paid	243,31		136,22		107,09	
Total Closed Trades	238		129		109	
Number Winning Trades	125		65		60	
Number Losing Trades	113		64		49	

Percent Profitable	52,52%		50,39%		55,05%	
Avg Trade	15,62	0,16%	14,23	0,14%	17,26	0,17%
Avg Winning Trade	77,54	0,78%	74,74	0,75%	80,57	0,81%
Avg Losing Trade	50,72	0,51%	45,09	0,45%	58,07	0,58%
Largest Winning Trade	260,60	2,61%	186,62	1,87%	260,60	2,61%
Largest Losing Trade	-144,82	1,45%	-100,12	1,00%	-144,82	1,45%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$9,692 and gross loss of \$5,731 creating a profit factor of 1,69 and an overall profit of \$3,717 which equals to 37.18% of the initial capital. Of that \$3,717 profit \$1,836 was from long positions which equals to 18.36% and \$1,881 from short positions which equals to 18.82% showing that overall long and short positions made an almost equal profit. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$361.

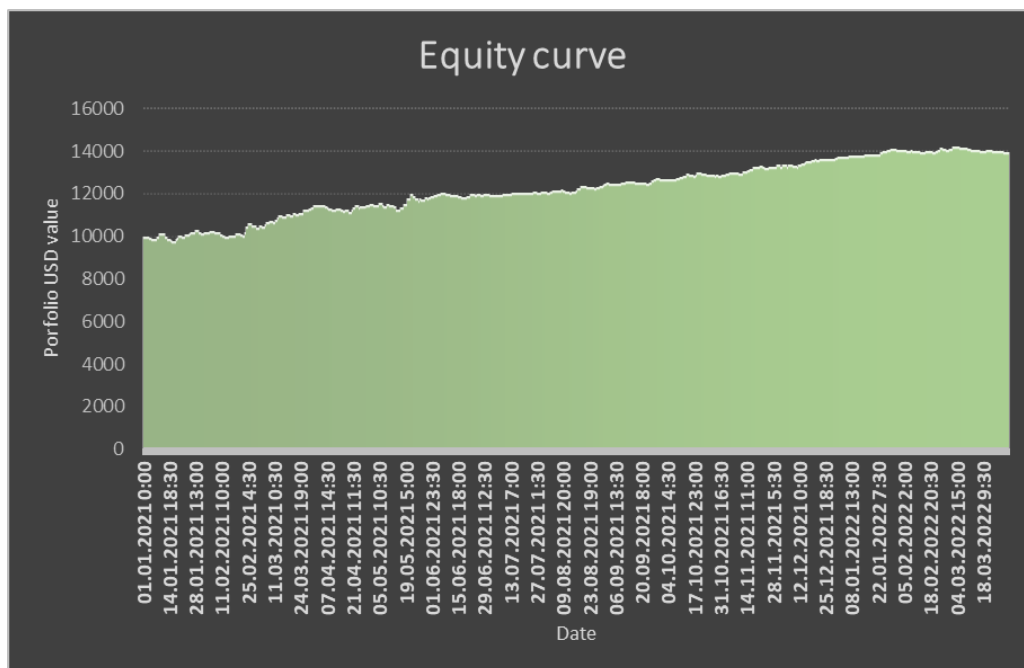
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$243.

The strategy executed 238 trades in total from which 129 were long positions and 109 short. From the total, 125 were profitable – 65 longs and 60 shorts, this led to an overall profitability of 52.52%, 50.39% for the side of long positions and 55.05 % which shows that longs performed worse than shorts by a small amount.

The average profit of a trade was around \$15,5. The average profit of a winning one was \$77 and \$50 for the losing one, this metric shows that when wins came, they were bigger than losses and because the overall trade profitability was over 52% the strategy brought in more profits than losses making it overall profitable.

The largest winning trade that led to a gain of \$260 was a short position and the largest losing position that led to a loss of \$144 was also a short position so the largest win brought in twice as much money than what the biggest loss cost.

Figure 46: VWAP close Bitcoin in-sample equity curve



The figure above shows the development of the equity curve where we can see a steady rise from the beginning to almost the end where it is going sideways which indicates a good performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$13,717 showing a rise of over 37%.

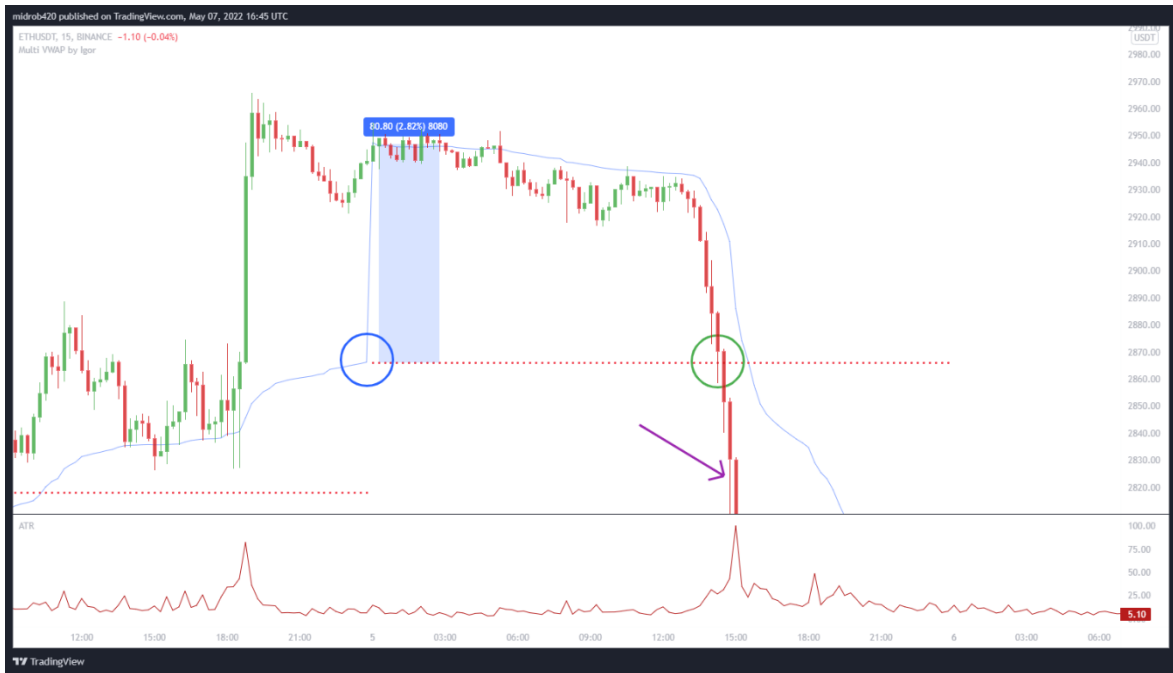
6.5.2 Ethereum

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$46,143.

Since this strategy uses only limit orders an exchange with low maker fees would be the ideal place to test it, therefore Bybit was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.01%. Those 0.01% are accounted for in the backtesting.

In the following figure we can see the strategy in action. The required distance marked by the blue rectangle is reached by the price opening way above the VWAP daily close marked by the blue circle. A limit order is then placed on the red line which is the previous day VWAP close. When the price reaches the VWAP close a long position is entered which is marked by the green circle. Unfortunately, the price goes the other way, and the stop loss is hit marked by the purple arrow. In the bottom part of the picture, we can see the ATR which is used to calculate the stop loss and take profit.

Figure 47: VWAP close strategy on Ethereum



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Distance filter: 80
- Stop loss: 0,5
- Take profit: 1

The following table presents the results of the out-sample on Ethereum. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 18: VWAP close Ethereum in-sample results

ETHEREUM	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	2925,01	29,25%	1523,35	15,23%	1401,66	14,02%
Gross Profit	7555,62	75,56%	3956,97	39,57%	3598,65	35,99%
Gross Loss	4422,62	44,23%	2312,91	23,13%	2109,72	21,10%
Max Drawdown	330,49	3,30%				
Profit Factor	1,71		1,71		1,71	
Commission Paid	207,99		120,72		87,28	
Total Closed Trades	195		112		83	
Number Winning Trades	93		51		42	
Number Losing Trades	102		61		41	
Percent Profitable	47,69%		45,54%		50,60%	
Avg Trade	15,00	0,15%	13,60	0,14%	16,89	0,17%
Avg Winning Trade	81,24	0,81%	77,59	0,78%	85,68	0,86%
Avg Losing Trade	43,36	0,43%	37,92	0,38%	51,46	0,51%

Largest Winning Trade	272,76	2,73%	266,91	2,67%	272,76	2,73%
Largest Losing Trade	-188,18	1,88%	-85,08	0,85%	-188,18	1,88%

The table above shows that the strategy was overall profitable after the 455 days it was running. It made a gross profit of \$7,555 and gross loss of \$4,422 creating a solid profit factor of 1,71 and an overall profit of \$2,925 which equals to 29.25% of the initial capital. Of that \$2,925 profit \$1,523 was from long positions which equals to 15.23% and \$1,401 from short positions which equals to 14.02% showing that overall long positions made more profit. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$330.

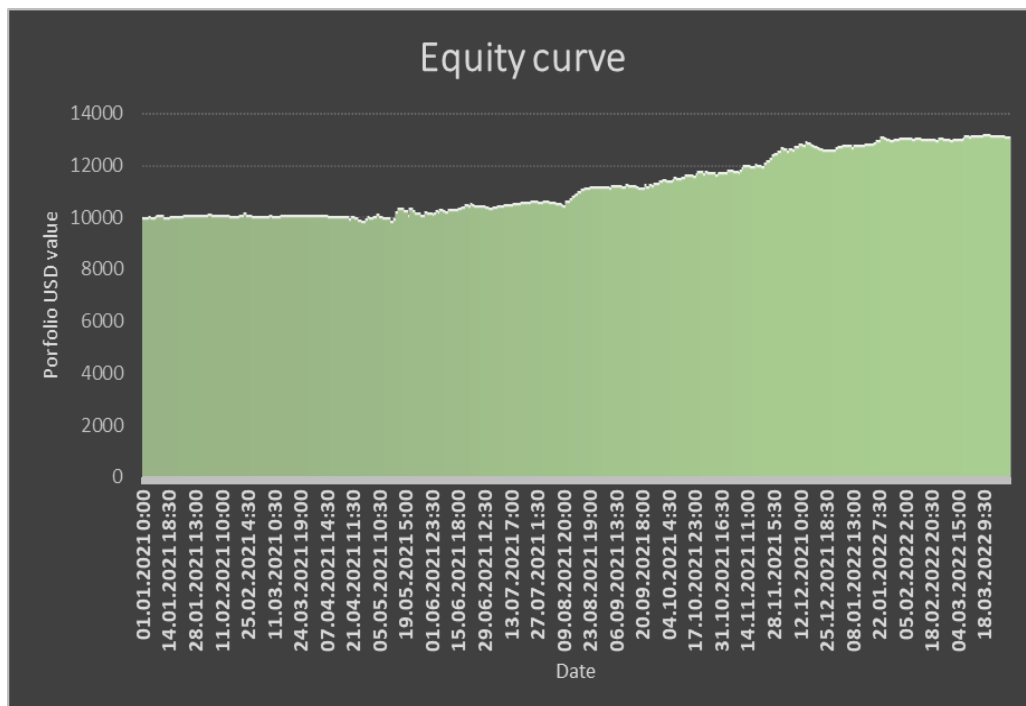
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$208.

The strategy executed 195 trades in total from which 112 were long positions and 83 short. From the total, 93 were profitable – 51 longs and 42 shorts, this led to an overall profitability of 47.69%, 45.54% for the side of long positions and 50.96% for the shorts so short. We can see that even though shorts had a higher winrate they brought in less profit.

The average profit of a trade was nearly \$15. The average profit of a winning one was \$81 and \$43 of the losing one, this metric shows that average profit of a winning trade was almost double to what the loss was making the strategy overall profitable even though it had a winrate around 50% the bigger wins simply outweighed the smaller losses by their.

The largest winning trade that led to a gain of \$277 was a short position and the largest losing trade that led to a loss of \$188 was also a short position. So again, the biggest win outweighed the loss.

Figure 48: VWAP close Ethereum in-sample equity curve



The figure above shows the development of the equity curve where we can see slight stagnation in the beginning which is then continued by a steady rise until the end. It shows that the trader's capital is growing over the time without any greater losses which is positive the only downside is the start where the strategy didn't do much. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$12,925 showing a rise of round 29%.

6.5.3 Polkadot

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$46,143.

Since this strategy uses only limit orders an exchange with low maker fees would be the ideal place to test it, therefore Bybit was chosen as the best available option for this strategy because it has the lowest taker fees of all exchanges presented which is 0.01%. Those 0.01% are accounted for in the backtesting.

In the following figure we can see the strategy in action. The required distance marked by the blue rectangle is reached by the price opening way below the VWAP daily close marked by the blue circle. A limit order is then placed on the red line which is the previous day VWAP close. When the price reaches the VWAP close a short position is entered which is marked by the red circle. The trade goes smoothly, and the take profit marked with an orange arrow is hit. In the bottom part of the picture, we can see the ATR which is used to calculate the stop loss and take profit.

Figure 49: VWAP close strategy on Polkadot



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Distance filter: 0,6
- Stop loss: 0,3
- Take profit: 1

The following table presents the results of the out-sample on Polkadot. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 19: VWAP close Polkadot in-sample results

POLKADOT	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	584,50	5,84%	1345,49	13,45%	-760,99	-7,61%
Gross Profit	8102,40	81,02%	5020,01	50,20%	3082,39	30,82%
Gross Loss	7222,46	72,22%	3511,28	35,11%	3711,18	37,11%
Max Drawdown	1145,77	11,46%				
Profit Factor	1,12		1,43		0,83	
Commission Paid	295,44		163,25		132,19	
Total Closed Trades	282		151		131	
Number Winning Trades	73		43		30	
Number Losing Trades	209		108		101	
Percent Profitable	25,89%		28,48%		22,90%	
Avg Trade	2,07	0,02%	8,91	0,09%	-5,81	-0,06%
Avg Winning Trade	110,99	1,11%	116,74	1,17%	102,75	1,03%
Avg Losing Trade	34,56	0,35%	32,51	0,33%	36,74	0,37%
Largest Winning Trade	352,10	3,52%	352,10	3,52%	199,58	2,00%
Largest Losing Trade	-166,43	1,66%	-97,99	0,98%	-166,43	1,66%

The table above shows that the strategy was overall profitable after the 455 days it was running but didn't make much overall profit. It made a gross profit of \$8,102 and gross loss of \$7,222 creating a profit factor of 1,12 and an overall profit of \$584 which equals to 5.84% of the initial capital. Of that \$584 profit \$1,345 was from long positions which equals to 13.45% and -\$760 from short positions which equals to -7.61% showing that long positions brought in profit whereas short positions weren't overall profitable. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$1,145 which is a lot compared to the overall profit.

The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$295.

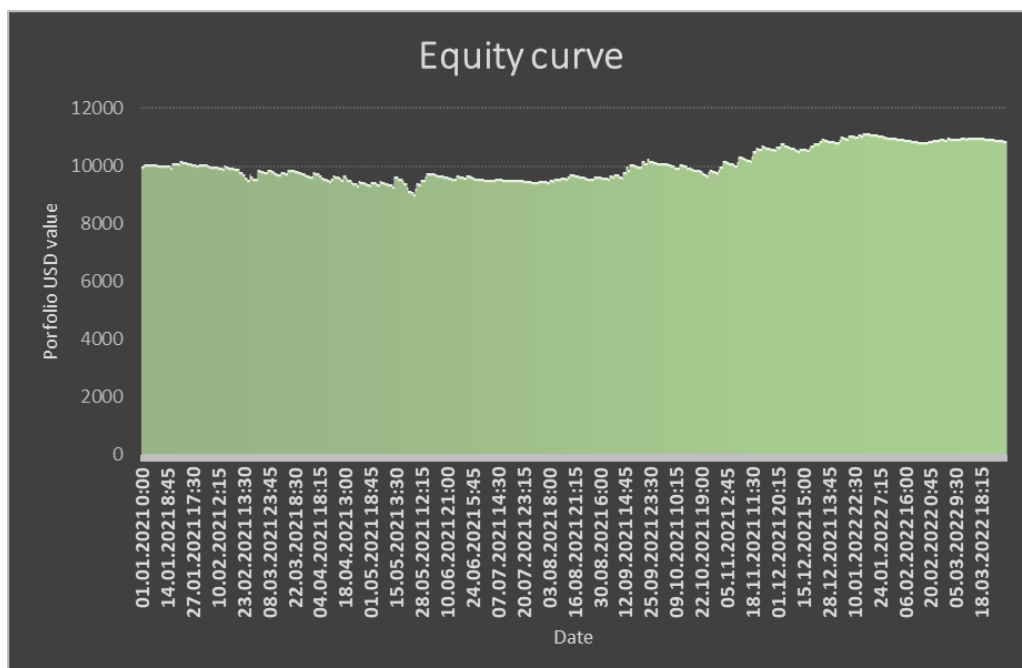
The strategy executed 282 trades in total from which 151 were long positions and 131 short. From the total, only 73 were profitable – 43 longs and 30 shorts, this led to an overall profitability of 25.89% which isn't a good result, 28.48% for the side of long positions and 22.90% for the shorts. The winrate percentage is quite low so the strategy would need a

really big difference between the average winning and losing trade to make it overall profitable.

The average profit of a trade was nearly \$2. The average profit of a winning one was \$110 and \$34 for the losing one, this metric shows that when wins came, they were much bigger than losses. Even though the average wins brought almost three times more money than the losses the overall winrate was so poor that the strategy realized only small gains.

The largest winning trade that led to a gain of \$352 was a long position whereas the largest losing position that led to a loss of \$166 was a short position. Even here the fact that the win is two times bigger than the loss didn't help the strategy to make any significant gains.

Figure 50: VWAP close Polkadot in-sample equity curve



The figure above shows the development of the equity curve where we can see that it is going down in the beginning and the strategy losing money which is then continued by a rise with few declines and at the end the curve is a little bit higher than at the start which indicates a poor performance of the strategy since the equity curve is downtrending for almost half of the tested time. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$10,584 showing small rise of only 5.84%.

6.5.4 Decentraland

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. This number was reached by dividing the initial capital by the backtesting period price average which for the 01.01.2021 to 31.03.2022 period was \$46,143.

Since this strategy uses only limit orders an exchange with low maker fees would be the ideal place to test it, therefore Bybit was chosen as the best available option for this

strategy because it has the lowest taker fees of all exchanges presented which is 0.01%. Those 0.01% are accounted for in the backtesting.

In the following figure we can see the strategy in action. The required distance marked by the blue rectangle is reached by the price opening way below the VWAP daily close marked by the blue circle. A limit order is then placed on the red line which is the previous day VWAP close. When the price reaches the VWAP close a short position is entered which is marked by the red circle. The trade goes well and the take profit marked by the orange arrow is hit making the trade a win. In the bottom part of the picture, we can see the ATR which is used to calculate the stop loss and take profit.

Figure 51: VWAP close strategy on Decentraland



own processing according to (TradingView)

After optimizing the strategy, the best results were found using the parameters below:

- Distance filter: 0,04
- Stop loss: 0,5
- Take profit: 1,05

The following table presents the results of the out-sample on Decentraland. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 20: VWAP close Decentraland in-sample results

DECENTRALAND	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	32,30	0,32%	326,44	3,26%	-294,14	-2,94%
Gross Profit	12653,27	126,53%	6891,89	68,92%	5761,38	57,61%

Gross Loss	12345,23	123,45%	6431,06	64,31%	5914,17	59,14%
Max Drawdown	2211,15	22,11%				
Profit Factor	1,02		1,07		0,97	
Commission Paid	275,74		134,39		141,35	
Total Closed Trades	209		103		106	
Number Winning Trades	77		41		36	
Number Losing Trades	132		62		70	
Percent Profitable	36,84%		39,81%		33,96%	
Avg Trade	0,15	0,00%	3,17	0,03%	-2,77	-0,03%
Avg Winning Trade	164,33	1,64%	168,09	1,68%	160,04	1,60%
Avg Losing Trade	93,52	0,94%	103,73	1,04%	84,49	0,84%
Largest Winning Trade	956,74	9,57%	956,74	9,57%	741,94	7,42%
Largest Losing Trade	-657,31	6,57%	-657,31	6,57%	-354,41	3,54%

The table above shows that the strategy made barely any overall profit after the 455 days it was running. It made a gross profit of \$12,653 and gross loss of \$12,345 creating a profit factor of 1,02 and an overall profit of \$32 which equals to 0,32% of the initial capital. Of that \$32 profit \$326 was from long positions which equals to 3.26% and -\$294 from short positions which equals to -2.94% showing that overall long positions only made profit and shorts were negative. The maximum drawdown which refers to the maximum amount of how much the strategy was down from its peak before reaching a new high was \$2,211.

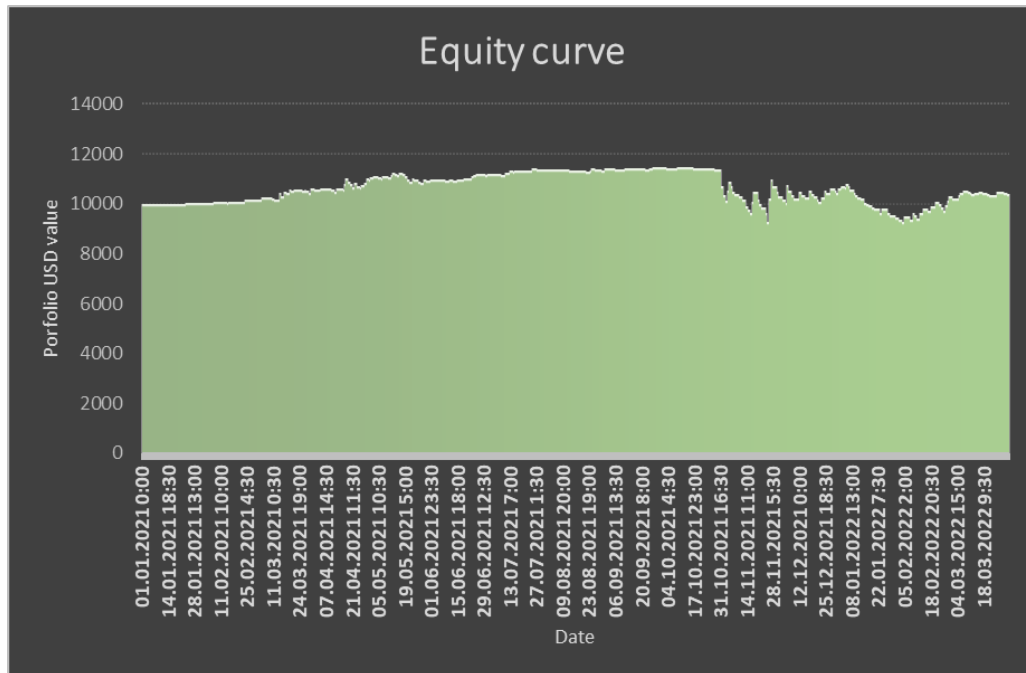
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$276.

The strategy executed 209 trades in total from which 103 were long positions and 106 short. From the total, 77 were profitable – 41 longs and 36 shorts, this led to an overall profitability of 36.84%, 39.81% for the side of long positions and 33.96% for the shorts. The overall winrate is poor and the only thing that could help the strategy to make more profit is if the average win was at least twice as big as the average loss.

The average profit of a trade was \$0,15 which already looks bad. The average profit of a winning one was \$164 and \$93 for the losing one, this metric shows that even though the average wins brought in more money, the poor winrate was the reason the strategy wasn't overall profitable.

The largest winning and losing trades were both long positions. The winning long led to a gain of \$956 and the losing one to a loss of -\$657.

Figure 52: VWAP close Decentraland in-sample equity curve



The figure above shows the development of the equity curve where we can see a slow uptrend in the beginning showing a good start but then the curve falls of a cliff and continues down ever after returning to where it started indicating an overall bad performance during the testing period. The development of the capital is recorded in the period from 01.01.2021 to 31.03.2022. The initial capital was \$10,000 which over the course of a year and three months has increased to \$10,032 showing a rise of 0,32%.

6.5.5 In-sample results summary

The following table presents the results of the tested strategy on all mentioned cryptocurrencies for the 455-day period. The first slope of each cryptocurrency presents the combined results of long and short positions in USD and the second one in percentage value.

Table 21: VWAP close in-sample results comparison

	BITCOIN		ETHEREUM		POLKADOT		DECENTRALAND	
	All \$	All %	All \$	All %	All \$	All %	All \$	All %
Net Profit	3717,93	37,18%	2925,01	29,25%	584,50	5,84%	32,30	0,32%
Gross Profit	9692,29	96,92%	7555,62	75,56%	8102,40	81,02%	12653,27	126,53%
Gross Loss	5731,05	57,31%	4422,62	44,23%	7222,46	72,22%	12345,23	123,45%
Max Drawdown	361,19	3,61%	330,49	3,30%	1145,77	11,46%	2211,15	22,11%
Profit Factor	1,69		1,71		1,12		1,02	
Commission Paid	243,31		207,99		295,44		275,74	
Total Closed Trades	238		195		282		209	
Number Winning Trades	125		93		73		77	
Number Losing Trades	113		102		209		132	
Percent Profitable	52,52%		47,69%		25,89%		36,84%	
Avg Trade	15,62	0,16%	15,00	0,15%	2,07	0,02%	0,15	0,00%
Avg Winning Trade	77,54	0,78%	81,24	0,81%	110,99	1,11%	164,33	1,64%
Avg Losing Trade	50,72	0,51%	43,36	0,43%	34,56	0,35%	93,52	0,94%
Largest Winning Trade	260,60	2,61%	272,76	2,73%	352,10	3,52%	956,74	9,57%
Largest Losing Trade	-144,82	1,45%	-188,18	1,88%	-166,43	1,66%	-657,31	6,57%

As mentioned before the initial capital invested in each cryptocurrency tested was \$10,000 and the number of traded contracts for each trade was adjusted to make the results comparable. The figure above shows that money wise the most profitable cryptocurrency for this strategy was Bitcoin making \$3,717 which was about 40% more than the amount of what the second largest recorded profit was. We can also see that the smaller market cap cryptocurrencies which are Polkadot and Decentraland made significantly less profits than what Ethereum and Bitcoin did but had higher gross profit and loss because they were more volatile than BTC and ETH therefore the percentage gains were much bigger but so were the losses. The highest profit factor was achieved on Bitcoin and Ethereum which reached around 1,7 this was the best ration between the gross profit and loss making those two the overall best performing cryptocurrencies for this strategy.

The highest commissions paid for trading was on Polkadot because the strategy opened 282 trades and the commission for each one was 0.01%.

When it comes to the number of trades, the strategy made much more buy and sell signals on Polkadot compared to the others which had similar trade counts.

When it comes to winrate Bitcoin and Ether were the best performers both reaching around 50% winrate and since this strategy is a scalping one which trades small price movements with risk to reward ratio between one to three the overall winrate matters for the strategy

to be profitable. The strategy on Polkadot had the highest difference between the average winning trade profit and loss. The win was three times bigger than the loss and thanks to that the strategy didn't end up losing money even though its overall winrate was the lowest compared to other cryptocurrencies since. Decentraland had a higher winrate but worse difference between the average trade win and loss profit, therefore it had the worst results profit wise making only \$32 after a year of trading which shows the strategy either needs adjusting or it simply isn't suitable to use on Decentraland.

The biggest winning (\$956) but also losing (\$657) trade was recorded on Decentraland, because of its extreme levels of volatility creating huge take profits and stop losses. The other tested assets had similar results to each other with the win being close to double the amount of the loss.

From the in-sample results we can say that the strategy suits Bitcoin and Ether the most.

6.6 VWAP close strategy out-sample

As mentioned above the out-sample period was between 01.04.2022 to 30.04.2022. The out-sample was made to verify whether the created strategy with its rules, signals and parameters is still reliable even in different market conditions. Same number of contracts and initial capital was used as in the in-sample to bring the out-sample as close as possible to the in-sample and to make the results easily comparable.

The out-sample data was processed the same way as in the in-sample and that is by using Excel and its logical functions and operators.

The following picture is the Bitcoin chart on the 4-hour timeframe with the marked out-sample period between the blue lines. During the month the price of Bitcoin was decreasing and with it were the prices of all other cryptocurrencies as well and faster. The 3 other tested altcoins did the exact same as Bitcoin with only fast and sharp movements to the upside which were then swiftly retraced even lower than where they originally began. Therefore, we can say that the market trend for the month was to the downside therefore, long positions were much riskier than short ones but, in this strategy, the overall trend shouldn't matter that much since it is focused on lower time frames.

Figure 53: Bitcoin out-sample price development



own processing according to (TradingView)

Bitcoin

The initial capital for trading Bitcoin was \$10,000 and for each trade 0,22 contracts were bought or sold. The same exchange was used (Bybit) to have the same trading fees as before (0.01%) which are accounted for.

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 22: VWAP close Bitcoin out-sample results

BITCOIN	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	11,14	0,11%	89,98	0,90%	-78,85	-0,79%
Gross Profit	218,79	2,19%	157,01	1,57%	61,78	0,62%
Gross Loss	195,70	1,96%	61,58	0,62%	134,13	1,34%
Max Drawdown	125,12	1,25%				
Profit Factor	1,12		1,27		1,15	
Commission Paid	11,96		5,46		6,50	
Total Closed Trades	13		6		7	
Number Winning Trades	6		4		2	
Number Losing Trades	7		2		5	
Percent Profitable	46,15%		66,67%		28,57%	
Avg Trade	0,86	0,01%	15,00	0,15%	-11,26	-0,11%
Avg Winning Trade	36,47	0,36%	39,25	0,39%	30,89	0,31%
Avg Losing Trade	27,96	0,28%	30,79	0,31%	26,83	0,27%

Largest Winning Trade	46,38	0,46%	46,38	0,46%	34,50	0,35%
Largest Losing Trade	-31,52	0,32%	-31,52	0,32%	-31,11	0,31%

The table above shows that the strategy ended up in small profit after the 30 days it was running. It made a gross profit of \$218 and gross loss of \$195 creating a profit factor of 1,12 and an overall profit of \$11 which equals to 0,11% of the initial capital. The total profit was entirely made out of long positions which brought in \$90, the short positions were not profitable and brought in a loss of \$-79. We can tell that the overall trend of the market doesn't matter in this strategy because even though the trend during the out-sample was down, short positions were overall negative whereas longs brought in profit. The maximum drawdown was as high as \$125.

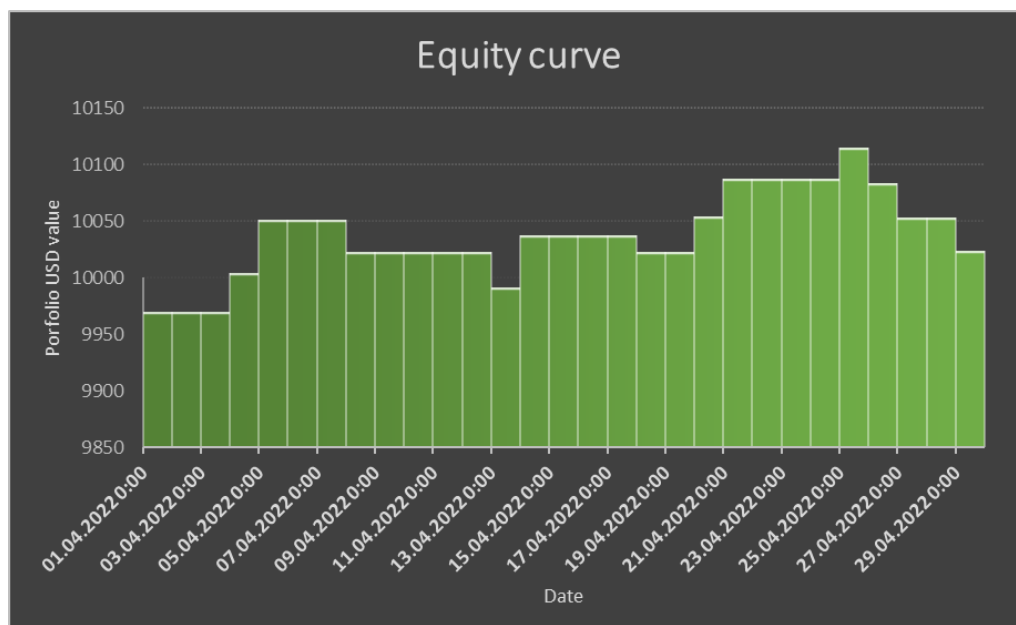
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$12.

The strategy executed 13 trades in total from which 6 were long positions and 7 short. From the total, 6 were profitable – 4 longs and 2 shorts, this led to an overall profitability of 46.15%, 66.67% for the side of long positions and 28.57% for the shorts. The difference between the winrate of longs is almost double which indicates a weakness in the strategy. Winrate wise the strategy performed worse on the side of short positions and better on the side of longs compared to the in-sample.

The average profit of a trade was \$0,86. The average profit of a winning one was \$36 and \$28 for the losing one. Because the winrate was around 50% the average winning trade is bigger than the loss the strategy is profitable. The ration between the average wining and losing trade is almost the same as in the in-sample.

Both the largest winning and losing trades were long positions, the win led to a gain of \$46 and the loss to a loss of \$31.

Figure 54: VWAP close Bitcoin out-sample equity curve



The figure above shows the development of the equity curve where we can see a decrease at the beginning and then a slow grind up which at the end is followed by a drop this indicates an average performance of the strategy during the tested period of time. The development of the capital is recorded in the period from 01.04.2021 to 30.04.2022. The initial capital was \$10,000 which over the course of a month has increased to \$10,011 showing a rise of around 0.11%.

Ethereum

The initial capital for trading Bitcoin was \$10,000 and for each trade 3,57 contracts were bought or sold. The same exchange was used (Bybit) to have the same trading fees as before (0.01%) which are accounted for.

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 23: VWAP close Ethereum out-sample results

ETHEREUM	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	-283,66	-2,84%	-131,60	-1,32%	-152,06	-1,52%
Gross Profit	180,69	1,81%	0,00	0,00%	180,69	1,81%
Gross Loss	453,17	4,53%	125,89	1,26%	327,28	3,27%
Max Drawdown	395,35	3,95%				
Profit Factor	0,40		N/A		0,55	
Commission Paid	11,18		5,72		5,47	
Total Closed Trades	10		5		5	
Number Winning Trades	4		0		4	
Number Losing Trades	6		5		1	
Percent Profitable	40,00%		0,00%		80,00%	
Avg Trade	-28,37	-0,28%	-26,32	-0,26%	-30,41	-0,30%
Avg Winning Trade	45,17	0,45%	N/A	N/A	45,17	0,45%
Avg Losing Trade	75,53	0,76%	25,18	0,25%	327,28	3,27%
Largest Winning Trade	68,37	0,68%	-21,21	-0,21%	68,37	0,68%
Largest Losing Trade	-327,28	3,27%	-28,69	0,29%	-327,28	3,27%

The table above shows that the strategy was not profitable overall after the 30 days it was running. It made a gross profit of \$180 and gross loss of \$453 creating a profit factor of 0,4 which is a very poor result and an overall loss of \$283 which equals to 2.84% of the initial capital. The loss of \$283 was made out of \$131 from long positions which equals to 1.32% and \$152 from short positions which equals to 1.52% showing neither the longs or shorts made any profit overall and the maximum drawdown was \$395 so the cryptocurrency that had the second-best results in the in-sample performed the worst in the out-sample.

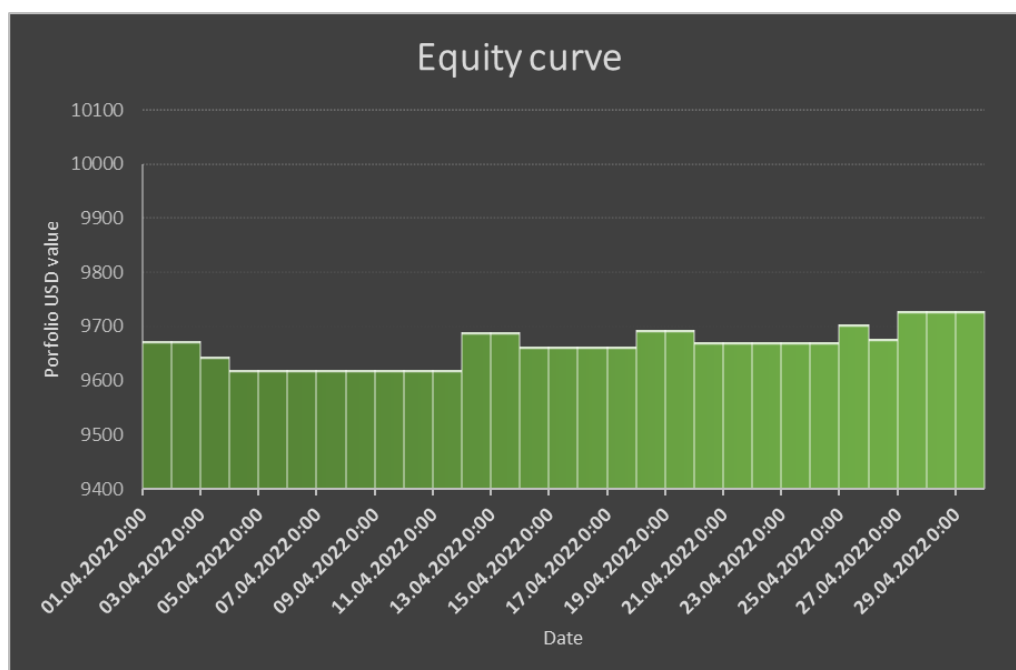
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$11.

The strategy executed 10 trades in total from which 5 were long positions and 5 short. From the total, 4 were profitable – 0 longs and 4 shorts, this led to an overall profitability of 40%, 0% for the side of long positions and 80% for the shorts. We can see that the performance of longs was miserable and even though shorts had an impressive 80% winrate they still didn't make any profit.

The average profit of a trade was -\$28. The average profit of a winning one was \$45 and \$75 of the losing one, this metric shows that not only the winrate was poor but also the average profit which was negative.

The largest winning and also losing trade were short positions that led to a gain of \$68 and a loss of \$327. Even in this metric the strategy performed poorly.

Figure 55: VWAP close Ethereum out-sample equity curve



The figure above shows the development of the equity curve where we can see that the strategy began by taking a big loss which it slowly started to recover but very poorly. Overall, the strategy didn't work well in the out-sample on Ethereum. The development of the capital is recorded in the period from 01.04.2022 to 30.04.2022. The initial capital was \$10,000 which over the course of month decreased to \$9,717 showing a decrease of around 3%.

Polkadot

The initial capital for trading Bitcoin was \$10,000 and for each trade 366 contracts were bought or sold. The same exchange was used (Bybit) to have the same trading fees as before (0.01%) which are accounted for.

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 24: VWAP close Polkadot out-sample results

POLKADOT	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	97,15	0,97%	57,05	0,57%	40,10	0,40%
Gross Profit	131,80	1,32%	78,35	0,78%	53,45	0,53%
Gross Loss	31,10	0,31%	19,08	0,19%	12,02	0,12%
Max Drawdown	19,08	0,19%				
Profit Factor	4,24		4,11		4,45	
Commission Paid	3,55		2,22		1,33	
Total Closed Trades	5		3		2	
Number Winning Trades	3		2		1	
Number Losing Trades	2		1		1	
Percent Profitable	60,00%		66,67%		50,00%	
Avg Trade	19,43	0,19%	19,02	0,19%	20,05	0,20%
Avg Winning Trade	43,93	0,44%	39,17	0,39%	53,45	0,53%
Avg Losing Trade	15,55	0,16%	19,08	0,19%	12,02	0,12%
Largest Winning Trade	53,45	0,53%	42,93	0,43%	53,45	0,53%
Largest Losing Trade	-19,08	0,19%	-19,08	0,19%	-12,02	0,12%

The table above shows that the strategy ended up in profit after the 30 days it was running. It made a gross profit of \$131 and gross loss of \$31 creating an amazing profit factor of 4,24 and an overall profit of \$97 which equals to 0.97% of the initial capital. Of that \$97 profit, \$57 was from long positions which equals to 0.78% and \$40 from short positions which equals to 0.40% showing that both longs and shorts made profit so even though the in-sample performance wasn't that impressive the outsample was good. The maximum drawdown was as \$19.

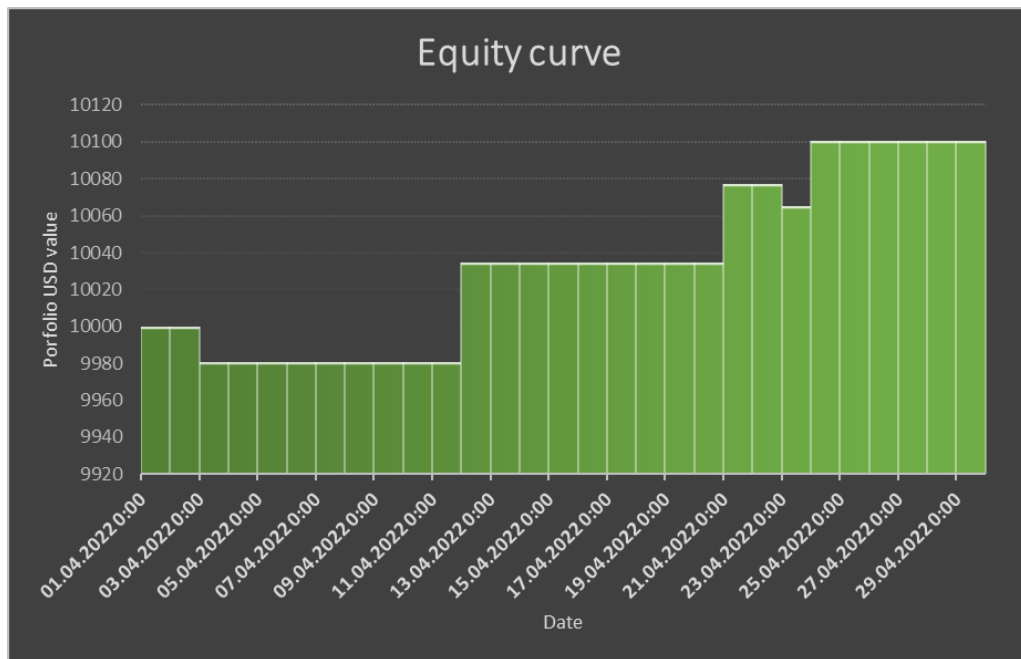
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$3,5.

The strategy executed 5 trades in total from which 3 were long positions and 2 short. From the total, 3 were profitable – 2 longs and 1 short, this led to an overall profitability of 60%, 66.67% for the side of long positions and 50% for the shorts.

The average profit of a trade was \$19. The average profit of a winning one was \$44 and \$15 of the losing one, this metric shows that the 60% winrate combined with the average profit being 3 times bigger than the loss gave good results.

The largest winning trade was a short position that led to a gain of \$53 and the largest loss was a long position with a loss of \$19. Therefore, the biggest gain would be enough to cover two and half times for the largest loss.

Figure 56: VWAP close Polkadot out-sample equity curve



The figure above shows the development of the equity curve where we can see a slight drop in the beginning that is then continued by a steady rise until the end indicating an overall good performance. The strategy worked even better in the out-sample than in the in-sample. The development of the capital is recorded in the period from 01.04.2022 to 30.04.2022. The initial capital was \$10,000 which over the course of month increased to \$10,097 showing an increase of around 0.97%.

Decentraland

The initial capital for trading Bitcoin was \$10,000 and for each trade 6660 contracts were bought or sold. The same exchange was used (Bybit) to have the same trading fees as before (0.01%) which are accounted for.

The following table presents the results of the out-sample on Bitcoin. The first two slopes present the combined results of long and short positions in USD and percentage value. The next two slopes present the results of long positions only in USD and percentage value and the last two slopes are the results of short positions only in USD and percentage value.

Table 25: VWAP close Decentraland out-sample results

DECENTRALAND	All \$	All %	Long \$	Long %	Short \$	Short %
Net Profit	-56,36	-0,56%	-34,72	-0,35%	-21,64	-0,22%
Gross Profit	456,72	4,57%	196,63	1,97%	260,08	2,60%
Gross Loss	493,37	4,93%	222,03	2,22%	271,33	2,71%
Max Drawdown	295,07	2,95%				
Profit Factor	0,93		0,89		0,96	
Commission Paid	19,71		9,32		10,39	
Total Closed Trades	13		6		7	
Number Winning Trades	5		2		3	

Number Losing Trades	8		4		4	
Percent Profitable	38,46%		33,33%		42,86%	
Avg Trade	-4,34	-0,04%	-5,79	-0,06%	-3,09	-0,03%
Avg Winning Trade	91,34	0,91%	98,32	0,98%	86,69	0,87%
Avg Losing Trade	61,67	0,62%	55,51	0,56%	67,83	0,68%
Largest Winning Trade	107,52	1,08%	107,52	1,08%	87,16	0,87%
Largest Losing Trade	-80,48	0,80%	-66,34	0,66%	-80,48	0,80%

The table above shows that the strategy was not profitable overall after the 30 days it was running. It made a gross profit of \$456 and gross loss of \$493 creating a profit factor of 0,93 and an overall loss of \$56 which equals to 0.56% of the initial capital. The loss of \$56 was made out of -\$34 from long positions which equals to -0.35% and -\$21 from short positions which equals to 0.22% showing neither the longs nor shorts made any profit overall and the maximum drawdown was \$295. Decentraland had overall poor results in the in-sample and the results of the out-sample were no different.

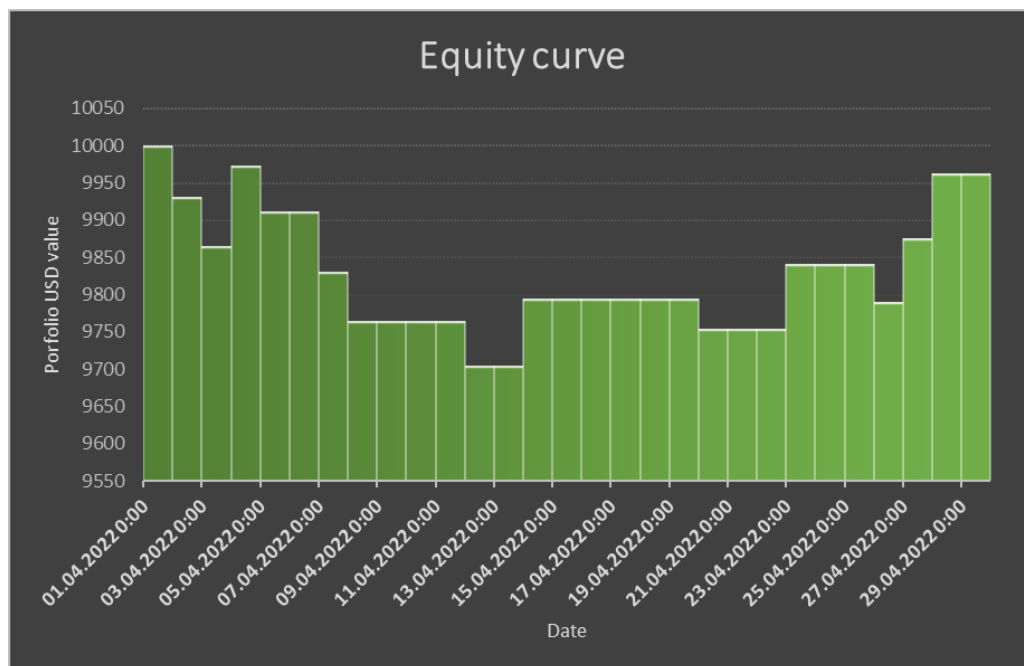
The amount paid to the exchange as trading fees for executing orders at predefined price was 0.01% for each trade making the total cost of opening all the positions \$20.

The strategy executed 13 trades in total from which 6 were long positions and 7 short. From the total, 5 were profitable – 2 longs and 3 shorts, this led to an overall profitability of 38.46%, 33.33% for the side of long positions and 42.86% for the shorts. The results show the poor performance of longs and shorts.

The average profit of a trade was -\$4. The average profit of a winning one was \$91 and \$61 of the losing one. Even though the average wins were bigger than losses the overall winrate was poor, so the strategy wasn't profitable.

The largest winning trade was a long position that led to a gain of \$107 and the largest loss was a short position with a loss of \$80. So, the biggest gain was almost the same as the biggest loss.

Figure 57: VWAP close Decentraland out-sample equity curve



The figure above shows the development of the equity curve which has a U shape starting with losses and then climbing back up indicating a poor performance. Overall, the strategy didn't work well in the in-sample and even the out-sample on Decentraland. The development of the capital is recorded in the period from 01.04.2022 to 30.04.2022. The initial capital was \$10,000 which over the course of month decreased to \$9,943 showing a decrease of around 0.56%.

VWAP close out-sample results summary

The following table presents the results of the tested strategy on all mentioned cryptocurrencies for the 30-day period. The first slope of each cryptocurrency presents the combined results of long and short positions in USD and the second one in percentage value.

Table 26: VWAP close out-sample results comparison

	BITCOIN		ETHEREUM		POLKADOT		DECENTRALAND	
	All \$	All %	All \$	All %	All \$	All %	All \$	All %
Net Profit	11,14	0,11%	-283,66	-2,84%	97,15	0,97%	-56,36	-0,56%
Gross Profit	218,79	2,19%	180,69	1,81%	131,80	1,32%	456,72	4,57%
Gross Loss	195,70	1,96%	453,17	4,53%	31,10	0,31%	493,37	4,93%
Max Drawdown	125,12	1,25%	395,35	3,95%	19,08	0,19%	295,07	2,95%
Profit Factor	1,12		0,40		4,24		0,93	
Commission Paid	11,96		11,184318		3,55		19,707693	
Total Closed Trades	13		10		5		13	
Number Winning Trades	6		4		3		5	
Number Losing Trades	7		6		2		8	
Percent Profitable	46,15%		40,00%		60,00%		38,46%	
Avg Trade	0,86	0,01%	-28,37	-0,28%	19,43	0,19%	-4,34	-0,04%
Avg Winning Trade	36,47	0,36%	45,17	0,45%	43,93	0,44%	91,34	0,91%
Avg Losing Trade	27,96	0,28%	75,53	0,76%	15,55	0,16%	61,67	0,62%
Largest Winning Trade	46,38	0,46%	68,37	0,68%	53,45	0,53%	107,52	1,08%
Largest Losing Trade	-31,52	0,32%	-327,28	3,27%	-19,08	0,19%	-80,48	0,80%

The initial capital invested in each cryptocurrency tested in the out-sample was \$10,000 and the number of traded contracts for each trade was the same as in the in-sample to make the results comparable. The figure above shows that profit wise the best performing cryptocurrency for this strategy was surprisingly Polkadot making \$97 which was around nine times more than what the second largest profit amount was which was Bitcoin and its \$11. Polkadot had also the best profit factor which was as high as 4,24 which is a fantastic result, but that might have changed if the strategy executed more than 5 trades. However, Ethereum's performance was a big disappointment compared to how the in-sample went because it performed well but in the out-sample it was by far the worst performer overall. Decentraland behaved the same way as in the in-sample executing a lot of trades and ending around where it started.

When it comes to the number of trades, they are pretty much the same ratio wise as in the in-sample, the major difference is Polkadot which had the most trades in the in-sample, whereas here it had the least. The strategy made the most buy and sell signals on Decentraland and Bitcoin.

When it comes to the overall winrate there were big changes between the best performers. Bitcoin and Decentraland had almost the exact same winrate as before. The big change was Polkadot whose winrate changed from 25% to 60% making it the best assets in this category. The results of Ethereum were about a 7% worse than in the in-sample.

Average profit and loss wise all tested assets had higher average profit than loss during the in-sample period, this change for Ethereum being the worst performer with the average win being three-quarters of the loss. The rest behaved the same as in the in-sample, so Polkadot was the winner with the difference between the average win being three times bigger than the loss and also having the best average trade overall results.

The biggest profit (\$107) but also loss (\$80) was recorded same as in the in-sample on Decentraland, because of its extreme levels of volatility creating huge take profits and stop losses based upon it. From the rest of the tested assets only Polkadot remained as the one with the largest win being over double the amount of the loss.

Overall, the VWAP close strategy proved to be profitable in both the in-sample and the out-sample only on two of the four tested assets which were Bitcoin and Polkadot ranking it as an average trading strategy that is very situational, needs improvement and is not usable on every asset.

Conclusion

The bachelor's thesis was about creating a trading strategy based on the tools of technical analysis for selected cryptocurrencies. Its goal was to create a profitable trading system, according to which a trader could trade. The main cryptocurrencies the bachelor's thesis focused on were Bitcoin, Ethereum, Polkadot and Decentraland.

The first chapters of the theoretical part explained the use case of blockchain itself and in cryptocurrencies and briefly talked about the cryptocurrency space and how it does or doesn't affect our real lives. It also included a detailed analysis about a few selected cryptocurrencies about their characteristics, price development and use cases.

The next chapters presented the differences and similarities between the traditional and cryptocurrency markets in trading, topics like market cycles, volatility, software, automation, and correlation between the two were discussed. After finding the differences in trading the cryptocurrency exchanges and traditional market brokers were compared in aspects like trading fees, offered services, security and a trader profile was created for each one.

The last theoretical chapter talked about the tools and indicators of technical analysis, from which some were then applied in the practical part to the charts of the mentioned cryptocurrencies. The last part of this chapter focused on risk and money management and how a trader should behave while trading the markets, and how important it is to set a correct set of rules when creating of a suitable trading strategy.

The practical part of this work was devoted to the creation of various trading strategies by applying the previously mentioned tools and indicators of technical analysis to the charts of the selected cryptocurrencies – Bitcoin, Ethereum, Polkadot and Decentraland and backtesting the results on new data to prove the profitability of the created strategies. The data and charts were obtained from tradingview.com website, and subsequently modified for the purposes of this work. The values were processed using MS Excel and TradingView, where mathematical and logical functions the software offers were used. The aim of this work was to find a profitable trading strategy for the cryptocurrency markets.

In conclusion, after creating, optimizing, and testing the two strategies the results proved that the overall performance of the RSI + MA strategy was good and profitable across all the tested assets in both the in and out-samples whereas the VWAP close was profitable only on two out of the four assets tested proving the RSI + MA to be the better, more consistent, and overall, a much suitable strategy ready for trading.

List of abbreviations

APR	Annual Percentage Rate
APY	Annual Percentage Yield
ATR	Average True Range
BIT	Bit Token
BNB	Binance Coin
BTC	Bitcoin
CFD	Contract for differences
CME	Chicago Mercantile Exchange
DCA	Dollar-Cost Averaging
DOT	Polkadot
EMA	Exponential Moving Average
ETF	Exchange-Traded Fund
ETH	Ethereum
IB	Initial Balance
KCS	Kucoin Token
KYC	Know Your Customer
MA	Moving Average
NFT	Non-Fungible Token
MANA	Decentraland
OI	Open Interest
P2P	Peer to Peer
PoS	Proof of Stake
PoW	Proof of Work
TP	Take Profit

RSI	Relative Strength Index
S&P 500	Standard & Poor's 500
SL	Stop Loss
SMA	Simple Moving Average
UI	User Interface
USD	United States Dollar
USDT	United States Dollar Token
VR	Virtual Reality
VWAP	Volume Weight Average Price

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ATAS professional trading and analytic platform. ATAS analysis program: time&sales (time and sales), smart tape, atas allows to analyze point volumes

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Binance is one of the largest cryptocurrency exchanges in the world, responsible for \$7.7 trillion crypto exchange volume in 2021. It was founded in 2017 by Changpeng Zhao, who previously worked for Blockchain.info and as CTO of OKCoin. Launched in Hong Kong, it quickly became one of the leading exchanges, but from the get-go faced increasingly restrictive regulations which forced it to migrate servers and headquarters to Malta. Binance has one of the widest selections of alt-coins available on its exchange and even launched its own coin, Binance Coin, which is currently the fifth most valuable coin by market cap behind

Bitcoin, Ethereum, USD Tether and Cardano. It has also launched several security tools aimed at improving trust in crypto, these include Binance Smart Chain, which

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Building robots for trading. What should a beginner know?, 2019. *Atas.net*. Online. [Accessed 11 March 2022]. Available from: <https://atas.net/atas-possibilities/charts/building-robots-for-trading/>

Did trading start to be too time consuming? Have you had thoughts about building a robot? What to start with? We will answer all questions of beginner robot-builders.

Co to je Proof-of-Stake a jak funguje?, [no date]. Online. [Accessed 8 March 2022]. Available from: <https://diit.cz/clanek/co-je-proof-stake-jak-funguje>

Mnozí z vás zřejmě slyšeli o tom, že síť Ethereum má projít změnou z Proof-of-Work na Proof-of-Stake. Jde o největší přechod, který kdy svět kryptoměn zažil. Druhá největší kryptoměna kompletně změní způsob stanovování konsensu. Jenže co to je Proof-of-Stake?

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A Computer Science portal for geeks. It contains well written, well thought and well explained computer science and programming articles, quizzes and practice/competitive programming/company interview Questions.

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Discover how crypto exchanges incentivize users with native cryptocurrency exchange tokens which can lower trading fees, enhance liquidity, and improve trust.

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Staking is a way to put your crypto to work and earn rewards on it.

Decentraland: Je virtuální svět plný kasín a klubů opravdu decentralizovaný?, [no date]. Online. [Accessed 16 March 2022]. Available from: <https://diit.cz/clanek/decentraland-je-virtualni-svet-plny-kasin-klubu-opravdu-decentralizovany>
Decentraland je blockchain aplikace, ve které si můžete ve virtuálním světě vytvořit cokoliv se vám zlíbí. A to bez cenzury. Jenže je tato hra opravdu decentralizovaná nebo jde jen o marketingový tah?

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By Tobias Adrian, Tara Iyer and Mahvash S. Qureshi There's a growing interconnectedness between virtual assets and financial markets. Crypto assets such as Bitcoin have matured from an obscure asset class with few users to an integral part of the digital asset revolution, raising financial stability concerns. Crypto assets are no

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Bitcoin halving is when the rate of new bitcoins entering circulation is cut in half, which occurs approximately every four years.

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Leverage results from using borrowed capital as a source of funding when investing to expand a firm's asset base and generate returns on risk capital.

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Most people think of crypto mining simply as a way of creating new coins. Learn More....

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Ethereum Ecosystem analytics by and for the community. Explore and share data from Ethereum Mainnet, Matic, Optimism and xDai for free.

ORDANO, Esteban, MEILICH, Ariel, JARDI, Yemel and ARAOZ, Manuel, [no date]. A blockchain-based virtual world. . P. 15. Decentraland is a virtual reality platform powered by the Ethereum blockchain. Users can create, experience, and monetize content and applications. Land in Decentraland is permanently owned by the community, giving them full control over their creations. Users claim ownership of virtual land on a blockchain-based ledger of parcels. Landowners control what content is published to their portion of land, which is identified by a set of cartesian coordinates (x,y). Contents can range from static 3D scenes to interactive systems such as games.

Otevírací časy burz - SIGNAL TRADE, [no date]. Online. [Accessed 1 March 2022]. Available from: <http://www.signaltrade.cz/oteviraci-casy-burz/>

Overbought vs. Oversold and What This Means for Traders, [no date]. Online. [Accessed 30 March 2022]. Available from: <http://finance.yahoo.com/news/overbought-vs-oversold-means-traders-030200320.html>
Overbought vs. Oversold and What This Means for Traders

Play-to-Earn Crypto Gaming: Beginner’s Guide, 2022. *FinanceFeeds*. Online. [Accessed 20 March 2022]. Available from: <https://financefeeds.com/play-earn-crypto-gaming-beginners-guide/>
The gaming industry is slated to undergo tremendous changes over the coming years. Rather

than players making publishers rich, developers are exploring opportunities to let gamers make money. As a result, Play-to-earn gaming is a very appealing concept, although one may need to adjust their expectations first and foremost.

Polkadot Network, [no date]. *Polkadot Network*. Online. [Accessed 1 March 2022 a]. Available from: <https://polkadot.network/>
Polkadot empowers blockchain networks to work together under the protection of shared security.

Polkadot Network, [no date]. *Polkadot Network*. Online. [Accessed 16 March 2022 b]. Available from: <https://polkadot.network/auctions/>
Find out how Polkadot parachain slot auctions and crowdloans raise the bar for blockchain projects.

Poplatky eToro – jaké jsou a jak se počítají, [no date]. *eToro*. Online. [Accessed 5 May 2022]. Available from: <https://www.eto.com/cs-cz/trading/fees/>
Přečtěte si o poplatcích eToro. Otevření účtu a vložení peněz je bezplatné, ale některé akce jsou zpoplatněny a tyto poplatky jsou na stránkách přehledně uvedeny.

PRICEWATERHOUSECOOPERS, [no date]. Making sense of bitcoin, cryptocurrency and blockchain. *PwC*. Online. [Accessed 2 March 2022]. Available from: <https://www.pwc.com/us/en/industries/financial-services/fintech/bitcoin-blockchain-cryptocurrency.html>

Understand how Blockchain works, what cryptocurrency is and the potential blockchain applications beyond Bitcoin and cryptocurrency to improve business processes.

PROFESSOR, 2022. How to gain exposure to crypto without buying it. *MoneySense*. Online. 31 January 2022. [Accessed 22 March 2022]. Available from: <https://www.moneysense.ca/save/investing/crypto/how-to-gain-exposure-to-crypto-without-buying-it/>

If you're not ready to dive into cryptocurrency ownership but still want some exposure to digital coins, here's how to invest indirectly.

Proof of Work (PoW) Consensus, 2019. *GeeksforGeeks*. Online. [Accessed 2 March 2022]. Available from: <https://www.geeksforgeeks.org/proof-of-work-pow-consensus/>
A Computer Science portal for geeks. It contains well written, well thought and well explained computer science and programming articles, quizzes and practice/competitive programming/company interview Questions.

Proof-of-stake (PoS), [no date]. *ethereum.org*. Online. [Accessed 4 March 2022]. Available from: <https://ethereum.org>
An explanation of the proof-of-stake consensus protocol and its role in Ethereum.

Pros and Cons of Automated Trading Systems, [no date]. *Investopedia*. Online. [Accessed 15 March 2022]. Available from: <https://www.investopedia.com/articles/trading/11/automated-trading-systems.asp>
What are some of the pros and cons of automated trading systems? Read on to find out how these systems can make trading easier, not to mention some of their pitfalls.

Regulation T (Reg T), [no date]. *Investopedia*. Online. [Accessed 23 March 2022]. Available from: <https://www.investopedia.com/terms/r/regulationt.asp>

Regulation T, or Reg T, governs cash accounts and the amount of credit that broker-dealers can extend to investors for the purchase of securities.

Relative Strength Index (RSI), [no date]. *Investopedia*. Online. [Accessed 29 March 2022]. Available from: <https://www.investopedia.com/terms/r/rsi.asp>
The Relative Strength Index (RSI) is a momentum indicator that measures the magnitude of recent price changes to analyze overbought or oversold conditions.

RIVERO, Nicolás, [no date]. The biggest corporate holder of bitcoin is not Square or Tesla. *Quartz*. Online. [Accessed 22 March 2022]. Available from: <https://qz.com/2118492/the-biggest-corporate-holder-of-bitcoin-is-not-square-or-tesla/>
At least 26 publicly traded companies hold bitcoin, including automakers, video game developers, fossil fuel investors, mortgage lenders, insurers, and one massive whale that owns 56% of all corporate crypto.

Samsung Debuts on the Decentraland Ethereum Metaverse – Trustnodes, [no date]. Online. [Accessed 16 March 2022]. Available from: <https://www.trustnodes.com/2022/02/09/samsung-debuts-on-the-decentraland-ethereum-metaverse>

SERGEENKOV, Andrey, 2021. What Does Liquidation Mean and How to Avoid It? Online. 13 August 2021. [Accessed 23 March 2022]. Available from: <https://www.coindesk.com/markets/2021/08/13/what-does-liquidation-mean-and-how-to-avoid-it/>
Liquidation happens when a trader has insufficient funds to keep a leveraged trade open.

Simple Moving Average (SMA), [no date]. *Investopedia*. Online. [Accessed 30 March 2022]. Available from: <https://www.investopedia.com/terms/s/sma.asp>
A simple moving average (SMA) calculates the average of a selected range of prices, usually closing prices, by the number of periods in that range.

Smart Contracts: What You Need to Know, [no date]. *Investopedia*. Online. [Accessed 16 March 2022]. Available from: <https://www.investopedia.com/terms/s/smart-contracts.asp>
Smart contracts are self-executing contracts with the terms of the contract between buyer and seller directly written into lines of code.

SOUKUP, Tomáš, 2021. KYC: Proč po vás makléři a burzy chtějí osobní údaje? » Finex.cz. *Finex.cz*. Online. 15 April 2021. [Accessed 29 March 2022]. Available from: <https://finex.cz/kyc-verifikace-na-burzach-a-u-brokeru/>
Jednou z nejčastějších otázek, na kterou se nás uživatelé ptají, je ověřování totožnosti u brokerů a kryptoměnových burz. Řada lidí má ze sdílení svých dokladů a osobních údajů obavy. Verifikace je přitom standardní součástí všech solidní obchodních platforem.

STALLINGS, William, 2017. *Cryptography and network security: principles and practice*. . Seventh edition. Boston: Pearson. ISBN 978-0-13-444428-4. TK5105.59 .S713 2017

Stock Market Map, [no date]. Online. [Accessed 1 March 2022]. Available from: https://finviz.com/map.ashx?t=sec_all

The 15 Biggest NFT Sales of All Time, 2022. *nft now*. Online. [Accessed 18 March 2022].

Available from: <https://nftnow.com/features/top-10-nft-sales-2021/>
To celebrate the most expensive NFTs, we've compiled a list of the biggest NFT sales – from Beeple to CryptoPunks and everything in-between.

The Bitcoin and Crypto Market Cycles: What You Need to Know, 2018. *CryptoPotato*. Online. [Accessed 23 March 2022]. Available from: <https://cryptopotato.com/what-are-crypto-market-cycles/>

What are market cycles in Bitcoin and other cryptocurrencies, and how do they impact the prices over the long-term? A comprehensive deep dive guide.

The idea and a brief history of cryptocurrencies, 2021. *The Guardian Nigeria News - Nigeria and World News*. Online. [Accessed 1 March 2022]. Available from: <https://editor.guardian.ng/news/the-idea-and-a-brief-history-of-cryptocurrencies/>

Cryptocurrencies are a type of digital currency decentralized and not controlled by any government. The history of cryptocurrencies can be traced back to the 1980s, when they were called cyber currencies. These coins started gaining in popularity in 2008 with the introduction of Bitcoin, which was created by an anonymous programmer or group of programmers [...]

The New York Stock Exchange | NYSE, [no date]. Online. [Accessed 23 March 2022]. Available from: <https://www.nyse.com/index>

Top 12 Smart Contract Use Cases, 2021. *101 Blockchains*. Online. [Accessed 16 March 2022]. Available from: <https://101blockchains.com/smart-contract-use-cases/>

If you are new to smart contracts, you might not be sure of its capability. This article will try to clear all your confusions by sharing the best smart

Trading Bot 101: KuCoin Trading Bot Now Supports 4 Strategies! How To Find The Bot Fits You Most? | KuCoin, [no date]. Online. [Accessed 11 March 2022]. Available from: <https://www.kucoin.com/blog/kucoin-trading-bot-now-supports-4-strategies-how-to-find-the-bot-fits-you-most>

Launched in early 2021, KuCoin Trading Bot has been up and running for some time. This product has gained considerable attention and interest as to whether in a

Trading Fee Rate | Cryptocurrency Exchange Fee | KuCoin, [no date]. Online. [Accessed 5 May 2022]. Available from: <https://www.kucoin.com/vip/level>
Find out KuCoin trading fee rate for different trading volume and investor type.

Trading Fee Structure, [no date]. *Bybit Official Help*. Online. [Accessed 5 May 2022]. Available from: <https://help.bybit.com/hc/en-us/articles/360039261154-Trading-Fee-Structure>

Trading Fee StructureVIP LevelSpot TradingPerpetual & Futures TradingUSDC Options TradingTaker Fee RateMaker Fee RateTaker Fee RateMaker Fee RateTaker Fee Rate...

TradingView – Track All Markets, [no date]. *TradingView*. Online. [Accessed 1 March 2022]. Available from: <https://www.tradingview.com/>
Where the world charts, chats and trades markets. We're a supercharged super-charting platform and social network for traders and investors. Free to sign up.

TradingView Trading Platform capabilities and features, [no date]. *TradingView*. Online.

[Accessed 8 March 2022]. Available from: <https://www.tradingview.com/features/> TradingView provides latest stock, futures, index, Forex and Bitcoin data in a useful format that works on any device.

Trality - Create & follow trading bots, [no date]. Online. [Accessed 1 March 2022]. Available from: <https://www.trality.com/> Trality is the platform for anyone to create and invest through automated crypto trading bots. Creators can build the sophisticated bots in our browser-based Python editor. Followers can copy-trade on bots via an easy-to-use mobile app.

TRETINA, Kat, 2021. How To Buy Cryptocurrency. *Forbes Advisor*. Online. 12 July 2021. [Accessed 20 March 2022]. Available from: <https://www.forbes.com/advisor/investing/how-to-buy-cryptocurrency/> If you're new to the world of crypto, figuring out how to buy Bitcoin, Dogecoin, Ethereum and other cryptocurrencies can be confusing at first. Thankfully, it's pretty simple to learn the ropes. You can start investing in cryptocurrency by following these five easy steps.1. Choose a Broker or Cryp

Úvod do peer-to-peer obchodování: co je P2P obchodování a jak funguje místní bitcoinová směnárna? | Blog Binance, [no date]. *Binance Blog*. Online. [Accessed 20 March 2022]. Available from: <https://www.binance.com/cs/blog/p2p/%C3%BAvod-do-peertopeer-obchodov%C3%A1n%C3%AD-co-je-p2p-obchodov%C3%A1n%C3%AD-a-jak-funguje-m%C3%ADstn%C3%AD-bitcoinov%C3%A1-sm%C4%9Bn%C3%A1rna-421499824684901839>

Uživatelé, kteří dávají přednost decentralizovanému přístupu k nákupu a prodeji kryptoměn, se často uchylují k peer-to-peer obchodování. Peer-to-peer obchodování nebo taky P2P obchodování, jak se běžně nazývá, umožňuje přímo propojit kupující a prodávající bez potřeby zprostředkovatele nebo prostředníka. Zjistěte, jak P2P směnárný fungují a jaké má P2P obchodování výhody a nevýhody.

Volume Weighted Average Price (VWAP) Indicator Explained, 2016. *SpeedTrader*. Online. [Accessed 29 March 2022]. Available from: <https://speedtrader.com/volume-weighted-average-price-vwap-explained/> The volume weighted average (VWAP) is a technical indicator that is popular amongst day traders. VWAP serves as a moving average that is particularly significant for intraday traders.

Welcome to Decentraland, [no date]. Online. [Accessed 1 March 2022]. Available from: <https://decentraland.org/> Create, explore and trade in the first-ever virtual world owned by its users.

What Are Launchpads In Crypto And Most Popular Ones, 2022. Online. [Accessed 28 March 2022]. Available from: <https://www.jumpstartmag.com/what-are-launchpads-in-crypto-and-which-are-the-most-popular-ones/> What Are Launchpads in Crypto and Which Are the Most Popular Ones As a startup new to the crypto space, you might be overwhelmed by the different ways,

What is automated trading, [no date]. *IG*. Online. [Accessed 11 March 2022]. Available from: <https://www.ig.com/en/trading-platforms/algorithmic-trading/what-is-automated-trading> Find out what automated trading is, how it works, and the benefits of using an automated

trading system.

What is cryptocurrency and how does it work?, 2022. *www.kaspersky.com*. Online. [Accessed 1 March 2022]. Available from: <https://www.kaspersky.com/resource-center/definitions/what-is-cryptocurrency>

Cryptocurrency is a digital currency using cryptography to secure transactions. Learn about buying cryptocurrency and cryptocurrency scams to look out for.

What Is Ethereum Blockchain; and its Key Use Cases?, [no date]. *Gemini*. Online. [Accessed 16 March 2022]. Available from:

<https://www.gemini.com/cryptopedia/ethereum-smart-contracts-tokens-use-cases>,
<https://www.gemini.com/cryptopedia/ethereum-smart-contracts-tokens-use-cases>

The Ethereum blockchain has proven itself as the place for original and ambitious decentralized applications, from DAOs to NFTs to DeFi.

What is Gapping? (And Playing the 4 Types of Gap), [no date]. *My Trading Skills*. Online. [Accessed 9 March 2022]. Available from: <https://mytradingskills.com/what-is-gapping>

You may have seen it in the market but what is gapping and why do the markets gap? we also reveal how you can play the gaps. See inside.

What is Investment Position Sizing?, [no date]. *Investopedia*. Online. [Accessed 1 April 2022]. Available from:

<https://www.investopedia.com/terms/p/positionsizing.asp>

Position sizing refers to the size of a position within a particular portfolio, or the dollar amount that an investor is going to trade.

What is Open Interest? Definition of Open Interest, Open Interest Meaning, [no date]. *The Economic Times*. Online. [Accessed 29 March 2022]. Available from: <https://economictimes.indiatimes.com/definition/open-interest>

Open Interest definition - What is meant by the term Open Interest? meaning of IPO, Definition of Open Interest on The Economic Times.

What is the Metaverse? Crypto Games with NFTs, [no date]. *Gemini*. Online. [Accessed 4 April 2022]. Available from: <https://www.gemini.com/cryptopedia/what-is-metaverse-crypto-nft-game-blockchain>, <https://www.gemini.com/cryptopedia/what-is-metaverse-crypto-nft-game-blockchain>

<https://www.gemini.com/cryptopedia/what-is-metaverse-crypto-nft-game-blockchain>

Learn about immersive virtual worlds known as crypto metaverses. Discover how the metaverse helps drive blockchain games — and NFT games in particular.

What is yield farming in DeFi, and how does it work?, 2021. Online. [Accessed 20 March 2022]. Available from: <https://forkast.news/what-is-yield-farming-how-it-works/>

One of DeFi's hottest trends, yield farming can bring riches but is not for crypto beginners or those who can't afford to lose. Forkast.News explains why.

Why is Cryptocurrency Rising in Popularity?, 2022. *Supply Chain Game Changer™*. Online. [Accessed 2 March 2022]. Available from:

<https://supplychaingamechanger.com/why-is-cryptocurrency-rising-in-popularity/>

Cryptocurrency is rising in popularity and is constantly in the news. What is this digital money and why is it important for the future?

WOOD, Dr Gavin, [no date]. POLKADOT: VISION FOR A HETEROGENEOUS MULTI-CHAIN FRAMEWORK. P. 21.

Present-day blockchain architectures all suffer from a number of issues not least practical means of extensibility and scalability. We believe this stems from tying two very important parts of the consensus architecture, namely canonicity and validity, too closely together. This paper introduces an architecture, the heterogeneous multi-chain, which fundamentally sets the two apart.

XTB - Trade your way, [no date]. *XTB.com*. Online. [Accessed 31 March 2022]. Available from: <https://www.xtb.com/en/learn-to-trade/chart-types>

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